Together First, Apart Later? When Optimizers Start to Learn Different Functions

Project report, MIT 6.7910 Statistical Learning Theory, Fall 2026

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Draft, October 31, 2026

Abstract

From the same initialization and the same sequence of mini-batches, we train small transformer language models with SGD with momentum, AdamW and Muon, and ask when their learned functions start to differ. We measure the distance between the next-token predictions of two runs on held-out text, compare runs at equal training loss so that speed is not mistaken for difference, and calibrate against the distance between two runs of the same optimizer that differ only in data order. For each pair of optimizers we look for a persistent split, a loss level from which their distance stays above this floor, with a bootstrap over data orders, and test what coincides with the changes in distance: sharpness reaching its stability threshold, the end of warmup, or a loss level. The answer to the question of the title is no. After warmup, at equal training loss, two optimizers are about as far apart as two data orders of one optimizer, in all three language-model settings. The exceptions involve SGD: on TinyStories-10M it is up to about twice as far from AdamW or Muon at losses 4–2.5 (1.4–1.5 times on average) and within 1.1 times below 2.5, while on TinyStories-30M every pair stays within 1.1 times; on Shakespeare, where SGD barely trains, AdamW–SGD stays at 1.3 to 1.7 times the floor and Muon–SGD has the only persistent split, which is lost with another metric or a stricter threshold. Within warmup the ratio to the floor reaches 30 to 200, but there the runs still share their initialization, the floor is near zero, and the ratios are inflated. At equal steps the optimizers do differ, mostly because they progress at different speeds: on TinyStories Muon's loss is below AdamW's at every logged step after warmup, and AdamW needs 1.35 and 1.22 times as many steps to reach its final validation loss. At equal steps AdamW and Muon on the same data order are at most about a fifth closer than two data orders of one optimizer, because they share their recent batches; at equal loss this vanishes. On TinyStories-10M every learning rate from a quarter to four times each optimizer's rate stays within 0.8–1.1 times the floor at equal loss. On TinyStories AdamW sits at its adaptive edge of stability for much of training whether or not the optimizers differ, SGD reaches its threshold only early in training, and once the learning rate decays the sharpness levels off below the rising threshold; across learning rates the top Hessian eigenvalue of all three optimizers falls about as $1/\eta$, and Muon's spectral-norm sharpness comes near its stability bound only around the end of warmup. At a constant learning rate the break-even point marks where runs at different rates separate for none of the three optimizers. On CIFAR-10 we replicate the break-even effects of Jastrzebski et al. (2020) on sharpness and gradient noise, but at equal loss, away from the initial loss, two learning rates differ by at most about twice the floor. When we switch optimizer at nine times during training, the run ends at the function of the new optimizer whenever that optimizer is fast enough to get there in the remaining steps, which on TinyStories is always the case when it is the faster of the two, so the endpoint is not decided early. Weight space keeps a mark that function space does not: after a change of optimizer a fork needs a later fork step to stay linearly connected to its parent than after a change of data order (except for Muon on Shakespeare), the loss barrier between AdamW and Muon extends beyond the first block, where that between two data orders lies, and the weights carry each optimizer's geometry, mixed after a swap according to how far each optimizer moved them. On TinyStories the hidden representations of AdamW and Muon differ after the first block and little after the others, and their functions differ no more than with another data order even off the data distribution. A second initialization reproduces the results, and on a linear model with known, different limits the same test does find a split, near interpolation; AdamW and Muon trained into memorization on Shakespeare still do not split. We explain these results with a simple model: predictors of equal loss within an n-gram class lie within a distance of the class optimum set by their excess loss, whatever the optimizer; symmetries of the weights let each optimizer reach the same function at its own weights; and the stability thresholds, for Muon one on a spectral-norm sharpness, cap the curvature without choosing between functions of equal loss. A toy bigram model reproduces the main effects. Finally, per-layer learning rates set to even out the curvature across layers help only SGD, as a self-tuned higher rate: for AdamW and Muon each layer's curvature along its step adapts to its own rate, so stability cannot allocate the rate across layers.

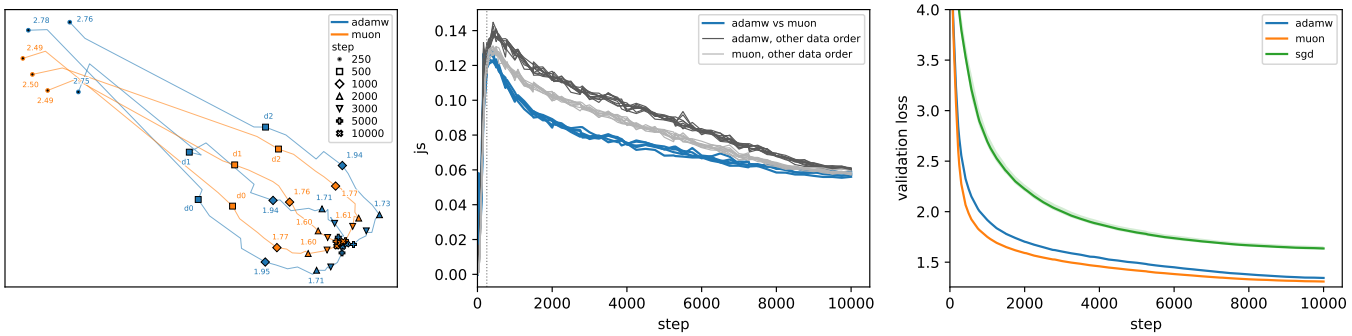


Figure 1: TinyStories-10M. Left: the AdamW and Muon runs on three data orders (d0–d2, at step 500) from step 250 on: densMAP of the Jensen–Shannon distance ($\sqrt{\text{JS}}$) between all pairs of logged steps, each run thinned to points at least 0.28 apart so its late steps do not crowd out the other runs. Black circles: step 250; markers: fixed steps, with each run's training loss written at steps 250, 1000 and 2000. densMAP keeps neighbourhoods but not distances. Middle: divergence between AdamW and Muon at equal steps (blue) and the data-order floor (grey); dotted: step 250, where the left panel starts. Right: validation loss of AdamW, Muon and SGD, mean over five data orders (three for SGD, at 0.06), band from min to max. Hyperparameters in Tables 1 and 5.

1 Introduction

The optimizer decides which of the many solutions with equal training loss a network converges to. In linear models this choice is understood: a first-order method acts as steepest descent in some norm and converges in direction to the max-margin solution in that norm (Gunasekar et al., 2018), which is ℓ_2 for gradient descent (Soudry et al., 2018), ℓ_∞ for Adam (Zhang et al., 2024a; Xie and Li, 2024) and the spectral norm for Muon (Fan et al., 2025). These are statements about the limit of infinite training, approached slowly. In deep networks the comparison has been made at the end of training: adaptive methods were found to generalize worse (Wilson et al., 2017), then to learn richer features (Vasudeva et al., 2025); models with the same error can compute different functions (Cortesi et al., 2026); and on language models most optimizers reach a similar loss (Zhao et al., 2025).

We ask *when* the difference between optimizers appears. From the same initialization and the same data order, do SGD with momentum, AdamW and Muon first follow the same path in function space and only later separate? If so, where is the split, what happens in the optimization at that moment, and is the final difference already decided there?

The timing matters for two reasons. First, it says which theory is relevant. If optimizers separate early, the asymptotic margin results do not describe how the difference arises, and early events in training are candidates instead. In vision, Jastrzebski et al. (2020) show that SGD runs with different learning rates separate in prediction space early in training, and tie this to a *break-even point*; and Frankle et al. (2020) find that training becomes stable to the noise of SGD early, after which copies of a network trained with different data orders stay in one linearly connected region. Edge-of-stability events (Cohen et al., 2021; Cohen et al., 2022; Andreyev and Beneventano, 2024) are the natural candidate for such an early event, and their thresholds differ by optimizer. Second, it matters in practice. Switching optimizers during training (Keskar and Socher, 2017) assumes the second optimizer can still change the solution; a split point bounds when that is possible.

Approach. We train nanoGPT models (Karpathy, 2022) on character-level Shakespeare (10.8M parameters) and on TinyStories (14M and 30M parameters), with SGD with momentum, AdamW, and Muon (Jordan et al., 2024), each at a learning rate chosen by a sweep, from a shared initialization, with five data orders per optimizer. The distance between two runs is the Jensen–Shannon divergence and the top-1 disagreement of their next-token predictions on fixed held-out contexts. We compare runs at equal steps and at equal training loss, and calibrate against a noise floor: the distance between two runs of the same optimizer with different data orders. A SimpleCNN on CIFAR-10 replicates the break-even analysis of Jastrzebski et al. (2020) as a check of the pipeline.

Contributions.

1. **Divergence curves** between every pair of optimizers in three settings, at equal steps and at equal training loss, against a data-order noise floor (Section 5.2). After warmup, at equal loss, two optimizers are about as far apart as two data orders of one optimizer: within 1.1 times on TinyStories-30M, up to about twice for SGD against AdamW or Muon at losses 4–2.5 on TinyStories-10M, and further only for the pairs with SGD on Shakespeare, where SGD barely trains. The larger ratios within warmup are inflated by a floor near zero.
2. **A test for a split point**: episodes† in which the distance exceeds the floor and a persistent split†, located as

a loss level with a bootstrap over data orders (Sections 4.2 and 5.2). There is a persistent split in 1 of 9 pairs (Muon–SGD on Shakespeare, from loss 2.82, in 96.5% of resamples, lost with another metric or a stricter threshold), none between AdamW and Muon.

3. **AdamW against Muon, and optimizer against data order**† at equal steps (Fig. 1, Section 5.3): Muon is faster, and AdamW and Muon on the same data order are at most about a fifth closer than two data orders of one optimizer, because they share their recent batches; at the end of TinyStories-30M the AdamW–Muon distance equals the data-order floor. The n-gram references show in which order the optimizers pass the unigram and bigram solutions.
4. **What coincides with the changes in distance**: sharpness against its optimizer-specific stability threshold, batch sharpness, the end of warmup (Section 5.4). The edge of stability does not explain them: on TinyStories AdamW is at its threshold for long stretches whether or not the optimizers differ, and SGD only early in training. With the cosine schedule the sharpness levels off and does not follow the threshold up as the learning rate decays, and across learning rates† the top Hessian eigenvalue of all three optimizers falls about as $1/\eta$. Muon's spectral-norm sharpness† comes near its bound only around the end of warmup. Muon with its auxiliary AdamW matched to AdamW† still memorizes on Shakespeare, so the memorization comes from its updates of the hidden matrices.
5. **A causal test by swapping** optimizers at nine fork times with warm optimizer states, scored by where the fork lands between the two pure runs, and a basin-level test by linear mode connectivity (Section 5.6). On TinyStories-10M the optimizer that trains last decides the function whenever it is fast enough to: forks from SGD to AdamW or Muon land on the new optimizer even after 4000 steps, while forks into SGD end closer to their parent than to pure SGD from step 1500 on (step 5000 on TinyStories-30M, where SGD's rate is higher). In weight space the optimizer leaves a mark that function space does not show: forks that change the optimizer stay linearly connected to their parent only from a later fork step than forks that change the data order (except for Muon on Shakespeare), and between pure runs† the barrier between AdamW and Muon extends beyond the first block, where that between two data orders lies.
6. **The anatomy of the sharp direction**: which parameter blocks carry the top Hessian eigenvector and its curvature, by optimizer and over training (Section 5.7). It differs by optimizer (embeddings for Muon on Shakespeare, value–output matrices for Muon on TinyStories, head or gains for AdamW on TinyStories). At the edge of stability the sharp direction lies at the input end of the network: on TinyStories-10M, SGD's in blocks 1–2 (66%), that of the operator that sets AdamW's threshold in the embeddings (68%), and Muon's, of H, which is not its threshold operator, in blocks 1–3 (84%). Except in AdamW's embeddings, most of the eigenvalue comes from curvature between layers (66% for SGD, 71% for Muon).
7. **A replication of the break-even point** on CIFAR-10 and its transfer to the language models (Section 5.9). The sharpness and gradient-noise effects replicate, but at equal training loss, away from the initial loss, two learning rates differ by at most about twice the floor. At a constant rate†, reaching the stability threshold does not mark where the paths of different rates split, for any of the three optimizers.
8. **Curvature-balanced per-layer rates**† (Section 5.8): rates that equalize the layers' shares of the top eigenvector of the stability operator, or the curvature along each layer's step. They help only SGD, mostly as a self-tuned higher rate; for AdamW and Muon every layer sits at its own edge whatever its rate, so stability cannot allocate rates across layers.
9. **Controls**† (Section 5.11): a second initialization moves the function as much as a new data order and reproduces the episodes; SGD's loss gap on rare tokens is a matter of speed; learning rates from a quarter to four times each optimizer's rate stay within 0.8–1.1 times the floor at equal loss; and on a linear model, where theory predicts different limits, the same detector finds a persistent split near interpolation, while AdamW and Muon trained into memorization on Shakespeare still do not split.
10. **A model of the results**† (Section 6): at equal loss, predictors of one n-gram class lie within a distance of the class optimum set by the excess loss, whatever the optimizer. The bound says nothing far above the class optimum, as in warmup, where the update geometries pick different routes, or where the class has no best predictor; near it, the distance between optimizers is mostly data-order noise. Weight symmetries let each optimizer end at its own point of the set of weights computing one function, which explains the barriers. Each optimizer has a stability threshold (for Muon on a spectral-norm sharpness, Islamov et al., 2026), and a stability cap on curvature does not choose between functions of equal loss. A toy bigram model reproduces these effects exactly.

Planned and added analyses. The project plan comprised the learning-rate sweeps, the divergence curves at equal steps and at equal loss against the data-order floor, the first exceedance t^* with a bootstrap over data orders, its comparison with the sharpness thresholds and the end of warmup, the swaps with warm optimizer states scored by distance and function metrics, linear mode connectivity against the fork time, the anatomy of the top eigenvector, and the CIFAR-10 break-even check. Everything else was added after the first results, mostly to test explanations of them, and is marked with † in the headings and captions: the episodes and the persistent split (as t^* turned out to

be degenerate), the decomposition into optimizer and data order and the n-gram references, the edge of stability of Muon's auxiliary AdamW and the matched control, the robustness checks, the anatomy of AdamW's preconditioned operator, the floor-normalized swap scores and the barriers to pure runs, the scale-free weight metrics ($\|\theta\|_\infty/\|\theta\|_2$ and the stable rank) and their shifts under swaps, the representation similarity, the layer-wise barriers, the comparison off the data distribution, the break-even runs on the language models and with two batch sizes, the learning-rate sweeps with sharpness measurements and the curvature between layers, the controls of Section 5.11 (a second initialization, token frequency, a linear positive control, Shakespeare runs trained into memorization, and other learning rates), and the model of the results in Section 6 with its toy model, Muon's spectral-norm sharpness, the break-even runs of the three optimizers at a constant rate, the stochastic edge of stability, and the curvature-balanced per-layer rates.

Code: `https://github.com/qqgjyx/MIT-6.7910` (`code/`).

2 Related work

Implicit bias of optimizers. For linear models the limit of training depends on the geometry of the optimizer: steepest descent in a norm converges to the max-margin solution in that norm (Gunasekar et al., 2018), giving ℓ_2 for gradient descent (Soudry et al., 2018), ℓ_∞ for Adam (Zhang et al., 2024a), an ℓ_∞-constrained KKT point for AdamW with weight decay (Xie and Li, 2024), and the spectral norm for Muon (Fan et al., 2025). Away from the limit, finite step sizes act as implicit regularizers: for SGD the modified loss penalizes the gradient norm (Smith et al., 2021), while for Adam the corresponding term can have the opposite sign (Cattaneo et al., 2024). Empirical comparisons in deep networks are made at the end of training (Wilson et al., 2017; Vasudeva et al., 2025; Zhao et al., 2025; Cortesi et al., 2026). We look at the whole trajectory instead.

Edge of stability. With full-batch gradient descent, the top Hessian eigenvalue rises until it reaches $2/\eta$ and then stays near it while the loss keeps decreasing (Cohen et al., 2021); this is explained by self-stabilization (Damian et al., 2023) and described by central flows (Cohen et al., 2025). Adaptive methods reach an analogous *adaptive edge of stability*, where the top eigenvalue of the preconditioned Hessian sits at $(2 + 2\beta_1)/((1 - \beta_1)\eta)$ (Cohen et al., 2022). For mini-batch SGD, Andreyev and Beneventano (2024) show that the quantity that stabilizes at $2/\eta$ is the *batch sharpness*, the curvature of mini-batch losses along their own gradients, while $\lambda_{\max}$ stays below; Andreyev et al. (2026) study the same regime with momentum. Islamov et al. (2026) extend the edge of stability to non-Euclidean gradient descent, such as the spectral gradient descent behind Muon, where a sharpness measured in the spectral norm sits at $2\|\nabla L\|_*/\eta$. Kalra and Barkeshli (2024) relate learning-rate warmup to these sharpness dynamics. Transformer Hessians differ strongly across parameter blocks (Zhang et al., 2024b), which motivates our block decomposition.

Break-even point and early training. Jastrzebski et al. (2020) argue that early in training SGD reaches a break-even point, after which the learning rate and batch size set the curvature and the gradient noise along the rest of the trajectory, and show that SGD runs with different learning rates separate early in prediction space. Frankle et al. (2020) find that training becomes stable to SGD noise early: two copies trained on from the same checkpoint with different data orders end in one linearly connected region once the checkpoint is past an early point. We use both as templates, comparing optimizers instead of learning rates or data orders.

Muon and optimizer switching. Muon (Jordan et al., 2024) updates each hidden weight matrix with its momentum after approximate orthogonalization by a Newton–Schulz iteration, and has been scaled to large language models (Liu et al., 2025). SWATS (Keskar and Socher, 2017) switches from Adam to SGD during training, scored by accuracy; our swaps are scored by where the function ends.

3 Setup

Models and data. Table 1 lists the four settings. The language models are nanoGPT (Karpathy, 2022) with pre-LayerNorm blocks, GELU MLPs of width $4d$, no biases, no dropout, an untied output head, and separate query, key and value projections (so that Muon orthogonalizes each and the block decomposition of Section 4.7 can separate the query-key from the value-output matrices). The context is 256 tokens. Character-level Shakespeare uses nanoGPT's 90/10 split and a 65-character vocabulary. TinyStories (Eldan and Li, 2023) (GPT-4 version) uses a 4096-token byte-level BPE trained on the training split, so that the head and the stored predictions stay small. The vision check is the SimpleCNN of Jastrzebski et al. (2020) (two 32-filter and two 64-filter 3×3 convolutions with max pooling after each pair, then a 128-unit dense layer, ReLU) on CIFAR-10 (Krizhevsky, 2009) without augmentation.

Shakespeare runs take 24.6M tokens (about 24 passes over the 1M training characters). Without dropout the validation

Table 1: Settings. L layers, model width d, vocabulary V. Parameters in parentheses exclude the token and position embeddings and the head. LM batches are sequences × tokens. "aux.": learning rate of Muon's auxiliary AdamW. CIFAR-10 settings in Section 4.9. LM schedules: linear warmup for w steps, then cosine decay to 10% of the peak learning rate at the last step. The CIFAR-10 runs use a constant learning rate. ‡TinyStories-10M SGD: 0.06 (data orders 0–2) for the SGD pairs of the divergence analysis (Figs. 1, 3 and 4, Table 2); every other SGD analysis on TinyStories-10M uses the earlier runs at 0.03 (data orders 0–4; Section 3, Learning rates).

Setting	Model (L, heads, d, V)	Params	Steps	Batch	w	η_{SGD}	η_{AdamW}	η_{Muon} (aux.)
Shakespeare-char	6, 6, 384, 65	10.8M (10.6M)	1,500	64×256	100	0.05	$3 \cdot 10^{-4}$	0.04 (10^{-3})
TinyStories-10M	6, 6, 384, 4096	13.9M (10.6M)	10,000	128×256	200	0.06‡	$2 \cdot 10^{-3}$	0.01 ($2 \cdot 10^{-3}$)
TinyStories-30M	8, 8, 512, 4096	29.5M (25.2M)	20,000	128×256	400	0.1	$2 \cdot 10^{-3}$	0.01 ($1.5 \cdot 10^{-3}$)
CIFAR-10	SimpleCNN, ReLU	0.36M	39,000	128	–	plain SGD, $\eta \in \{0.001, 0.01, 0.1\}$		

loss is lowest near step 1000 and rises after, so these runs stop at 1500 steps. The TinyStories runs take 328M and 655M tokens (about 0.6 and 1.2 epochs).

Optimizers. Write g_t for the mini-batch gradient at step t and $\eta_t = \eta\,\varphi(t)$ for the learning rate, with $\varphi(t) = (t+1)/w$ for $t < w$ and $\varphi(t) = 0.1 + 0.45\,(1 + \cos(\pi\,(t-w)/(T-w)))$ after, T being the last step.

- **SGD** with heavy-ball momentum $\mu = 0.9$ and no weight decay: $b_{t+1} = \mu b_t + g_t$, $\theta_{t+1} = \theta_t - \eta_t b_{t+1}$.
- **AdamW** with $(\beta_1, \beta_2) = (0.9, 0.95)$ ($(0.9, 0.99)$ for Shakespeare, as in nanoGPT), $\epsilon = 10^{-8}$, and decoupled weight decay 0.1 on every tensor with at least two dimensions (including embeddings and head) and none on LayerNorm gains: $\theta_{t+1} = (1 - \eta_t \lambda_{\text{wd}})\theta_t - \eta_t\,\hat{m}_{t+1}/(\sqrt{\hat{v}_{t+1}} + \epsilon)$, with bias-corrected moments $\hat{m}, \hat{v}$.
- **Muon** (Jordan et al., 2024) on every matrix inside the transformer blocks (query, key, value, attention output, both MLP matrices), with momentum $\beta = 0.95$, Nesterov, and no weight decay: $M_{t+1} = \beta M_t + (1-\beta)G_t$, $W_{t+1} = W_t - \eta_t \sqrt{\max(1, n_{\text{out}}/n_{\text{in}})}\,\text{NS}_5\big((1-\beta)G_t + \beta M_{t+1}\big)$, where NS_5 is five steps of the quintic Newton–Schulz iteration that approximates the orthogonal polar factor. Embeddings, head and gains are updated by an auxiliary AdamW with $(0.9, 0.95)$, $\epsilon = 10^{-8}$, no weight decay, and its own learning rate (Table 1). This matches the reference implementation (Jordan et al., 2024) in the Newton–Schulz settings, the Nesterov momentum of 0.95, the shape scaling and the split of parameters between Muon and AdamW.

Gradient clipping is off, since it would hide edge-of-stability dynamics. Training steps of the language models use bfloat16 autocast; all measurements (Section 4) are in float32, with TF32 matrix products on the GPU.

Learning rates. For each setting and optimizer we ran five learning rates (grids in Table 6) on data order 0, with the schedule and length of the main runs and no measurements, and kept the one with the best final validation loss (on the validation sequences of the probe, Section 4.1). Shakespeare is the exception: since its runs overfit, we kept the best *minimum* validation loss over training. SGD is the exception to the rule: in every setting its best rate on data order 0 diverged or blew up on other data orders (0.2 on Shakespeare, 0.1 on TinyStories-10M, 0.3 on TinyStories-30M), so SGD uses a lower rate: 0.05 on Shakespeare (the smallest of its grid) and the next lower rate on TinyStories, 0.03 and 0.1. On Shakespeare every SGD rate of the grid stops at about the same minimum validation loss (2.44–2.50, the bigram level), so we took the smallest and most stable one. On TinyStories-10M 0.03 turned out to be under-tuned: two of its five runs blew up (the training loss rose by 2.3–2.4 nats at steps 1540 and 1040 on orders 0 and 2) and ended at validation loss 1.90–1.96 against 1.77 for the others. We re-tuned SGD there on orders 0–2 at 0.03, 0.045, 0.06, 0.085 and 0.12 (Fig. 27): only 0.06 never blew up (a rise of more than 1 nat above the running minimum of the training loss), and it has the best mean final validation loss, 1.70 against 1.88 at 0.03. Blow-ups are not monotone in the rate (0.045 and 0.085 blew up, 0.06 did not) and are sensitive to numerical noise: rerun with the main logging on another node, order 2 at 0.06 skipped the early spike of its tuning run (step 70) and ended at 1.66 instead of 1.84, while orders 0 and 1 repeated exactly. The SGD pairs of the divergence analysis on TinyStories-10M use these runs at 0.06 (orders 0–2, final validation loss 1.62–1.66); every other SGD analysis there keeps the runs at 0.03. Runs whose training loss became non-finite were stopped and counted as diverged. Only the learning rates were tuned, in the same way for every optimizer; all other hyperparameters are fixed (Table 5).

Shared initialization and data order. Every run starts from the same initialization (seed 0). The mini-batch at step t depends only on the data order d and on t: for the language models it is B windows of 256 tokens at offsets drawn uniformly from the training split by a generator seeded with (d, t); for CIFAR-10 it is a slice of a per-epoch permutation seeded with (d, epoch), dropping the last incomplete batch. Two runs with the same d therefore see identical batches whatever their optimizer, and forks and resumed runs see the batches their parent would have seen. Batches used for measurements come from separate random streams and never change the training order. Each optimizer is run with data orders $d \in \{0, \ldots, 4\}$: 15 main runs per setting. The runs with $d = 0$ also keep the other optimizers' states in

shadow and save fork checkpoints (Section 4.5).

Control for Muon's auxiliary Adam.† On Shakespeare, Muon's auxiliary AdamW and the AdamW runs differ in learning rate (10^{-3} against $3 \cdot 10^{-4}$), β_2 (0.95 against 0.99) and weight decay (none against 0.1). A control suite repeats Muon with its auxiliary AdamW matched to AdamW (learning rate $3 \cdot 10^{-4}$, $\beta = (0.9, 0.99)$, weight decay 0.1, which here also applies to the gains), with data orders 0–4 ("matched Muon"). It differs from AdamW only in how the hidden matrices are trained (and in decay on the gains).

4 Methods

Step s always refers to the model θ_s after s updates ($s = 0$ is the initialization). Each run logs predictions at its prediction steps $\mathcal{S}$ and curvature at its measurement steps $\mathcal{M}$; both grids combine a linear and a log-spaced grid (Table 8).

4.1 Function distance and alignment

Distance. Every run stores, at each $s \in \mathcal{S}$, its next-token distributions on a fixed set of N held-out positions, identical across runs: for Shakespeare all 256 positions of 64 validation sequences ($N = 16{,}384$), for TinyStories 16 evenly spaced positions of 128 validation sequences ($N = 2{,}048$), and for CIFAR-10 the 10,000 test images. Write $p_i^X(s)$ for the distribution of run X at step s on position i. The distance between run X at step s and run Y at step s' is the mean Jensen–Shannon divergence in bits,

$$D_{\mathrm{JS}}(X_s, Y_{s'}) = \frac{1}{N}\sum_{i=1}^{N} \mathrm{JS}\big(p_i^X(s),\, p_i^Y(s')\big), \qquad \mathrm{JS}(p,q) = \tfrac{1}{2}\mathrm{KL}(p \,\|\, m) + \tfrac{1}{2}\mathrm{KL}(q \,\|\, m), \quad m = \tfrac{1}{2}(p+q), \tag{1}$$

with base-2 logarithms, so $D_{\mathrm{JS}} \in [0,1]$. The second distance is the top-1 disagreement $D_{\mathrm{dis}}(X_s, Y_{s'}) = 1 - \frac{1}{N}\sum_i \mathbf{1}\{\arg\max_v p_{i,v}^X(s) = \arg\max_v p_{i,v}^Y(s')\}$.

Alignment by step. $D(X_s, Y_s)$ at the steps $s \in \mathcal{S}$ that both runs log.

Pictures of trajectories.† To draw several runs at once we embed their logged steps in two dimensions with densMAP, a variant of UMAP that also keeps local density, applied to the Jensen–Shannon distance $\sqrt{\mathrm{JS}}$ (a metric) between all pairs of steps. densMAP and UMAP keep neighbourhoods, not distances, so distances are read from the curves at equal steps or equal loss, never from the embedding. Late in training the logged steps of a run lie close together and would crowd out the other runs, so each run is thinned to points at least a fixed distance apart (0.28 in $\sqrt{\mathrm{JS}}$ in Fig. 1).

Alignment by training loss. Optimizers progress at different speeds, and comparing at equal steps confounds "faster" with "different". Each run therefore also logs, at each $s \in \mathcal{S}$, its mean cross-entropy $\ell_X(s)$ on a fixed training probe, the same for all runs (64 training sequences, all positions, for the language models; 2,000 training images for CIFAR-10). We make it monotone with a running minimum, $\bar{\ell}_X(s) = \min\{\ell_X(s') : s' \in \mathcal{S},\ s' \le s\}$. Each step s of X is matched with the earliest step of Y whose loss is closest,

$$\pi_{XY}(s) = \min\Big(\arg\min_{s' \in \mathcal{S}} \big|\bar{\ell}_Y(s') - \bar{\ell}_X(s)\big|\Big), \qquad \text{kept if } \big|\bar{\ell}_Y(\pi_{XY}(s)) - \bar{\ell}_X(s)\big| \le \tau\, \bar{\ell}_X(s), \tag{2}$$

with $\tau = 0.02$. The kept matches give points $\big(\bar{\ell}_X(s),\, D(X_s, Y_{\pi_{XY}(s)})\big)$, and the *loss-matched distance* $D^{\mathrm{loss}}_{X,Y}(\ell)$ is their piecewise-linear interpolation in ℓ (where several steps share a loss value, the earliest is used), undefined outside the range of kept losses. The loss coordinate is that of the first run, X.

Noise floor. Two runs of the same optimizer from the same initialization with different data orders, $D^{\mathrm{loss}}_{O_d, O_{d'}}$ with $d \neq d'$, differ only through the sampling of mini-batches. A difference between optimizers counts only where it exceeds this floor.

4.2 Locating the split

Fix two optimizers A and B and write A_d, B_d for their runs with data order $d \in \mathcal{D} = \{0, \dots, 4\}$. Runs with the same d share initialization and batches, so they form the *cross* pairs; runs of one optimizer with different d form the *floor* pairs:

$$c_d(\ell) = D^{\mathrm{loss}}_{A_d, B_d}(\ell), \qquad f^O_{dd'}(\ell) = D^{\mathrm{loss}}_{O_d, O_{d'}}(\ell) \quad (O \in \{A, B\},\ d < d').$$

Loss grid. Let $\ell_1 > \ell_2 > \cdots > \ell_J$ be the distinct values of $\bar{\ell}_{A_0}(s)$, $s \in \mathcal{S}$, that lie in the range covered by all $2|\mathcal{D}|$ runs,

$$\Big[\max_X \min_{s\in\mathcal{S}} \bar{\ell}_X(s),\ \min_X \max_{s\in\mathcal{S}} \bar{\ell}_X(s)\Big].$$

Decreasing loss is the order of training.

Statistic. For a multiset S of data orders with distinct elements $U(S)$, let

$$\bar{c}_S(\ell) = \text{mean}_{d\in S}\ c_d(\ell), \qquad F_S(\ell) = \max\Big(\text{mean}_{d<d'\in U(S)}\ f^A_{dd'}(\ell),\ \text{mean}_{d<d'\in U(S)}\ f^B_{dd'}(\ell)\Big), \tag{3}$$

where the cross mean counts repeated orders with their multiplicity, means skip undefined values, and the maximum takes the defined side if only one is. The floor is thus the larger of the two optimizers' average seed-to-seed distances. A level is *above the floor* if the cross distance exceeds the floor by a margin,

$$o_S(j) = \mathbf{1}\big\{\bar{c}_S(\ell_j) > (1+m_\times)\, F_S(\ell_j) + m_+\big\}, \tag{4}$$

with $m_\times = 0.5$ and $m_+ = 0$; a comparison with an undefined value counts as false.

First exceedance. The first grid level from which the cross distance is above the floor at $k = 3$ consecutive levels,

$$t^*(S) = \ell_{j^*}, \qquad j^* = \min\big\{j :\ o_S(i) = 1 \ \text{ for } i = j, \ldots, j+k-1\big\}, \tag{5}$$

undefined if no such j exists. Since the grid is made of the losses of A_0 at its prediction steps, the k levels are consecutive distinct loss values of that run. This was the planned definition of the split, but it is degenerate: at initialization all runs compute the same function, so the floor is close to 0 for the first steps, while two different optimizers move apart from their first update. In every pair of optimizers t^* is one of the first grid levels, reached at steps 1–8 (Table 2). It records that two optimizers differ from the start, not that they split. We therefore also report:

Episodes and persistent split.† An *episode* is a maximal run of consecutive levels $\ell_{j_0}, \ldots, \ell_{j_1}$ with $o_S(j) = 1$ that contains at least k levels ($j_1 - j_0 + 1 \geq k$), reported as the loss range $[\ell_{j_0}, \ell_{j_1}]$. The *persistent split* $t^*_{\text{p}}(S)$ is the start ℓ_{j_0} of the last episode if that episode ends at the last level of the grid ($j_1 = J$), and undefined otherwise: the loss from which the cross distance stays above the floor down to the lowest loss that all runs reach. A persistent split is what "together first, apart later" predicts; an episode that ends before ℓ_J is a difference that later disappears. Figures also show the ratio $\bar{c}_{\mathcal{D}}(\ell)/F_{\mathcal{D}}(\ell)$ against the loss, with the threshold $1 + m_\times$.

Estimate and uncertainty. The point estimates are $t^* = t^*(\mathcal{D})$, $t^*_{\text{p}} = t^*_{\text{p}}(\mathcal{D})$ and the episodes of $\mathcal{D}$. For their uncertainty we draw 1000 bootstrap resamples S_b of $|\mathcal{D}|$ data orders with replacement and report, for t^* and t^*_{p} separately, the 2.5 and 97.5 percentiles of the defined values, together with the fraction of resamples in which one is found (a resample with a single distinct order has no floor and gives none). The distance is D_{JS}; D_{dis} and other values of $m_\times$, k and τ are checked in Section 5.2 (300 resamples each).

In steps. t^* is a loss level, which makes it comparable across optimizers that reach it at different steps. For each run X, $t^*_X = \min\{s \in \mathcal{S} : \bar{\ell}_X(s) \leq t^*\}$ is the first step at which X reaches it, and likewise $t^*_{\text{p},X}$ for the persistent split.

4.3 Optimizer against data order, and n-gram references†

Decomposition at equal steps. For two optimizers X, Y with data orders $d = 0, \ldots, n-1$ ($n = 5$) and $d^+ = (d+1) \bmod n$, we compare at each shared prediction step s four families of n pairs: $D(X^d_s, Y^d_s)$ (other optimizer, same data: the cross pairs above), $D(X^d_s, Y^{d^+}_s)$ (other optimizer, other data), and the floors $D(X^d_s, X^{d^+}_s)$ and $D(Y^d_s, Y^{d^+}_s)$ (same optimizer, other data), and plot the median and range over d. If the same-data distance is no larger than the floors, the change of optimizer changes the function no more than a change of data order; if it is smaller than the other-data distance, sharing the batches keeps the two optimizers closer.

N-gram references. From the counts $n(b)$ and $n(a, b)$ of tokens and of consecutive token pairs in the training split we build a unigram model $p_1(b) \propto n(b)+1$ and a bigram model smoothed towards it, $p_2(b \mid a) = \big(n(a,b)+\alpha\, p_1(b)\big)/\big(n(a)+\alpha\big)$ with $\alpha = 1$, evaluated at the validation probe positions (the bigram conditioned on the current token). For a run X we plot D_{JS} (1) between $p^X(s)$ and each reference against the training loss $\bar{\ell}_X(s)$. A minimum near zero means the run passes through that n-gram solution; two optimizers that pass through it at the same loss meet there in function space.

4.4 Sharpness and stability thresholds

At each $s \in \mathcal{M}$, before the update of that step, we measure on θ_s (in float32, with TF32 matrix products):

Sharpness. $\lambda_{\max}$ is the largest eigenvalue of the Hessian $H = \nabla^2 L_{\mathcal{H}}(\theta_s)$ of the mean cross-entropy on a fixed batch $\mathcal{H}$ of training examples (32 sequences of 256 tokens for the language models; 2,500 images, 5% of the training set, for CIFAR-10), computed by Lanczos with full reorthogonalization (20 iterations from a fixed random start) on Hessian-vector products. The curvature is thus that of the training loss on fixed training sequences, while the predictions compared in Section 4.1 are on held-out validation contexts.

SGD threshold. On a quadratic with curvature λ, heavy-ball momentum as above is stable if and only if $\eta\lambda < 2+2\mu$. We compare $\lambda_{\max}(\theta_s)$ with $(2+2\mu)/\eta_s$, which is $3.8/\eta_s$ for $\mu = 0.9$ and $2/\eta$ for the plain SGD of the break-even runs.

AdamW preconditioned sharpness. Following Cohen et al. (2022), let $P = (1-\beta_1^t)\,\mathrm{diag}(\sqrt{\hat{v}}+\epsilon)$, with $\hat{v} = v/(1-\beta_2^t)$ built from the optimizer's current second-moment state and step count t. The AdamW update is then $-\eta_t P^{-1} m$ with m the uncorrected first moment, so with P frozen and weight decay ignored it is heavy-ball momentum with coefficient β_1 and step $\eta_t(1-\beta_1)$ in the metric P. It is stable if $\lambda_{\max}(P^{-1}H) < (2+2\beta_1)/((1-\beta_1)\eta_t)$, which is $38/\eta_t$ for $\beta_1 = 0.9$. We compute the preconditioned sharpness $\lambda_{\max}(P^{-1/2}HP^{-1/2})$, which has the spectrum of $P^{-1}H$, by the same Lanczos procedure, for every run that trains with AdamW (including forks into AdamW).

Batch sharpness. Following Andreyev and Beneventano (2024, Def. 1),

$$\mathrm{BS}(\theta) = \frac{\sum_{j=1}^{n} g_j^\top H_j\, g_j}{\sum_{j=1}^{n} \|g_j\|^2}, \tag{6}$$

where g_j and H_j are the gradient and Hessian of the loss on mini-batch j, over $n = 8$ random training mini-batches of the training batch size (a fresh random stream at every measured step). For mini-batch SGD without momentum, Andreyev and Beneventano (2024) find that BS stabilizes at $2/\eta$. We measure it for every optimizer and compare it with $(2+2\mu)/\eta_s$ for SGD.

Muon. Muon's update on a hidden matrix is orthogonalized, with all singular values near one whatever the size of the gradient, so the quadratic argument above, where the step scales with the gradient, does not carry over, and $\lambda_{\max}(H)$ is not the quantity its threshold applies to. Islamov et al. (2026) show that for such scale-invariant updates the stability condition bounds a sharpness measured in the spectral norm, $S = \max\{\langle D, H[D]\rangle : \|D\| \le 1\}$, by $2\|\nabla L\|_*/\eta$, with $\|\cdot\|_*$ the dual (nuclear) norm, and that full-batch spectral gradient descent sits at this bound (Section 6). We measure† S over Muon's hidden matrices $D = (D_l)$, with $\|D\| = \max_l \|D_l\|_2/c_l$ and $c_l = \sqrt{\max(1, n_{\text{out}}/n_{\text{in}})}$ Muon's shape factor, so that Muon's step has norm η_t and the dual norm is $\sum_l c_l\|\nabla_l L\|_*$. On the Hessian batch $\mathcal{H}$, Frank–Wolfe steps $D \leftarrow (1-\gamma_k)D + \gamma_k\big(c_l\,\mathrm{polar}((H[D])_l)\big)_l$, $\gamma_k = 2/(k+2)$, 50 from each of 5 random starts, give the largest $\langle D, H[D]\rangle$ seen, a lower bound on S; we compare it with $2\sum_l c_l\|\nabla_l L\|_*/\eta_s$, with the gradient on the same batch. Momentum is ignored. We measure S at the saved checkpoints of the TinyStories-10M runs with data orders 0 and 1 and at the last step of the Muon sweep with sharpness measurements (below). Otherwise we report Muon's $\lambda_{\max}$ and batch sharpness without a threshold, and test against a threshold the part of the model that Muon leaves to its auxiliary AdamW (Section 4.4).

One-step loss change. $\Delta L(s) = \ell_X(s) - \ell_X(s+1)$ on the training probe, used in the break-even analysis.

Edge of stability of the Adam-managed block.† Muon trains the embeddings, head and gains with an auxiliary AdamW; write a for these coordinates. At the fork checkpoints of the $d = 0$ runs (Table 7), which hold the optimizer states, we compute the top eigenvalue of $P_a^{-1/2} H_{aa} P_a^{-1/2}$, where H_{aa} is the block of the Hessian on $\mathcal{H}$ restricted to a and P_a the preconditioner P defined above, built from the state of the run's own Adam for these coordinates (Muon's auxiliary AdamW, or AdamW itself), by the same Lanczos procedure. We compare it with $(2+2\beta_1)/((1-\beta_1)\eta_s)$, with η and β_1 of that Adam. This treats the block as if the rest of the model were frozen, so it is a check of the auxiliary AdamW on its own, not a threshold for Muon.

What coincides with the split. For a sharpness measure $\mathcal{Q}$ with threshold Θ, the measured steps at the edge of stability are $\mathcal{E}(X) = \{s \in \mathcal{M} : \mathcal{Q}_X(s) \ge (1-\delta)\,\Theta_X(s)\}$ with $\delta = 0.1$. We report the onset $s_{\mathrm{EoS}}(X) = \min \mathcal{E}(X)$, the last step of $\mathcal{E}(X)$, and the fraction $|\mathcal{E}(X)|/|\mathcal{M}|$ of measured steps at the edge of stability, for $\lambda_{\max}$ (SGD) and the preconditioned sharpness (AdamW), and the ratio $\mathcal{Q}_X(s)/\Theta_X(s)$ at the steps t_X^* and $t_{\mathrm{p},X}^*$. Since $\mathcal{M}$ is partly log-spaced, the fraction weights early training more than a fraction of steps would. Batch sharpness is shown in the figures but has no onset computed. We compare these with t_X^* and $t_{\mathrm{p},X}^*$, with the episodes, with the end of warmup w, and with the loss levels across settings.

Sharpness across learning rates.† To see how each optimizer's sharpness scales with the learning rate, we repeated part of the sweeps on data order 0 with the sharpness measurements of the main runs: on TinyStories-10M SGD at 0.01, 0.03 and 0.1, AdamW at 10^{-3}, $2 \cdot 10^{-3}$ and $4 \cdot 10^{-3}$, and Muon at 0.005, 0.01, 0.02 and 0.04; on Shakespeare the five rates of each optimizer in Table 6. The plateau of a run is the median of its sharpness over the steps after warmup with $\varphi(t) \geq 0.8$, and we fit a line to the logarithm of the plateau against that of the peak rate η. A slope of -1 means that the run sits at the same fraction of a threshold proportional to $1/\eta$ at every rate.

4.5 Swaps with warm optimizer states

Shadow states. During each $d = 0$ run of optimizer A, the other two optimizers are stepped on the same mini-batch gradients with learning rate 0. The state updates of SGD momentum, AdamW and Muon do not depend on the learning rate, and with learning rate 0 neither the update nor the decoupled weight decay moves the parameters; so their momenta, second moments and step counts evolve as if they were training, while the parameters follow A alone. (Our copy of Muon does not overwrite the gradient in place, so a shadow optimizer stepping after it sees the same gradient.)

Forks. At fork steps s (nine per setting, Table 7), the $d = 0$ run saves θ_s and the states of all three optimizers. The fork $A \to B$ at s loads θ_s and B's shadow state, keeps B's own hyperparameters and learning rate, and continues from s to T on the same schedule φ and the same batches. All six ordered pairs are forked at all nine steps. The control "A continued with the same batches" is the parent run itself. In addition, A is forked into A with data order 1 at the same steps, the control for mode connectivity (Section 4.6). This gives 81 forks per language-model setting. Forks log predictions and measurements like their parents.

Where a fork lands. Let F be the final model of the fork $A \to B$ at s, A the final model of the parent A_0, and B the final model of the pure run B_0 (from initialization, same batches). With D either distance of Section 4.1, the landing coordinate is

$$r = \frac{D(F, A)}{D(F, A) + D(F, B)}, \tag{7}$$

which is 0 at A's function and 1 at B's. It is a ratio of distances, not a projection: a fork far from both ends has r near $1/2$, so we report $D(F, A)$, $D(F, B)$ and $D(A, B)$ with it. We also give $D(F, B)$ over the final data-order floor of B, and $D(F, A)$ over that of A†: a fork within one floor of a pure run is as close to it as another data order is. For each function metric ϕ of Section 4.8 we also report the shift $(\phi(F) - \phi(A))/(\phi(B) - \phi(A))$, the fraction of the gap between the pure runs that the swap closes. If the function is decided at the split, forks with $s < t^*_A$ should end near $r = 1$ and forks with $s > t^*_A$ near $r = 0$. Since t^* turns out to be degenerate (Section 4.2), the time to compare fork steps with is the end of the early episode.

4.6 Linear mode connectivity

For two final weight vectors θ_1, θ_2 of the same architecture let $L(\alpha)$ be the mean cross-entropy of $(1 - \alpha)\theta_1 + \alpha\theta_2$ (every floating-point entry of the state interpolated) on the 64 sequences of the training probe. The barrier is

$$\mathrm{Bar}(\theta_1, \theta_2) = \max_{\alpha \in \{0, 0.1, \ldots, 1\}} \big[L(\alpha) - (1 - \alpha)L(0) - \alpha L(1)\big], \tag{8}$$

a loss version of the instability of Frankle et al. (2020). We compute it between the parent A_0 and each of its forks, as a function of the fork step: the forks $A \to A$ with data order 1 reproduce the instability analysis of Frankle et al. (2020) for each optimizer, and the forks $A \to B$ ask whether a change of optimizer at s leaves the solution in the same basin.

4.7 Anatomy of the top eigenvector

Partition the parameters into blocks: *emb* (token and position embeddings); for each layer i, qk_i (W_Q, W_K), ov_i (W_V and the attention output W_O) and mlp_i (both MLP matrices); *norms* (all LayerNorm gains); and *head*. For the CNN each layer is a block. Let v be the unit top eigenvector of H returned by Lanczos, and v_b the vector equal to v on the coordinates of block b and zero elsewhere. Since the blocks partition the coordinates, $v = \sum_b v_b$ and $\sum_b \|v_b\|^2 = 1$, and expanding the quadratic form gives

$$\lambda_{\max} = v^\top H v = \sum_b s_b + \chi, \qquad s_b = v_b^\top H_{bb} v_b, \qquad \chi = \sum_{b \neq c} v_b^\top H_{bc} v_c, \tag{9}$$

where H_{bc} is the block of H with rows in b and columns in c (and v_b is read as its restriction to block b where needed). We record, for each block, its share of the direction $\|v_b\|^2$ and its self-curvature s_b, and the cross-block remainder $\chi = v^\top Hv - \sum_b s_b$, with $v^\top Hv$ from one more Hessian-vector product (it equals the Lanczos estimate of $\lambda_{\max}$ up to the Lanczos error). Both s_b and χ can be negative, and $s_b \le \lambda_{\max}(H_{bb})\|v_b\|^2$: a block carries a large part of the sharpness only if the direction lies in it and its own diagonal block is curved along it, while a large χ means the sharp direction comes from interactions between blocks. The split of the project plan, $v_b^\top (Hv)_b$, equals $\lambda_{\max}\|v_b\|^2$ for an exact eigenvector and so repeats the direction share; this is why we use (9) instead. For AdamW runs we repeat† the decomposition for $P^{-1/2}HP^{-1/2}$ and its top eigenvector, whose diagonal blocks are $P_b^{-1/2}H_{bb}P_b^{-1/2}$. Figures sum blocks by type (all *qk*, all *ov*, all *mlp*) or by layer. Since χ counts every pair of distinct blocks, it includes the curvature between the attention and MLP matrices of one layer. For the curvature between layers† we recompute, at saved checkpoints, the whole matrix $v_\ell^\top A v_m$ over layers ℓ, m (embeddings, blocks $1, \dots, L$, and the head with the final LayerNorm), with $A = H$ for SGD and Muon and $A = P^{-1/2}HP^{-1/2}$ for AdamW; its entries sum to $\lambda_{\max}(A)$, and its off-diagonal part is the curvature between layers. The same matrix over sublayers splits each block into its attention and its MLP, each with the LayerNorm that feeds it.

4.8 Function metrics

Offline, on saved checkpoints, we compute three groups of metrics at fixed validation points (definitions in Section B.1). *Input derivatives* of the loss as a function of the input (the token embeddings for the language models), from first to third order: gradient norm, Jacobian norm of the softmax output (Novak et al., 2018), Frobenius norm and top eigenvalue of the input Hessian (Moosavi-Dezfooli et al., 2019), level-set curvature (Srinivas et al., 2022), and three third-order quantities. *Weight norms*: those linear theory predicts each optimizer keeps small (ℓ_2, ℓ_∞, spectral), and norm-based measures compared by Jiang et al. (2020). *Accuracy and robustness*: Gaussian input noise for the CNN (Ford et al., 2019) and one-token substitution sensitivity (Hahn et al., 2021) for the language models. *Representations*†: linear CKA (Kornblith et al., 2019) between the final hidden states of two runs after each transformer block and after the final LayerNorm, on 64 training sequences, for pairs of pure runs that differ only in the optimizer, only in the data order, or in the initialization.

4.9 Break-even replication

We follow the protocol of Jastrzebski et al. (2020): plain SGD (no momentum) at a constant learning rate, without warmup, as they do; our main language-model runs instead use warmup and cosine decay to 10% of the peak rate. Besides $\lambda_{\max}$ (their λ_H^1) and ΔL, at each measured step we draw $L = 25$ mini-batch gradients $g_1, \dots, g_L$ and estimate the gradient covariance $\hat{K} = \frac{1}{L}\sum_i (g_i - \bar{g})(g_i - \bar{g})^\top$ through the eigendecomposition of its $L \times L$ Gram matrix. We record its top eigenvalue λ_K^1, its smallest non-zero eigenvalue λ_K^* ($\hat{K}$ has rank at most $L-1$), $\operatorname{Tr}\hat{K}$, and $\|g\|/\|g_5\|$, the mean over i of $\|g_i\|/\|\Pi_5 g_i\|$ with Π_5 the projection on the top five eigenvectors of $\hat{K}$. Prediction trajectories are embedded in two dimensions (PCA, then UMAP; Section B).

Runs. CIFAR-10: learning rates 0.001, 0.01 and 0.1 at batch size 128 (39,000 steps, 100 epochs), and batch sizes† 30 and 300 at learning rate 0.05 (also 100 epochs), with $\hat{K}$ always from batch-128 gradients; two data orders each. Shakespeare and TinyStories-10M†: learning rates 0.03, 0.1 and 0.3 at the training batch size, with the setting's model and length; three data orders each. A rate of 1.0 diverged within 15 (Shakespeare) and 20 (TinyStories-10M) steps and was dropped. The split analysis of Section 4.2 applied to two learning rates instead of two optimizers measures the separation of trajectories in prediction space. The three optimizers† on TinyStories-10M: each with its own hyperparameters at $\frac{1}{4}$, $\frac{1}{2}$, 1, 2 and 4 times its tuned rate, 200 warmup steps and then a constant rate, 3000 steps, data order 0 (15 runs). Their break-even step is the first step after warmup at which the sharpness reaches 0.9 of its threshold ($\lambda_{\max}$ for SGD, the preconditioned sharpness for AdamW) or, for Muon, which has no threshold on $\lambda_{\max}(H)$, 0.9 of its maximum after warmup.

5 Results

5.1 Learning-rate sweeps

Figure 26 (Section D) shows the sweeps. On both TinyStories models Muon reaches the lowest validation loss, then AdamW, then SGD: best final validation loss 1.305, 1.339 and 1.914 on TinyStories-10M and 1.181, 1.201 and 1.340 on TinyStories-30M (SGD at its best rate of the sweep on data order 0; at the rates we use, 1.622 at the re-tuned 0.06 and

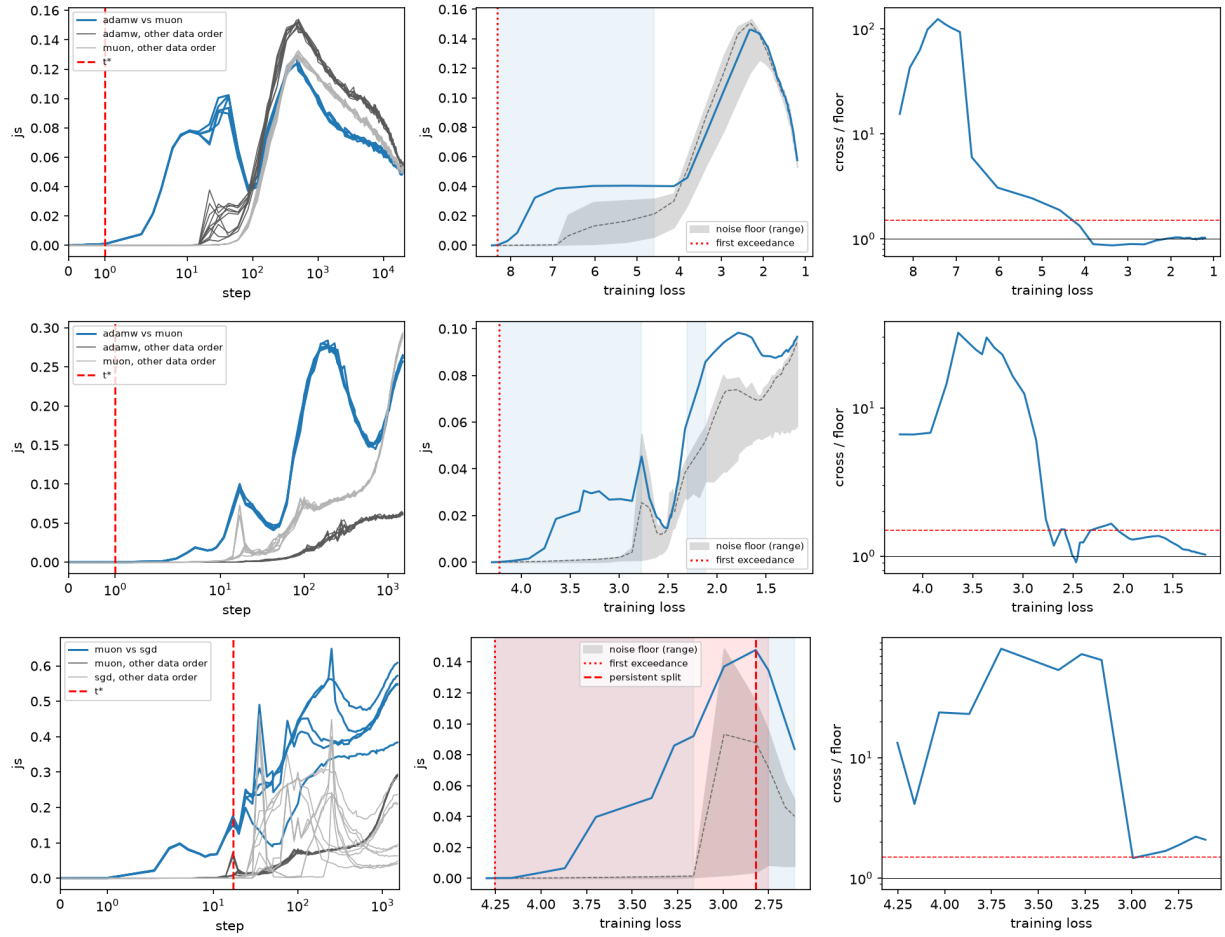


Figure 2: Function distance D_{JS} (bits) between optimizers. Rows: TinyStories-30M AdamW–Muon, Shakespeare AdamW–Muon, Shakespeare Muon–SGD (the only persistent split). Left: by step, for the 5 cross pairs (blue, same data order) and the 10 floor pairs of each optimizer (greys); the red line is the median step at which the runs of the first optimizer reach the persistent split, or the first exceedance t^* if there is none. Middle: loss-matched distance (blue) against training loss (axis reversed), with the range over floor pairs (grey band), the floor $F_{\mathcal{D}}$ (dashed), the episodes above the floor (shaded), t^* (dotted) and the persistent split (dashed, with its 95% interval shaded red). Right: cross distance over the floor, with the threshold $1 + m_{\times} = 1.5$ (red). The other pairs are in Figs. 29 and 30.

1.370), and in the main runs AdamW takes 1.35 and 1.22 times as many steps as Muon to reach its own final validation loss. On Shakespeare AdamW reaches the lowest, then Muon, then SGD: best minimum validation loss 1.54, 1.56 and 2.44 (2.50 for SGD at 0.05, about the loss at which all three optimizers pass the bigram model, Section 5.3). SGD at 1 and 3 diverged on both TinyStories models. Without dropout the Shakespeare runs overfit, Muon fastest: its validation loss is lowest near step 500 and ends at 2.84–2.88 in the main runs, against 1.54–1.55 for AdamW (Fig. 28).

5.2 Divergence curves and the split

Table 2 summarizes the split analysis for all pairs, Fig. 3 shows the loss-matched distances of every pair on TinyStories-10M, Fig. 4 the ratio to the floor in every setting, and Fig. 2 three pairs in detail.

After warmup, about the floor. After warmup, when their training loss is about the same, two runs with different optimizers are about as far apart as two runs of one optimizer with different data orders (Fig. 3). For AdamW–Muon the ratio to the floor lies in 0.95–1.38 (median 1.02) below loss 4.85 on TinyStories-10M, in 0.87–1.34 (median 1.01) below loss 4.59 on TinyStories-30M (both ranges include the end of warmup), and in 1.02–1.46 (median 1.11) below loss 2.12 on Shakespeare, after a second, short episode (2.30–2.12). On TinyStories the one exception is SGD on TinyStories-10M (at 0.06): it is up to 2.1 times as far from AdamW and 1.9 times as far from Muon at losses 4–2.5 (1.4 and 1.5 times on average), above the threshold of the split rule down to loss 3.97 against AdamW and 3.54 against Muon, which SGD reaches at steps 250 and 418, after warmup (Table 2), and within 1.1 times below loss 2.5, down to

Table 2: Split analysis with D_{JS}. Grid: range of the common loss grid. t^*: first exceedance (5), found in 99.8% of resamples (all those with at least two distinct data orders) for every language-model pair but Matched–Muon, and in 52% for CIFAR-10 (two data orders), with a 95% interval equal to the point; steps: the steps at which A_0 and B_0 reach it. Episodes: loss ranges above the floor. Persistent: the persistent split t^*_{p} and the fraction of resamples in which one is found. Max and end: the ratio $\bar{c}_{\mathcal{D}}/F_{\mathcal{D}}$ at its maximum and at the last grid level. "Matched"†: Muon with its auxiliary AdamW matched to AdamW (Section 3). Break-even rows compare plain SGD at two learning rates. TinyStories-10M SGD at 0.06 (three data orders, t^* found in the 88% of resamples with at least two distinct orders) and, in grey, at 0.03 as in the other analyses (five data orders).

Setting	Pair A–B	Grid	t^*	steps	Episodes†	Persistent†	Max	End
Shakespeare	AdamW–Muon	4.30–1.18	4.23	1 / 2	4.23–2.77, 2.30–2.12	none (0%)	32	1.02
	AdamW–SGD	4.30–2.62	4.23	1 / 1	4.23–3.10	none (34%)	98	1.69
	Muon–SGD	4.30–2.61	4.25	1 / 1	4.25–3.16, 2.82–2.61	**2.82** (96.5%)	80	2.09
	Matched–AdamW	4.30–1.19	4.26	1 / 1	4.26–2.85, 2.70–1.68	none (0%)	29	1.07
	Matched–Muon	4.30–0.23	none (4%)	–	none	none (0%)	1.28	0.98
TinyStories-10M	AdamW–Muon	8.39–1.35	8.26	1 / 7	8.26–4.85	none (0%)	111	0.99
	AdamW–SGD	8.39–1.69	8.26	1 / 3	8.26–6.32, 4.85–3.97	none (0%)	222	0.96
	Muon–SGD	8.39–1.67	8.38	1 / 1	8.38–3.54	none (0%)	240	0.95
	AdamW–SGD$_{0.03}$	8.39–2.04	8.26	1 / 3	8.26–6.32	none (0%)	162	1.03
	Muon–SGD$_{0.03}$	8.39–2.02	8.38	1 / 1	8.38–5.81	none (0%)	204	1.00
TinyStories-30M	AdamW–Muon	8.45–1.20	8.31	1 / 8	8.31–4.59	none (0%)	124	1.02
	AdamW–SGD	8.45–1.65	8.31	1 / 2	8.31–6.03	none (0%)	121	1.01
	Muon–SGD	8.45–1.65	8.44	1 / 1	8.44–6.82	none (0%)	200	1.00
CIFAR-10	η 0.001–0.01	2.30–1.09	2.27	4290 / 510	4 short within 2.27–1.13	none (0%)	19	1.47
	η 0.01–0.1	2.30–0.0004	0.50	11310 / 2730	0.50–0.41	none (0%)	1.75	1.24
Shakespeare	η 0.1–0.3	4.30–3.09	none (0%)	–	none	none (0%)	1.17	0.72

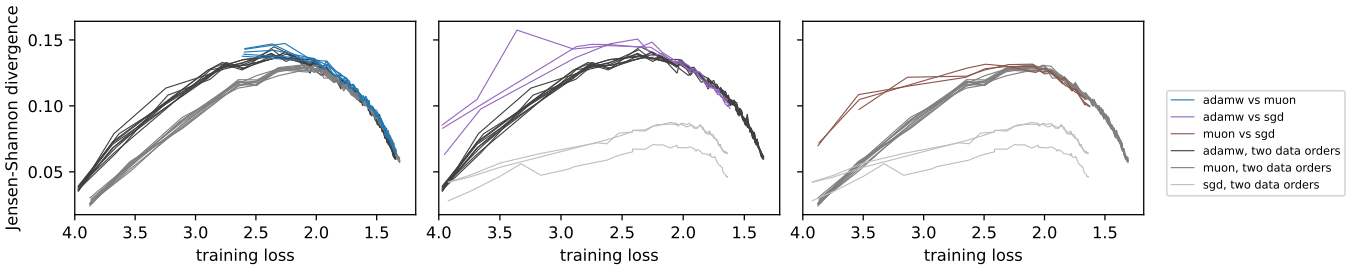


Figure 3: TinyStories-10M: Jensen–Shannon divergence between two optimizers at equal training loss, for each data order (colour), and between two data orders of the same optimizer (grey; one tone per optimizer). SGD at 0.06 on three data orders; its curves end at its lowest training loss. Hyperparameters in Tables 1 and 5.

the lowest common loss (1.69 and 1.67). With the earlier runs at 0.03 these ratios were smaller (up to 1.5 and 1.4 times at losses 4–2.5, 1.3 on average) because two of those runs blew up, which raised SGD's own data-order distance, the floor of these pairs; the two rates give about the same function at equal loss (0.76–0.95 times the floor at losses 4–2.5, 0.96–1.06 below, same data order). On TinyStories-30M every pair stays within 1.1 times the floor after warmup. At the lowest common loss all TinyStories pairs are at 0.95–1.03 times the floor. The largest absolute AdamW–Muon distances, 0.14–0.15 bits near loss 2.3 on TinyStories, occur where the floor is as large (Fig. 3). On Shakespeare the pairs with SGD stay above the floor (below).

Within warmup, large ratios over a near-zero floor.† The first exceedance t^* is the first or second level of the loss grid, reached at steps 1–8: two optimizers take different first steps while the floor is still near zero. Each pair then has one long episode above the floor at high loss. It ends at loss 4.59–6.82 on TinyStories and 2.77–3.16 on Shakespeare, which the $d = 0$ runs reach at steps 9–85, within warmup (Section 5.4), and the ratio peaks in it at 32–240. The exception is Muon–SGD on TinyStories-10M: with SGD at 0.06 its episode lasts until loss 3.54, which SGD reaches at step 418 (above); at 0.03 it ended at 5.81, at SGD step 221. These ratios are inflated: the runs still share their initialization, so the floor is near zero, and in absolute terms the cross distance in the episode is at most 0.034–0.045 bits for AdamW–Muon and 0.04–0.18 bits for pairs with SGD. We report these episodes as a detail of the first steps, not as a split.

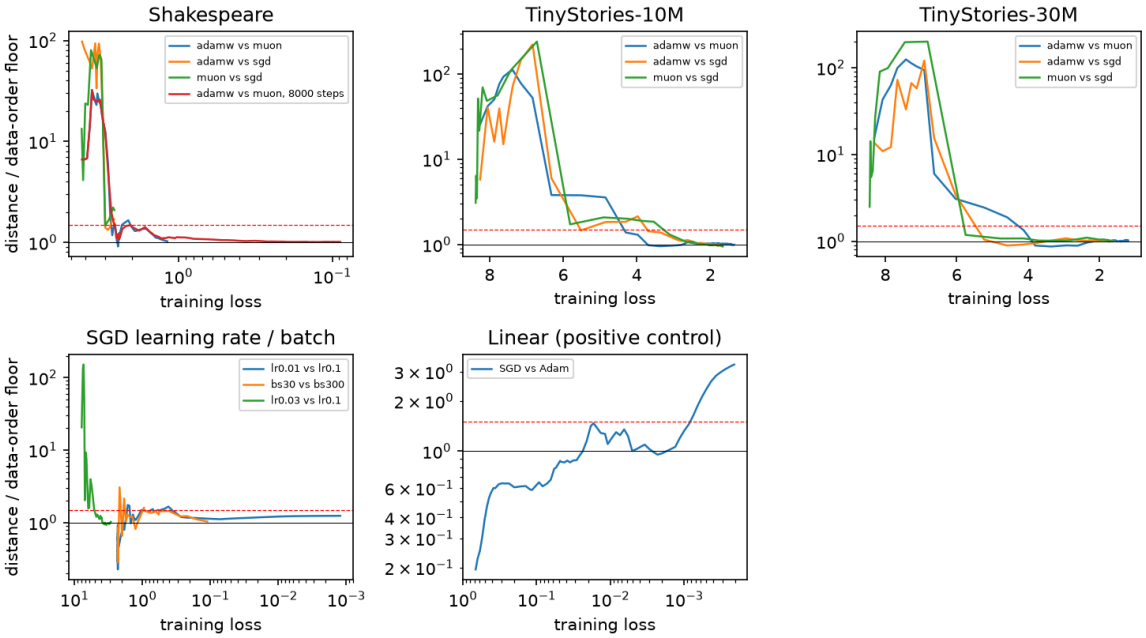


Figure 4: Function distance between two optimizers (or two SGD learning rates or batch sizes) at equal training loss, over the distance between two runs of one of them with different data orders (training progresses to the right). The red line is the threshold $1 + m_\times = 1.5$ of the split rule. In the language models the large ratios at high loss lie within warmup, where the runs still share the initialization and the floor is near zero; after warmup the ratio stays near 1, except for the pairs with SGD on Shakespeare, where SGD stalls (Muon–SGD is the one persistent split), and on TinyStories-10M, where SGD (at 0.06) stays at 1.4–2.1 times the floor between losses 4.9 and 3.4 and then returns to it. The learning rates and batch sizes of plain SGD on CIFAR-10 and TinyStories-10M (bottom left) behave alike. Red, top left: AdamW–Muon trained to memorization (8000 steps†). Bottom middle: a linear model on separable data, where gradient descent and Adam have different limits†; there the rule finds a persistent split, near interpolation.

One persistent split.† Only Muon–SGD on Shakespeare has a persistent split: a second episode from loss 2.82 to the end of the common grid at 2.61, found in 96.5% of resamples (95% interval 2.75–4.25). Muon reaches it at steps 14–17 and SGD at steps 43–550, depending on the data order. This pair tells us the least about how optimizers shape the function: the comparison stops at 2.61 because the slowest SGD run gets no further, SGD at 0.05 is barely stable there (loss spikes, Fig. 28; erratic floor, Fig. 2 bottom left), and four of its five runs end at training loss 2.32–2.58, about the bigram level (the fifth at 1.66). AdamW–SGD on Shakespeare is borderline: after its episode the ratio stays at 1.33–1.73, which is a persistent split in 34% of resamples but not in the full sample. On TinyStories SGD's own data-order noise is large (on TinyStories-30M one run ends at validation loss 1.61 against 1.37–1.39 for the others; on TinyStories-10M the runs at 0.03 end at 1.77–1.96, those at 0.06 at 1.62–1.66), so "no split" against SGD means no larger than SGD's own data-order noise.

By step the picture differs. At equal steps (left panels) the optimizers look different for much longer, because they progress at different speeds: on TinyStories-30M Muon's loss stays flat for about the first ten steps and then falls faster than AdamW's, and on Shakespeare Muon memorizes the training set while AdamW does not (Fig. 28). Section 5.3 separates the two effects for AdamW and Muon (Fig. 1).

Robustness.† We repeated the estimate for every pair and setting with D_{dis} instead of D_{JS}, and with D_{JS} at $m_\times \in \{0.25, 1\}$, $k \in \{2, 5\}$ and $\tau \in \{0.01, 0.05\}$, one change at a time (seven variants). On both TinyStories models no variant finds a persistent split for any pair, nor does any variant for AdamW and Muon on Shakespeare. The one persistent split, Muon against SGD on Shakespeare, is fragile: it is found with $k = 2$ (99% of resamples), $m_\times = 0.25$ and both tolerances, but not at the point estimate with D_{dis}, with $m_\times = 1$ (26% of resamples) or with $k = 5$ (42%). We read it as SGD ending slightly off the others where it stalls, not as a clear split.

5.3 AdamW against Muon, and optimizer against data order†

AdamW and Muon. Muon is faster than AdamW (Fig. 1, right). On TinyStories its loss is below AdamW's at every logged step after warmup in every data order; on TinyStories-10M its training loss is 2.50 against 2.76 at step

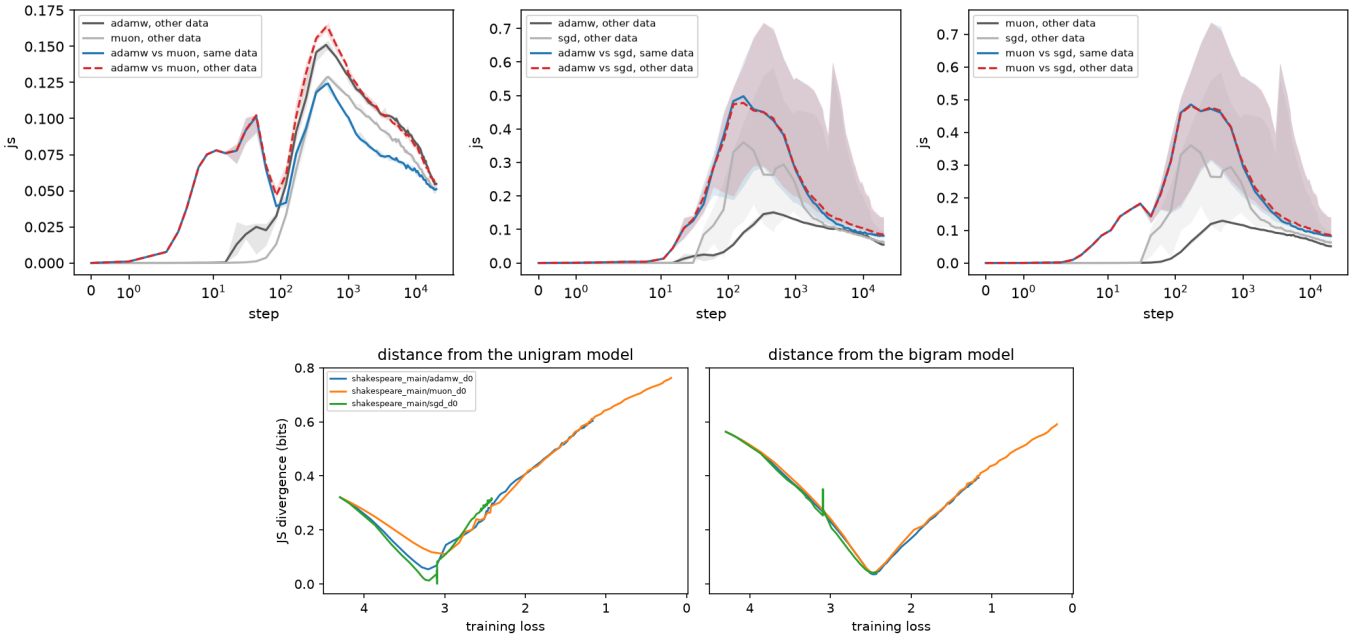


Figure 5: Top: optimizer against data order at equal steps on TinyStories-30M (Section 4.3): other optimizer on the same data order (blue) and on another data order (red, dashed), and the two floors (greys); median over the 5 data orders, range shaded. Bottom: D_{JS} of the $d = 0$ runs on Shakespeare from the unigram (left) and bigram (right) models of the training data, against training loss. Shakespeare decomposition and TinyStories n-gram distances in Fig. 31.

250 and 1.77 against 1.95 at step 1000; and AdamW needs 1.35 (TinyStories-10M) and 1.22 times (TinyStories-30M) as many steps to reach its own final validation loss. The speedup is in line with the 1.35 times reported by Jordan et al. (2024) and with Wen et al. (2025), and it does not come from a handicapped AdamW: our Muon follows the reference implementation, only the learning rates were tuned, in the same way for both, and the other difference is weight decay (none for Muon, 0.1 for AdamW; Section 3). Despite the gap in loss, AdamW and Muon trained on the same data order stay close (Fig. 1, left and middle): in the embedding each such pair are nearest neighbours, and on TinyStories-10M their divergence at equal steps is 0–23% below that between two data orders of one optimizer from step 250 to 3000, so at most about a fifth. They are closer because they share their recent batches; at equal loss this vanishes (Fig. 3). Towards the end of training all six runs of Fig. 1 gather in one cluster, as their divergences shrink.

Optimizer against data order. On TinyStories-30M (Fig. 5, top left), from step 332 on, the AdamW–Muon distance on the same data order is at or below both optimizers' floors. At step 20,000 it is 0.051 bits, against floors of 0.055 (AdamW) and 0.051 (Muon) and 0.056 for AdamW and Muon on different data orders: at the end of training, replacing AdamW by Muon changes the function no more than reshuffling the data. Sharing the data order does keep the two optimizers closer: the same-data distance is 20–32% below the other-data distance between steps 119 and 10,000, and 8% below at the end. Against SGD, the same-data distances at the end are 0.081 (AdamW) and 0.082 bits (Muon), above the median floors (0.051–0.063), and within 5% of the other-data distances; at equal steps this includes SGD being slower (final validation loss 1.37–1.39, one run 1.61, against 1.18–1.20 for AdamW and Muon). On Shakespeare (Fig. 31) the optimizer dominates at equal steps: at step 1500 the AdamW–Muon distance is 0.26 bits, four times AdamW's floor (0.06) but below Muon's own floor (0.29), which grows as Muon memorizes; same and other data orders give the same distance (0.263 and 0.264).

N-gram stages. On Shakespeare (Fig. 5, bottom) the three optimizers pass close to the bigram model at nearly the same training loss: the minimum distance to it is 0.036 (AdamW), 0.041 (Muon) and 0.041 bits (SGD), at loss 2.46, 2.47 and 2.48. At loss 2.47 the loss-matched AdamW–Muon ratio is 0.91. Before that the routes differ: SGD passes through the unigram model (0.001 bits at loss 3.09) and AdamW near it (0.054 at 3.20), while Muon stays further away (0.111 at 2.99). On TinyStories (Fig. 31) the optimizers come closest to the bigram model at loss 3.7–4.0, less closely than on Shakespeare (0.14–0.16 bits for AdamW and Muon, 0.20 for SGD); the AdamW–Muon ratio stays below 1.2 from loss 3.69 (TinyStories-10M) and 3.80 (TinyStories-30M) on. The episodes within warmup therefore cover the stage before the bigram statistics are learned, where the optimizers take different routes while the data-order floor is still near zero.

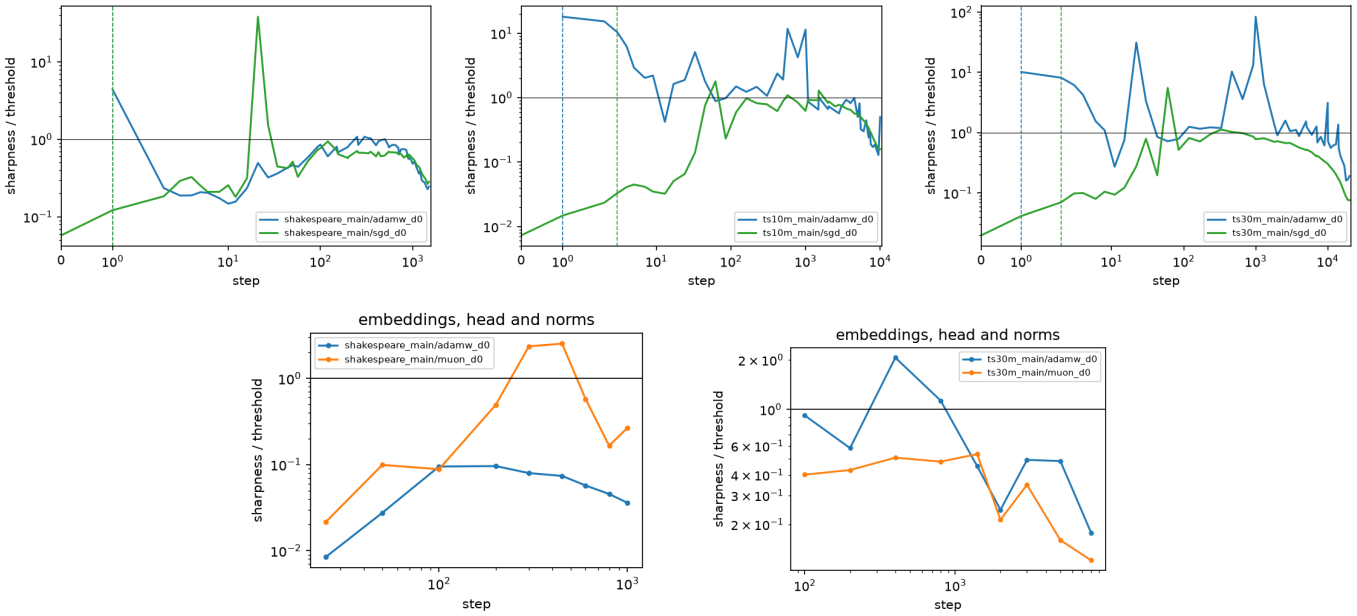


Figure 6: Top: sharpness over its stability threshold for the $d = 0$ runs on Shakespeare, TinyStories-10M and TinyStories-30M: $\lambda_{\max}\,\eta_s/(2+2\mu)$ for SGD and the preconditioned sharpness times $(1-\beta_1)\eta_s/(2+2\beta_1)$ for AdamW; 1 is the edge of stability, dashed lines the first exceedance t_X^* of the AdamW–SGD pair. Bottom†: the same ratio for the embeddings, head and gains alone (Section 4.4), at the fork checkpoints of AdamW and Muon, on Shakespeare (left) and TinyStories-30M (right). Raw sharpness, batch sharpness and thresholds in Fig. 32.

5.4 What coincides with the changes in distance

SGD reaches its threshold $(2+2\mu)/\eta_s$ only early in training, around and after the end of warmup (Fig. 6; Table 10 in Section D): in the $d = 0$ runs $\lambda_{\max}$ is within 10% of it, or above, only between steps 21 and 120, 62 and 1500, and 60 and 658, for warmups of 100, 200 and 400 steps, and at 2–16% of the measured steps over all data orders. After that, $\lambda_{\max}$ levels off and does not follow the threshold, which rises as the learning rate decays: the threshold is a ceiling, and the runs leave the edge of stability once the decay sets in. On Shakespeare the SGD runs have large transient spikes (maximum ratio 7.9–956 over the data orders), matching their loss spikes. Batch sharpness stays below $\lambda_{\max}$, and so below the threshold, apart from early transients (Fig. 32).

AdamW. The preconditioned sharpness is above its threshold at the first measured steps (ratio 3 to 21 at step 1), when the second-moment estimate rests on one or a few gradients, so $s_{\mathrm{EoS}} = 1$ in every run. After a dip it returns to the threshold or above (on TinyStories-30M with spikes to 10–80 times it around steps 500–1000) and stays there for long stretches, until the cosine decay raises the threshold away from it: the ratio is at least 0.9 at measured steps up to 425–600 of 1500 on Shakespeare, 4500–9500 of 10,000 on TinyStories-10M and 9000–19,000 of 20,000 on TinyStories-30M (11–16%, 50–65% and 40–69% of the measured steps). This is the adaptive edge of stability of Cohen et al. (2022).

Muon. Its threshold applies to the spectral-norm sharpness S (Section 4.4), which stays below its bound (Fig. 7; data orders 0 and 1 agree). The ratio peaks at 0.93 within warmup (step 50), is 0.80 when warmup ends and then falls, to 0.39 at step 3000, 0.27 at step 5000 and about 0.06 at the end: $\lambda_{\max}(H)$ levels off at about 14 by step 400, while S keeps falling, by about 3.5 times, until step 4000, and both rise again as the rate decays. At the end of the sweep runs (Fig. 33) S is 0.045, 0.055, 0.068 and 0.096 of the bound at peak rates 0.005, 0.01, 0.02 and 0.04: S falls as $\eta^{-1.58}$, the bound as $\eta^{-1.94}$ and $\lambda_{\max}(H)$ as $\eta^{-0.81}$. The bound assumes that the step is the polar factor U of the full gradient, so that $\langle \nabla L, U\rangle = \|\nabla L\|_*$. A stochastic Muon step follows the momentum and the batch noise and is several times less aligned with the population gradient, so the bound does not bind; this fits the stochastic edge of stability of Section 5.5. Muon's $\lambda_{\max}$ fluctuates widely on Shakespeare (between about 10 and 400) and is smooth on TinyStories (Fig. 32).

Across learning rates.† In the sweeps with sharpness measurements (Section 4.4) the plateau of the sharpness falls about as $1/\eta$ for all three optimizers. On TinyStories-10M the log-log slope against the peak rate is -0.96 for SGD (3 rates, at 0.8–0.9 of its threshold), -0.73 for AdamW's preconditioned sharpness (3 rates, at 1.0–1.5 of its threshold) and -0.86 for its raw $\lambda_{\max}$, and -0.84 for Muon's raw $\lambda_{\max}$ (4 rates). On Shakespeare AdamW's slope is -0.90, at its threshold, without its two largest rates, which blew up; SGD stalls at loss 2.5 at every rate, so its sweep says nothing;

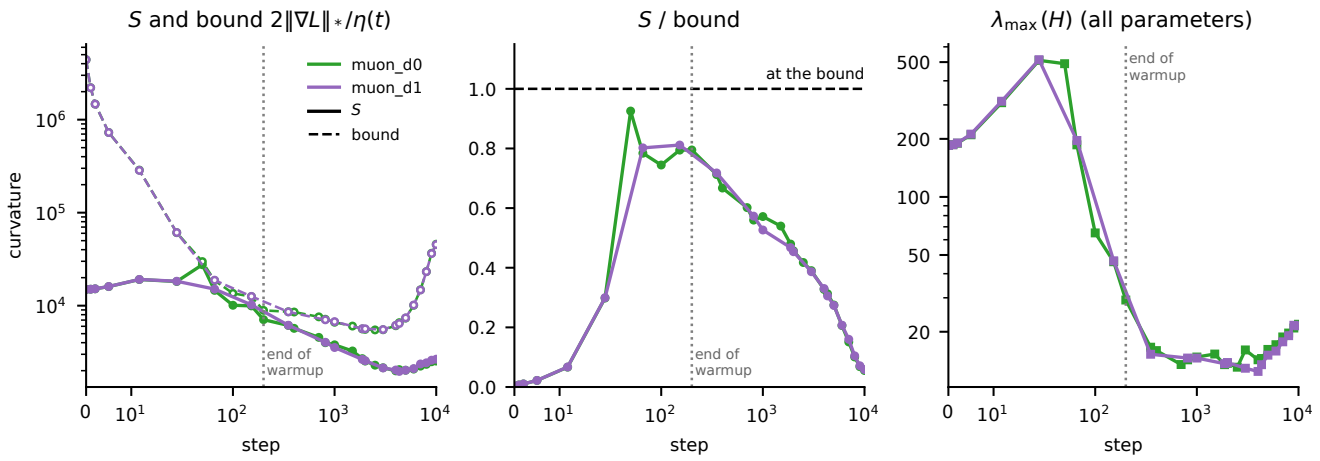


Figure 7: Muon's spectral-norm sharpness† S on TinyStories-10M, data orders 0 and 1 (Section 4.4). Left: S (solid) and its bound $2\sum_l c_l \|\nabla_l L\|_*/\eta_s$ (dashed); middle: their ratio, 1 at the bound; right: $\lambda_{\max}(H)$ over all parameters. Dotted: end of warmup. The sweep over learning rates is in Fig. 33.

and Muon's slope is -0.21 over four rates but -1.0 over the three smallest. As in the main runs, the sharpness levels off after warmup and does not follow the threshold up as the cosine decay lowers the rate. Muon's $\lambda_{\max}(H)$ thus scales like the thresholds of the other two, although its own threshold is on a spectral-norm sharpness (Section 6). Unlike Jastrzebski et al. (2020), who keep the learning rate constant, these runs use warmup and cosine decay to 10%; our CIFAR-10 replication uses a constant rate, as they do (Section 5.9).

Coincidences. No edge-of-stability event explains the changes in distance. The first exceedance, at steps 1–8, comes before the end of warmup and before SGD's onset; it coincides only with AdamW's onset at step 1, which reflects the preconditioner of the first steps rather than an edge of stability. The early episodes are short in steps: the $d = 0$ runs reach the loss at which the episode ends at steps 9–85 (221 for SGD at 0.03 in the TinyStories-10M Muon–SGD pair), within warmup in every pair but that one. With SGD at 0.06 (whose sharpness we did not analyse) the Muon–SGD episode ends at SGD step 418 and the second AdamW–SGD episode at step 250 (Section 5.2). For the pairs with SGD this is at SGD steps 20–45 (and 221), in the same range as SGD's onset in the $d = 0$ runs (steps 21, 62 and 60). But the AdamW–Muon pairs return to the floor in the same way without SGD, and AdamW, which reaches that loss at steps 29, 45 and 60, stays near its threshold long after the distance has returned to the floor. At the one persistent split (Muon–SGD, Shakespeare), each SGD run is below its threshold when it reaches loss 2.82 (ratio 0.18–0.80), and Muon's auxiliary AdamW is far below its own (ratio 0.02 at step 25, the first fork checkpoint after Muon reaches 2.82). As a loss level the end of the early episode is similar across the two TinyStories sizes (AdamW–Muon: 4.85 and 4.59) and close to the bigram stage (Section 5.3), which suggests describing it as a stage of learning rather than a step.

The block left to Muon's auxiliary AdamW.† On Shakespeare (Fig. 6, bottom left), the embeddings, head and gains of Muon exceed their threshold at the fork checkpoints at steps 300 and 450 (ratio 2.35 and 2.53) and are below it at the other seven (at most 0.57), while in the AdamW run the same block stays at most 0.10 times its threshold. These steps come just before Muon's validation loss starts to rise, and in this period Muon's top Hessian eigenvector lies almost entirely in the embeddings (Section 5.7). On TinyStories-30M it is the other way round: AdamW's block exceeds its threshold at steps 400 and 800 (2.06 and 1.13) and Muon's stays at most 0.53. In both settings the block that overshoots is trained by the Adam with the larger learning rate on these parameters (Shakespeare: Muon's 10^{-3} against AdamW's $3 \cdot 10^{-4}$; TinyStories-30M: AdamW's $2 \cdot 10^{-3}$ against Muon's $1.5 \cdot 10^{-3}$), so the learning rate plays a role; the control below shows that it is not the whole story.

Control.† Matching Muon's auxiliary AdamW to AdamW (Section 3) does not remove the memorization: matched Muon still reaches training loss 0.22 with validation loss 2.5, against 1.55 for AdamW, so the memorization comes from Muon's updates of the hidden matrices. With the same Adam, learning rate and threshold on the embeddings, head and gains as AdamW, matched Muon's block still reaches its threshold (ratio 1.14 and 1.08 at steps 600 and 1000, against at most 0.10 for AdamW; Fig. 39), later and less far than with Muon's default 10^{-3} (2.35 and 2.53 at steps 300 and 450). Its raw sharpness on this block reaches 236 and 579 at steps 600 and 1000, against at most 61 for AdamW. So Muon's updates of the hidden matrices make the embeddings sharper, and the larger learning rate of the default auxiliary AdamW brings the overshoot forward. In function space matched Muon is indistinguishable from

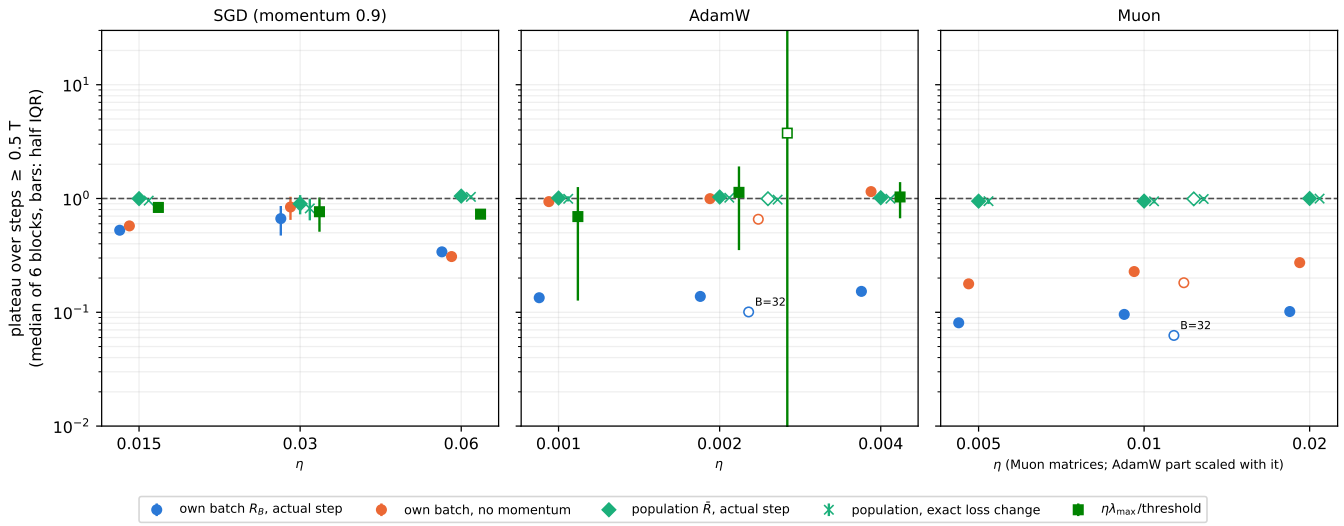


Figure 8: Stochastic edge of stability† on TinyStories-10M: constant rate after 200 warmup steps, 3000 steps, data order 0. Each ratio's plateau over the second half (median of 6 blocks; bars: half the interquartile range) against the rate; filled: batch 128, open and shifted right: batch 32. Blue: R_B on the step's own batch; orange: the same along the step without momentum ($-\eta g_B$, $-\eta D^{-1} g_B$, $-\eta\,\mathrm{polar}(g_B)$ on Muon's matrices); green: $\bar{R}$ on 256 fresh sequences (crosses: from the exact loss change); squares: $\eta\lambda_{\max}$ over its threshold on 32 fixed sequences (preconditioned for AdamW). SGD's tuned rate is 0.06; Muon's auxiliary AdamW takes 10^{-3}, $2 \cdot 10^{-3}$ and $4 \cdot 10^{-3}$. SGD at 0.03 blew up at step 2300, after $\bar{R}$ had been above 1 for 300 steps.

Muon (first exceedance in 4% of resamples, ratio at most 1.28; Table 2), and against AdamW it behaves as Muon does: two episodes (4.26–2.85 and 2.70–1.68), no persistent split, and a ratio of 1.07 at the lowest common loss (Fig. 29).

5.5 Stochastic edge of stability for AdamW and Muon†

To second order, a step s lowers the loss if and only if $R = s^\top H s/(-2g^\top s) < 1$, with g and H the gradient and Hessian before the step; this is the curvature along the update the optimizer actually takes, the quantity Lee and Lee (2026) find near 1 for full-batch training. On TinyStories-10M (data order 0) we measure it along the actual step (without decoupled weight decay, which changes AdamW's ratio by less than 1%) at three constant rates per optimizer (Fig. 8). With g_B and H_B on the step's own batch, R_B does not settle at 1: it falls throughout training, and over the second half of the run it is 0.13–0.15 for AdamW, 0.08–0.10 for Muon and 0.34–0.67 for SGD. With $\bar{g}$ and $\bar{H}$ on 256 fresh training sequences, which estimate the population loss, $\bar{R}$ settles within about 500 steps, at 1.01–1.03 for AdamW, 0.95–1.00 for Muon and 0.90–1.05 for SGD. It stays there at batch 32 (1.00 for AdamW, 0.99 for Muon) and under the cosine schedule (0.95, 0.92 and 1.05 over its second half), while $\eta\lambda_{\max}$ falls to about 0.3 of its threshold (Fig. 34). The two ratios differ by the alignment of the step with the gradient: exactly $R_B = \bar{R}\,\kappa\,\rho$, with the curvature ratio $\kappa = s^\top H_B s / s^\top \bar{H} s$ at 1.0–1.4 and the coherence $\rho = \bar{g}^\top s / g_B^\top s$ at 0.10–0.12 for AdamW, 0.06–0.07 for Muon and 0.30–0.66 for SGD, so R_B falls as the coherence does. AdamW and Muon thus sit at an edge of stability of the population loss along the step they take, not of their batch loss. For Muon neither part of the step is at 1 alone (about 0.84 on the hidden matrices and 0.45 on the auxiliary AdamW part); the cross term brings the full step to 1. $\bar{R} \approx 1$ is a weak descent: from the exact loss change, $1 - \bar{R}$ is the fraction of its first-order decrease of the population loss that a step achieves, between -2% and 6% for AdamW and Muon at a constant rate.

5.6 Swaps and mode connectivity

Swaps. On TinyStories-10M (Fig. 9, top) the forks fall into three groups. *AdamW↔Muon*: at every fork step from 50 to 4000 the fork ends within the floor of the pure run of the new optimizer (0.86–0.95 floors) and, increasingly with the fork step, also of the parent's (0.92 floors at step 50, 0.20 at step 4000), so r falls from 0.5 to 0.18. Since the two pure runs are themselves within the floor of each other at the end (Section 5.3), these swaps cannot reveal an imprint of either optimizer. *SGD→AdamW and SGD→Muon* land on the new optimizer (0.91–1.08 floors from its pure run) at every fork step up to 4000, and stay 1.7–2.0 floors from pure SGD: 4000 steps of SGD leave no trace that AdamW or Muon do not erase. *AdamW→SGD and Muon→SGD* land on pure SGD when forked by step 200 (0.91–1.01 floors); later forks end further from it (1.04–1.42 floors at steps 400–1000), and from step 1500 on closer to the parent than to

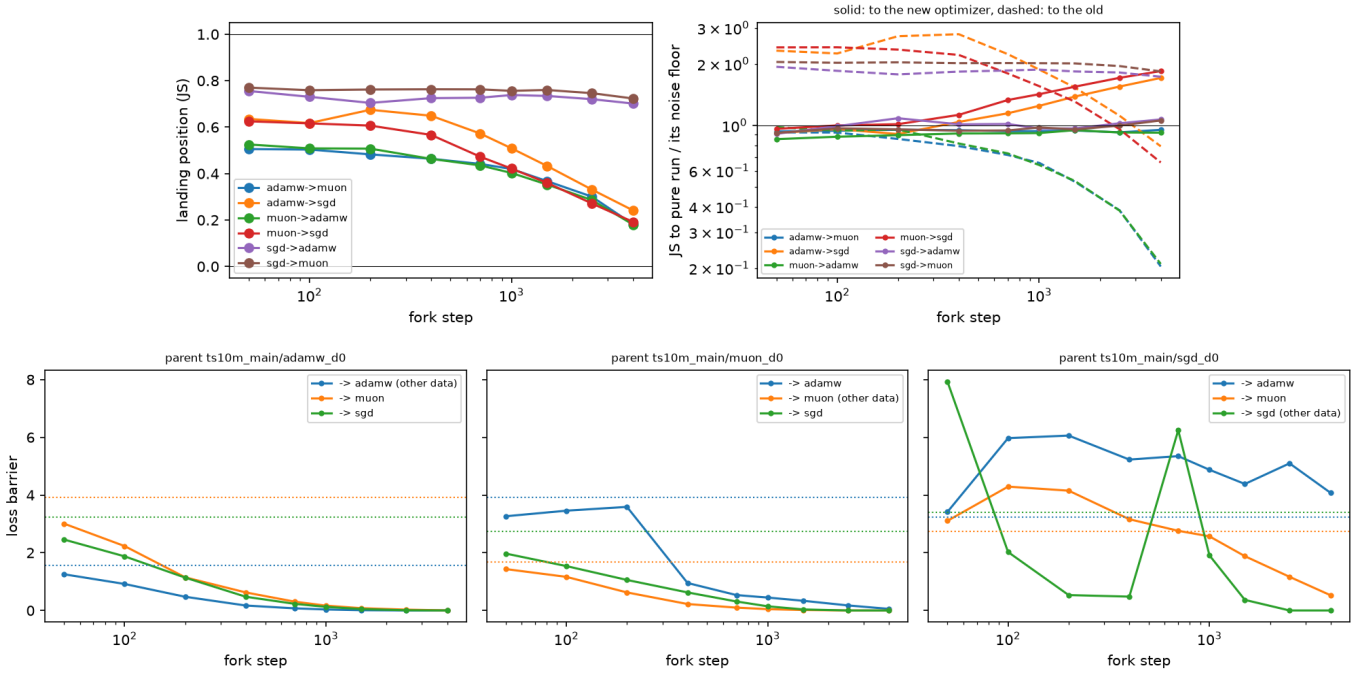


Figure 9: Swaps and mode connectivity on TinyStories-10M. Top left: landing coordinate r (7) with D_{JS} against the fork step, for the six ordered pairs; 0 is the final function of the parent (optimizer kept), 1 that of the pure run of the new optimizer (same batches). Top right†: the fork's distance to the pure run of the new optimizer (solid) and of the old one (dashed), each over that pure run's final data-order floor. Bottom: barrier (8) between the final weights of each $d = 0$ parent and of its forks against the fork step; "other data" are the forks that keep the optimizer and change the data order. Dotted†: barrier to the pure runs of the other optimizers and to the parent's optimizer with data order 1, which share only the initialization.

pure SGD (0.66–0.79 floors from the parent and 1.7–1.8 from pure SGD at step 4000): SGD at its stable rate is too slow to undo what the adaptive optimizer has already learned. In function space the last optimizer decides the endpoint whenever it is fast enough to do so, and none of the forks keeps an imprint of an early phase that the later optimizer could have reached on its own. Nearly all fork steps lie after the end of the early episodes (steps 9–85 in the $d = 0$ runs, Section 5.4), and r changes smoothly across them, so the end of the episode does not decide the endpoint either.

On Shakespeare (Fig. 42) the same rule holds, with Muon as the fast optimizer. AdamW→Muon and SGD→Muon land on pure Muon (0.93–1.05 floors) at every fork step from 25 to 1000. The reverse swaps do not land on the slower optimizer: from step 50 on, forks from Muon into SGD stay with Muon (0.43–1.03 of Muon's floor, 5.7–9.0 of SGD's), and forks from Muon into AdamW end 1.2 floors from pure AdamW at step 25 but 3.2 at step 1000. Muon's floor is large here (0.29 bits, against 0.06 for AdamW), as its runs memorize different things, so ending within it is a weak statement. Forks from SGD into AdamW keep a mark of SGD from step 50 on (3.2–5.7 floors from pure AdamW): AdamW at $3 \cdot 10^{-4}$ does not undo the early, unstable phase of SGD on Shakespeare in the remaining steps. Forks from AdamW into SGD are erratic (r between 0.01 and 0.93), like SGD's own runs there. In both settings, then, the optimizer that trains last decides the function when it is fast enough to undo the earlier optimizer's work in the remaining steps: on TinyStories whenever it is the faster one, and otherwise only for early forks.

Mode connectivity. Forks that keep the optimizer and change the data order (Fig. 9, bottom) become linearly connected to their parent at about the same time for AdamW and Muon: the barrier falls below 0.05 nats for forks from step 1000 (AdamW 0.035, Muon 0.046) and to 0 from step 2500, the stability point of Frankle et al. (2020) at 10–25% of training. For SGD the same barrier jumps between 0.4 and 7.9 up to step 1500, like its data-order noise (Section 5.2), and is 0 from step 2500. A change of optimizer needs a later fork to stay in the parent's basin: from the AdamW parent the Muon and SGD forks reach 0.03 and 0.01 nats only at step 2500, from the Muon parent the AdamW fork is still at 0.055 at step 4000, and from the SGD parent the AdamW fork is never connected (barrier 3.4–6.1 at every fork step) and the Muon fork only approaches it (0.53 at step 4000). Weight space thus keeps a mark of the optimizer that function space does not: the fork from AdamW to Muon at step 1000, which ends within the floor of both pure runs, has a barrier of 0.17 to its parent, against 0.035 for the fork that keeps AdamW and changes the data order. The pure runs, which share only the initialization, are not linearly connected for any pair (dotted lines); we compare them below.

On Shakespeare (Fig. 42, bottom) the forks that keep AdamW and change the data order are connected to their parent

from step 300 on (0.015, then 0), and even the pure AdamW runs with two data orders have barriers of only 0.11–0.17: at $3 \cdot 10^{-4}$ AdamW stays close to the initialization. Forks from AdamW into Muon or SGD keep barriers of 1.4–3.3 up to step 600, and those into Muon are still at 0.27 at step 1000; forks from SGD into Muon stay at 1.3–1.7 up to step 800. The exception is the Muon parent: its forks with another data order are connected only from step 800 (barrier 3.5 at step 25, 0.06 at 800), later than its forks into AdamW and SGD (0.9 and 0.7 at step 100, 0.01 at 800). Muon on Shakespeare is unstable to the data order for half of training, while it memorizes, and its forks into slower optimizers move the weights less after the fork. Apart from this case, a fork that changes the optimizer stays further from its parent in weight space than one that changes the data order, although it does not in function space.

Do swaps end in the new optimizer's basin?† A fork into B that ends at B's function need not end in the region of weight space of the pure B run. We computed the barrier between the final weights of each pure $d = 0$ run and of every fork into its optimizer (Fig. 45). On TinyStories-10M the forks into AdamW (2.9–8.0 nats) and into Muon (3.5–6.5) are further from the pure run than the same optimizer with another data order from the initialization (1.2–1.9 and 1.6–1.8 over ten pairs), at every fork step from 50 on, while the forks into SGD (2.4–3.4) are within SGD's own data-order barriers (0.1–4.1). The forks from AdamW into Muon, for instance, end within Muon's data-order floor in function space at every fork step, but 3.7–4.3 nats of barrier away from pure Muon. On Shakespeare the forks into AdamW (0.8–2.4) are far above AdamW's data-order barriers (0.11–0.17), while those into Muon (2.3–6.6) and SGD (0.9–2.3) are comparable to Muon's (3.6–6.6) and SGD's (1.0–3.2). As few as 50 steps of another optimizer at the start can move the endpoint further in weight space than a different order of all the data, while in function space they leave no trace.

TinyStories-30M. The larger model (Figs. 43 and 45) repeats these results, with one difference in the swaps into SGD. Forks between AdamW and Muon end within the floor of both pure runs at every fork step from 100 to 8000 (of 20,000), and r falls from 0.52 to 0.25. Forks from SGD land on the new optimizer (0.84–1.07 floors) at every fork step, but pure SGD ends only 1.1–1.4 floors from AdamW and Muon here, so these swaps say little. Forks from AdamW and Muon into SGD land near pure SGD up to step 3000 (0.92–1.21 floors), and only those at steps 5000 and 8000 end closer to the parent (0.57–1.00 floors from it) than to pure SGD, against step 1500 on TinyStories-10M: SGD's tuned rate is higher here (0.1 against 0.03), so it erases more. Forks that keep AdamW or Muon and change the data order are connected to their parent from step 2000 (0.022 and 0.040 nats, 0 from step 5000). Those that keep SGD have a barrier of only 0.14 at step 100 and below 0.06 from step 400, although the pure SGD runs with data orders 0 and 1 have a barrier of 8.8: SGD's sensitivity to the data order is confined to the first 100 steps. A change of optimizer again needs a later fork. From AdamW the SGD fork reaches 0.048 nats at step 3000, and the Muon fork 0.077 only at step 8000. From Muon the SGD fork is connected from step 3000 (0.068), but the AdamW fork is still at 0.19 at step 8000. From SGD the AdamW and Muon forks are never connected (0.69 and 0.29 at step 8000). The forks into Muon (4.4–5.7 nats) again end further from the pure run than another data order does (1.6–2.1), at every fork step. The forks into AdamW (3.5–18.8) do so only when forked at steps 200–400 from Muon and 200–5000 from SGD, since AdamW's own data orders are 3.3–6.8 nats apart here, and the forks into SGD (1.6–2.9, apart from 10.2 for Muon at step 100) stay within SGD's own (0.6–9.2).

Barriers between pure runs†. We computed the barrier between the final weights of every pair of pure runs that differ only in the optimizer or only in the data order, in full and with one group of parameters moved at a time (Fig. 10). In size, a change of optimizer is not consistently further than a change of data order. AdamW and Muon are further apart than two data orders of either on TinyStories-10M (3.5–4.1 nats against 1.2–1.9 and 1.6–1.8), but not on TinyStories-30M, where AdamW's data orders are 3.3–6.8 nats apart (AdamW–Muon 4.3–4.8), nor on Shakespeare, where Muon's are 3.6–6.6 apart (AdamW–Muon 2.1–2.7); SGD's data-order barriers vary so much (0.1–4.1 and 0.6–9.2 on TinyStories) that the pairs with SGD fall within them. Where the barrier sits does tell them apart. Between two data orders the single-layer barriers sit in the first block, up to 4.0 nats on average, and are at most 0.6 in any later block, or there are none at all (AdamW on Shakespeare, Muon on TinyStories-30M) although the full barrier is not small. Between AdamW and Muon on the same data they extend into the middle of the network: 0.8 and 0.4 nats in blocks 1–2 on TinyStories-10M, 2.3–0.5 in blocks 1–4 on TinyStories-30M and 1.1–1.9 in blocks 1–5 on Shakespeare. The pairs with SGD look like data orders, with the barrier mostly in the first block (and the embeddings on Shakespeare), and the head carries none (the loss is convex in the head). Two data orders, or SGD and an adaptive optimizer, thus end in solutions that differ in how the first block reads the embeddings, while AdamW and Muon also differ in how the middle blocks process them.

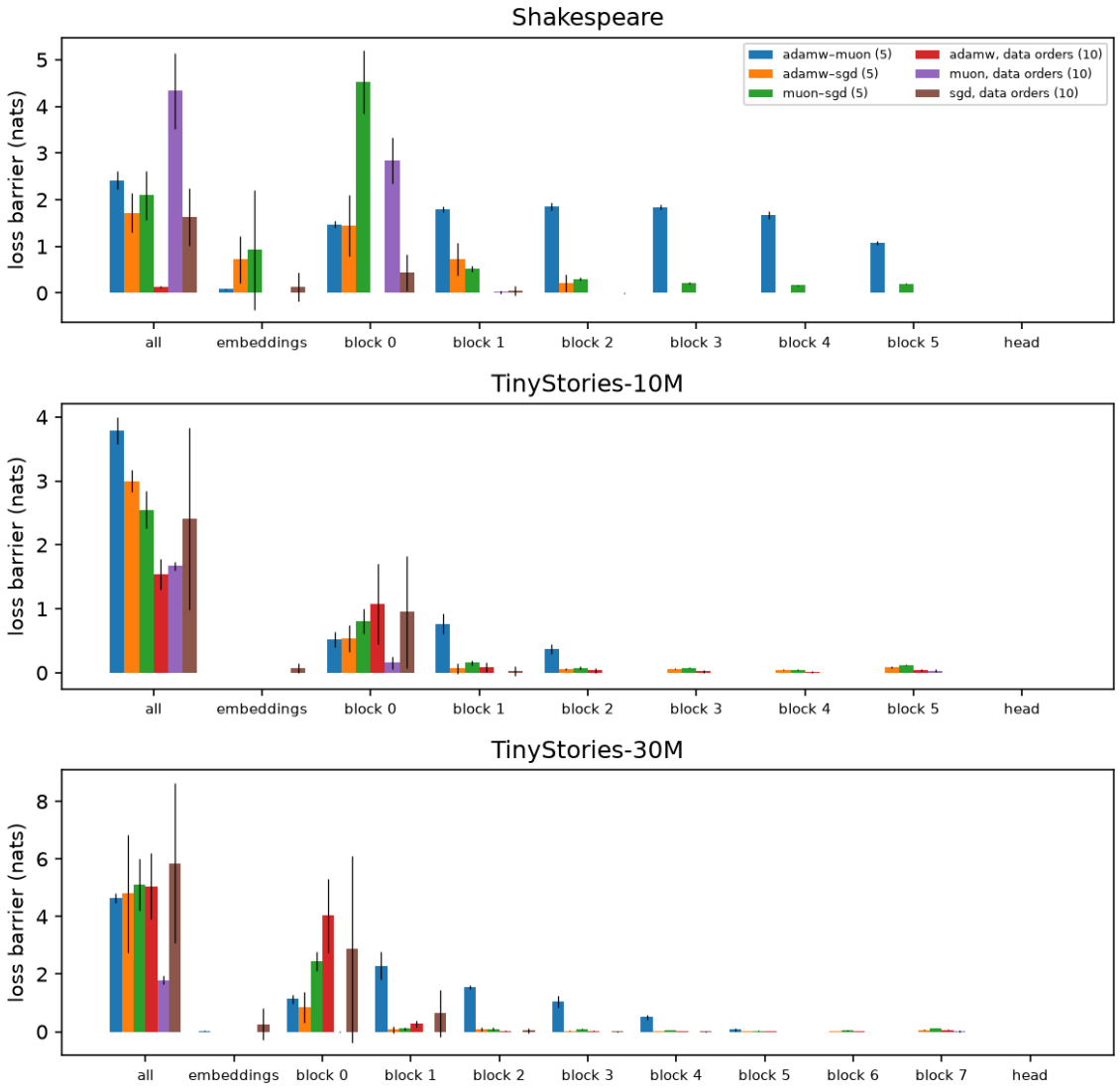


Figure 10: Loss barriers† between the final weights of pure runs, for every pair that differs only in the optimizer (5 per pair of optimizers) or only in the data order (10 per optimizer); mean and standard deviation over pairs. "All" is the ordinary barrier; for the other groups only that group of parameters moves along the straight line from the first run to the second, the rest stay at the first run's (Adilova et al., 2024).

5.7 Anatomy of the sharp direction

On TinyStories-30M (Fig. 11) the sharp direction differs by optimizer. Muon's lies mostly in the value–output matrices (about 60–70% of the direction) throughout training, with most of the rest in the MLP matrices. AdamW's lies in the head until about step 4000 and then largely in the gains (30–50%), with the value–output and MLP matrices. SGD's is spread over the MLP, value–output and head blocks. In all three the cross-block term χ, which here also counts the curvature between the attention and MLP matrices of one layer, is the largest part of the eigenvalue: about 75–85% for Muon and SGD throughout, and for AdamW about 25–40% while the head dominates and 60–70% after. The sharp direction is therefore not the top eigenvector of a single block but comes from curvature between blocks. On TinyStories-10M (Fig. 35) Muon's direction is again in the value–output matrices (60–70%), while AdamW's stays mostly in the head (60–75%). On Shakespeare Muon's direction lies almost entirely (85–97%) in the embeddings from step 200 on, with the embeddings' own curvature making up most of the eigenvalue: the sharp direction is in the block that Muon leaves to its auxiliary AdamW, including the period in which that block is over its threshold (Section 5.4). AdamW's direction on Shakespeare is spread over the MLP, value–output and embedding blocks, with about 70% of the eigenvalue from χ. The changes of dominant block (AdamW on TinyStories-30M, head to gains around step 4000–5000) happen long after warmup, when the loss-matched distance between optimizers is at the floor. For AdamW

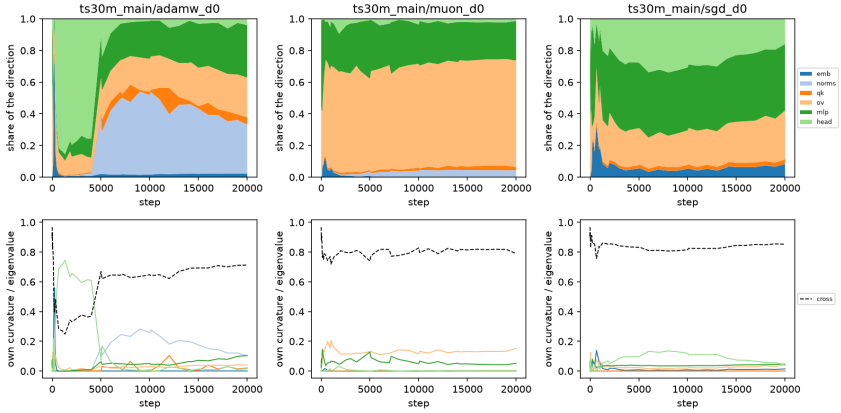


Figure 11: Anatomy of the top Hessian eigenvector of the $d = 0$ runs on TinyStories-30M (9). Top: share of the direction $\|v_b\|^2$ by block type. Bottom: each block type's own curvature s_b and the cross-block remainder χ (dashed), as fractions of the eigenvalue. Shakespeare and TinyStories-10M in Fig. 35.

the top eigenvector of the preconditioned† operator $P^{-1/2}HP^{-1/2}$, the one that sets its stability threshold, lies elsewhere than that of H (Fig. 36): mostly in the MLP matrices (40–60%), the attention matrices and the embeddings on both TinyStories sizes, rather than in the head or the gains, with short periods (the first 1000 steps and a few isolated measurements) in which it lies almost entirely in the embeddings and their own curvature makes up the eigenvalue. Outside those periods 80–95% of the preconditioned eigenvalue comes from curvature between blocks, even more than for H.

By layer, at the edge of stability. Restricted to the steps at which each optimizer is at its threshold and averaged over the five data orders (Fig. 12), the direction of the operator that sets the threshold lies at the input end of the network: in all three settings the embeddings and the first three blocks carry 75–90% of it, and the later blocks little. Which part of that end differs by optimizer. SGD, at its threshold only early in training, has its direction mostly in the first block (35–47%), with the embeddings and the second block; later, below the threshold, part of it moves to the head (21–30% on Shakespeare and TinyStories-30M). The direction that sets AdamW's threshold lies on TinyStories in the embeddings (59–66%, against 29–32% at the steps below the threshold), whose own curvature makes up most of the preconditioned eigenvalue (51–59%). The preconditioner P^{-1} is largest where gradients are rarest, as in the embedding rows of rare tokens, so this is where the preconditioned operator is sharpest; on Shakespeare, with 65 characters, it lies instead in the first three blocks. At the same steps the direction of AdamW's raw H is in the head (29–32%) and the first block. Muon's direction lies in the first three blocks on TinyStories (73–82%), in the matrices it trains; on Shakespeare it lies in the embeddings (63%), which its auxiliary AdamW trains. The first block, where most of the sharp direction of SGD and Muon lies, is also where the single-layer barriers between data orders sit (Section 5.6).

Curvature between layers.† On TinyStories-10M we recomputed the split of the eigenvalue over pairs of layers at the saved checkpoints at the threshold (Fig. 13). The sharp direction lies at the input end: SGD's in blocks 1–2 (66%), AdamW's, of $P^{-1/2}HP^{-1/2}$, in the embeddings (68% at its two steps at the threshold, both of d0; 66% over all its saved steps), and Muon's, of H, which is not its threshold operator, in blocks 1–3 (84%). The share of the eigenvalue that comes from curvature between layers is 66% for SGD and 71% for Muon; for AdamW it is 36% overall, since most of its eigenvalue is the embeddings' own curvature, and 62% outside the embeddings. These shares are lower than the cross-block term χ above, which also counts the curvature between the attention and MLP matrices of one layer. Split into sublayers (Fig. 14), SGD's direction lies mostly in the MLPs (52%, against 37% in attention; 28% in the first MLP), Muon's in attention (60% against 37%; 26% in the first MLP, 23% in the second attention and 16% in the first), and AdamW's in the embeddings (68%) and then the first attention (13%). Between sublayers comes 76% of the eigenvalue for SGD, 80% for Muon and 40% for AdamW, and no cell is negative. The layer that holds the direction keeps changing long after warmup (AdamW's between the embeddings and block 1 throughout training, Muon's from block 1 to 2 after step 6000), so it does not mark a split.

5.8 Curvature-balanced per-layer learning rates†

The sharp direction lies in a few layers (Fig. 13), so a rate shared by all layers is limited by them. We tested whether per-layer rates that even out the curvature train faster, like other layer-wise rates (You et al., 2017; Yang et al., 2021; Wang et al., 2025) (Section E).

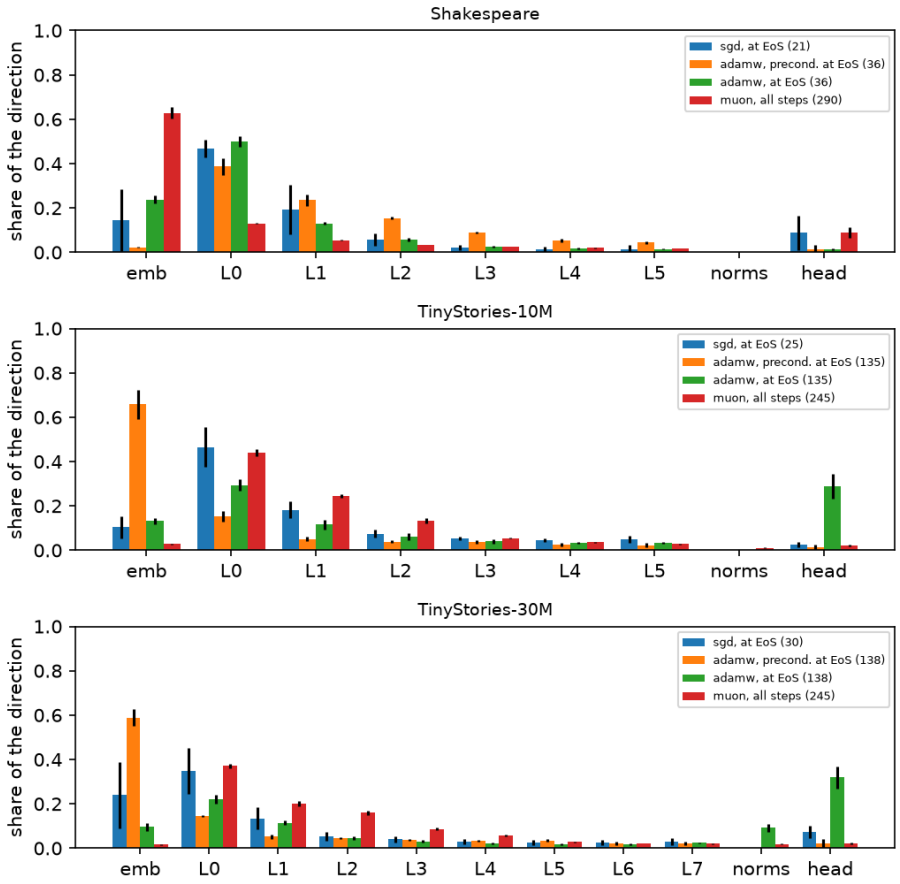


Figure 12: Share of the top eigenvector by layer at the edge of stability (sharpness at least 0.9 of the threshold), mean and standard deviation over the five data orders; in brackets, the number of measured steps. For AdamW, of $P^{-1/2}HP^{-1/2}$ (which sets its threshold) and of H at the same steps; for Muon, of H (threshold: spectral-norm sharpness), all steps. Li: transformer block i; norms: the gains of all LayerNorms.

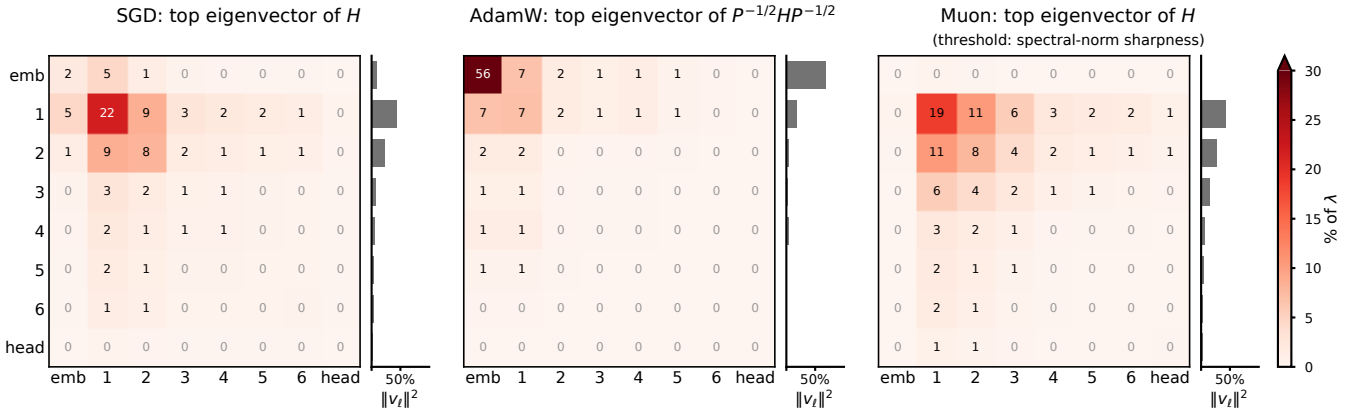


Figure 13: Split of the top eigenvalue by layer† on TinyStories-10M at the edge of stability: cell (ℓ, m) is $v_\ell^\top A v_m$ as a percentage of the eigenvalue, with A the operator in each title (diagonal: curvature within a layer; off-diagonal: between layers); bars: share of the direction $\|v_\ell\|^2$. Mean over data orders of each run's mean over its saved checkpoints at the threshold (all saved steps for Muon). The AdamW panel uses 2 steps of d0, because the preconditioned operator needs the optimizer state, which only the fork checkpoints keep.

Method. Layer a (the embeddings, each block, and the final LayerNorm with the head) trains at $\eta_t \kappa m_a$. With M the diagonal matrix with κm_a on the coordinates of layer a, and $A = H$ for SGD and $P^{-1/2}HP^{-1/2}$ for AdamW, the operator $\hat{H} = \eta_t M^{1/2} A M^{1/2}$ is stable below $2 + 2\mu$ (SGD) or $(2 + 2\beta_1)/(1 - \beta_1)$ (AdamW). The share of layer a in

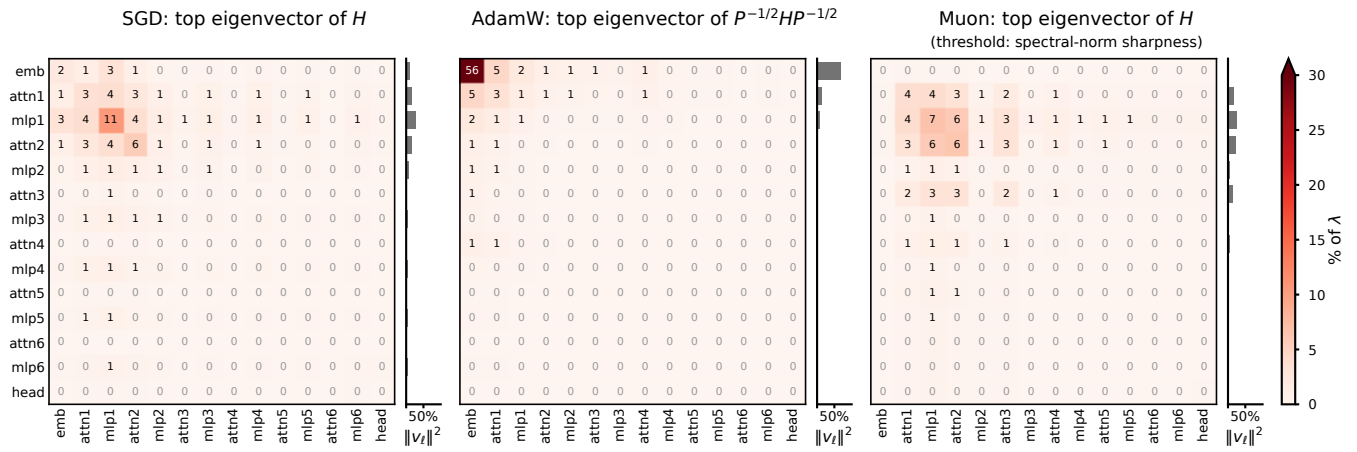


Figure 14: As Fig. 13, with each block split into its attention and its MLP (each with the LayerNorm that feeds it). Same steps; the AdamW panel again uses 2 steps of d0.

Table 3: Curvature-balanced per-layer rates†: final validation loss minus that of the baseline on the same data order (on Shakespeare, best minus best, since the runs overfit), for data orders 0, 1, 2 or as a range over them (for B, over its targets on data order 0); single values are data order 0. In brackets, the speedup: the baseline's steps to its best validation loss over the balanced run's steps to reach it. –: not run.

Optimizer, variant	TinyStories-10M	TinyStories-30M	Shakespeare
SGD B1	$-0.27, -0.21, -0.33$ (2.2–2.7×)	-0.06, blew up, -0.06 (1.5×)	$+0.03, -0.88, -0.17$
SGD B0.9-fe-fresh	$-0.29, -0.16, -0.25$ (1.8–2.0×)	$-0.05, -0.15$ (1.4, 1.7×)	–
SGD A-fe	$-0.19, -0.06, -0.18$ (1.4–2.1×)	$+0.04, -0.16, +0.03$	$+0.00, -0.05, +0.17$
AdamW A-fe	+0.006	+0.003	+0.013 to +0.022
AdamW A; B	+0.015 to +0.018; +0.06 to +0.16	–	–
Muon A	−0.004	cancelled	+0.003 to +0.012
Muon B	+0.02 to +0.04	–	–
v2: AdamW, Muon	+0.094, +0.029	–	–
v2, own block: AdamW, Muon	+0.061, +0.020	–	–

its top eigenvector v, $\omega_a = \|v_a\|^2$, is the row sum of layer a in a heatmap like Fig. 13 (each off-diagonal cell split in half between its two layers) and the elasticity $\partial \log \lambda_{\max}(\hat{H}) / \partial \log m_a$; equal shares are the stationarity condition for maximizing $\prod_a m_a$ at fixed $\lambda_{\max}(\hat{H})$. Variant A only redistributes the rate, with the geometric mean of the m_a fixed at 1 and $\kappa = 1$; variant B also scales κ until $\lambda_{\max}(\hat{H})$ sits at a target fraction of the threshold (B1, B0.9). Every 20 steps a warm-started block Rayleigh–Ritz iteration updates v on the Hessian batch ("-fresh": on 32 fresh sequences), and the multipliers move towards equal shares; they stay at 1 during warmup and within $[0.1, 10]$ after. "-fe" keeps the embeddings at the schedule's rate, since the 32 sequences contain only part of the vocabulary. Muon is balanced in the same way on its spectral-norm sharpness. A second version (v2, for AdamW and Muon) equalizes across layers the curvature along the step each layer takes, $R_a = s_a^\top (Hs)_a / (-2 g_a^\top s_a)$ on 256 fresh sequences (Section 5.5).

Results. Balancing helps only SGD (Table 3, Fig. 15, top). SGD B1 beats the baseline on every data order of TinyStories-10M, 2.2–2.7 times faster, and on two of three orders of TinyStories-30M and Shakespeare. At a good rate, though, redistribution adds nothing: plain SGD at 0.12, four times the baseline rate, reaches 1.552 and 1.554 on data orders 0 and 2 (and blows up on order 1), and A-fe at 0.12 reaches 1.567 and 1.569, so B's gain is mostly a rate tuned on the fly that stays below an instability. The SGD baseline rate is being re-tuned, so the gains over it may shrink. For AdamW and Muon, A stays within about 0.02 of the baseline everywhere and B is worse. B can also fail: several B runs blew up at an instability the 32 fixed sequences did not show, and low targets collapse κ to its floor (Section E). At equal training loss every balanced run is about as close to the baseline on the same data order as another data order is, or closer (Fig. 49): the per-layer rates change the speed, not the function.

Where the rate goes. After balancing the shares are even: over steps 5000–10,000, 12–15% per block and the head and 3% in the embeddings for SGD B1, against 0–44% for the baseline (Fig. 13), and 9–24% per block and the head for Muon A (Fig. 47). The final multipliers rise with depth (Fig. 15, bottom): block 1 ends at 0.1–0.25 times the

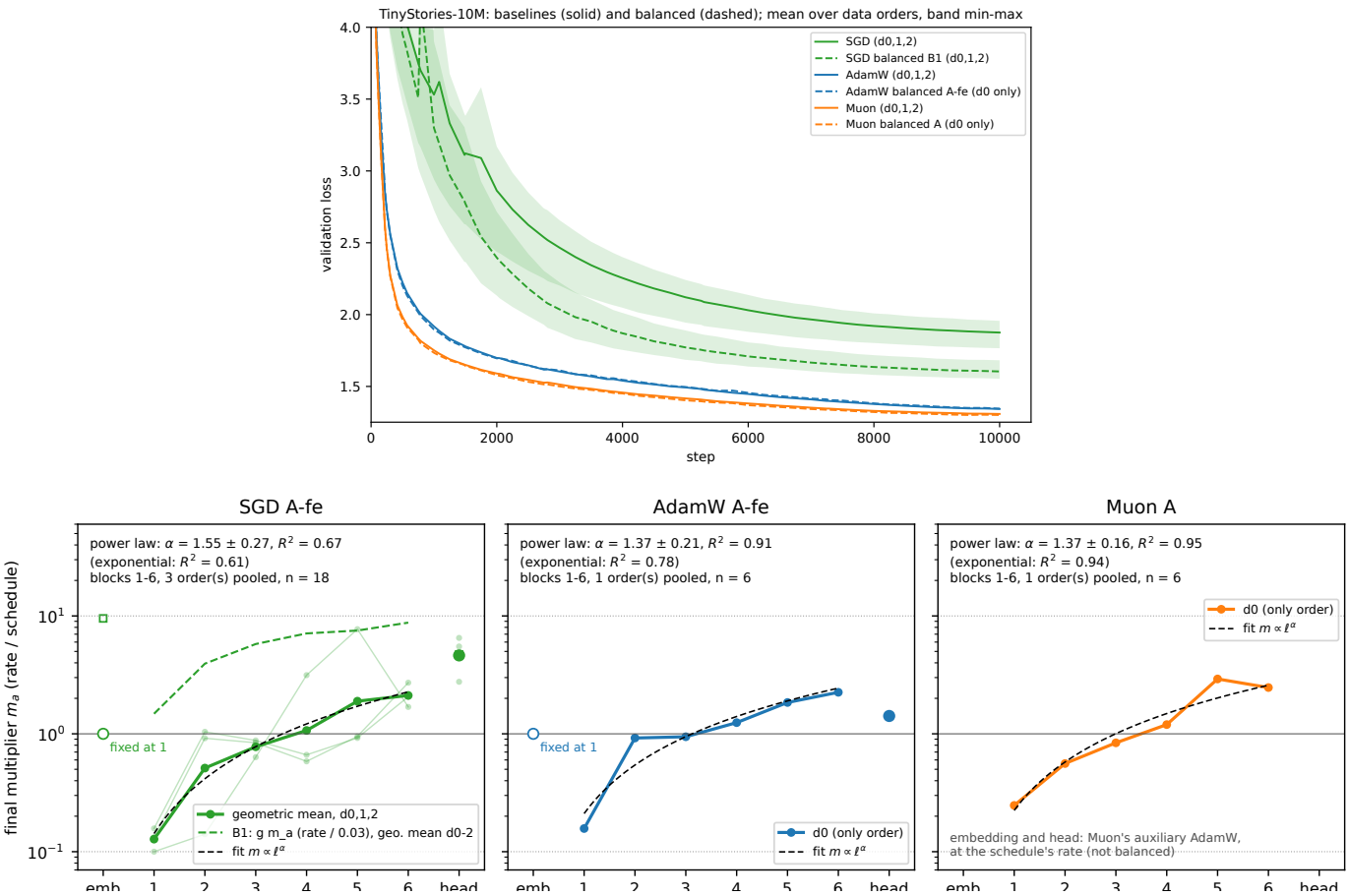


Figure 15: Curvature-balanced rates† on TinyStories-10M. Top: validation loss of the baselines (solid) and of SGD B1, AdamW A-fe and Muon A (dashed), mean over data orders, band from minimum to maximum. Bottom: final multiplier m_a of each block in the runs that only redistribute the rate (A, A-fe), with a power-law fit $m_a \propto \ell^\alpha$ over blocks 1–6; the embeddings are fixed at 1, and Muon's embeddings and head are trained by its auxiliary AdamW at the schedule's rate. Dotted: the caps.

schedule's rate and blocks 5–6 at about 2–3 times, with power laws $\alpha = 1.55 \pm 0.27$ (SGD, $R^2 = 0.67$; its single data orders scatter a lot), 1.37 ± 0.21 (AdamW, $R^2 = 0.91$) and 1.37 ± 0.16 (Muon, $R^2 = 0.95$). Yet these final multipliers, fixed from the start without balancing, are worse than the baseline: AdamW reaches 1.370–1.384 against 1.339–1.346, Muon 1.353–1.360 against 1.305–1.314, and SGD diverges.

Each layer at its own edge. v2 shows why. R_a settles at 0.8–1.2 in every layer from step 100 on, whatever the layer's rate, even with rates 100 times apart (Fig. 50). With the layer's own block in the numerator, $s_a^\top H_{aa} s_a$ as in block Jacobi, it settles at 0.25–0.3, the curvature between layers being about 70% of the numerator. Each layer's curvature along its step adapts to its own rate, so no quantity of this kind can choose the allocation, and the rule random-walks the rates into the caps. Curvature-based per-layer rates thus help only SGD, as a self-tuning rate; for AdamW and Muon, stability cannot allocate the rate across layers.

5.9 Break-even replication

CIFAR-10. The effects reported by Jastrzebski et al. (2020) replicate (Fig. 16). The larger the learning rate, the lower the sharpness in the early phase: λ_H^1 levels off at about 1000, 200 and 20 for $\eta = 0.001$, 0.01 and 0.1, that is at or below $2/\eta$, and the maximum of λ_K^1 decreases with the learning rate in the same way. The larger rates also give a better-conditioned gradient covariance (higher λ_K^*/λ_K^1) and a larger $\|g\|/\|g_5\|$. The relation between λ_H^1 and λ_K^1 and ΔL is in Fig. 37. In prediction space the three runs share a path from the initialization; $\eta = 0.1$ leaves it first, then $\eta = 0.01$. At equal steps the distance between $\eta = 0.01$ and 0.1 is far above the data-order floor (bottom right, left panel). At equal training loss, however, it is 1.0–1.75 times the floor at every loss below 1.8 (median 1.24); it exceeds 1.5 times the floor at three consecutive levels only once (loss 0.50–0.41) and there is no persistent split. For $\eta = 0.001$ against 0.01 the ratio is 19 near the initialization, where the floor is near zero, and 0.57–2.08 below loss 1.5 (median

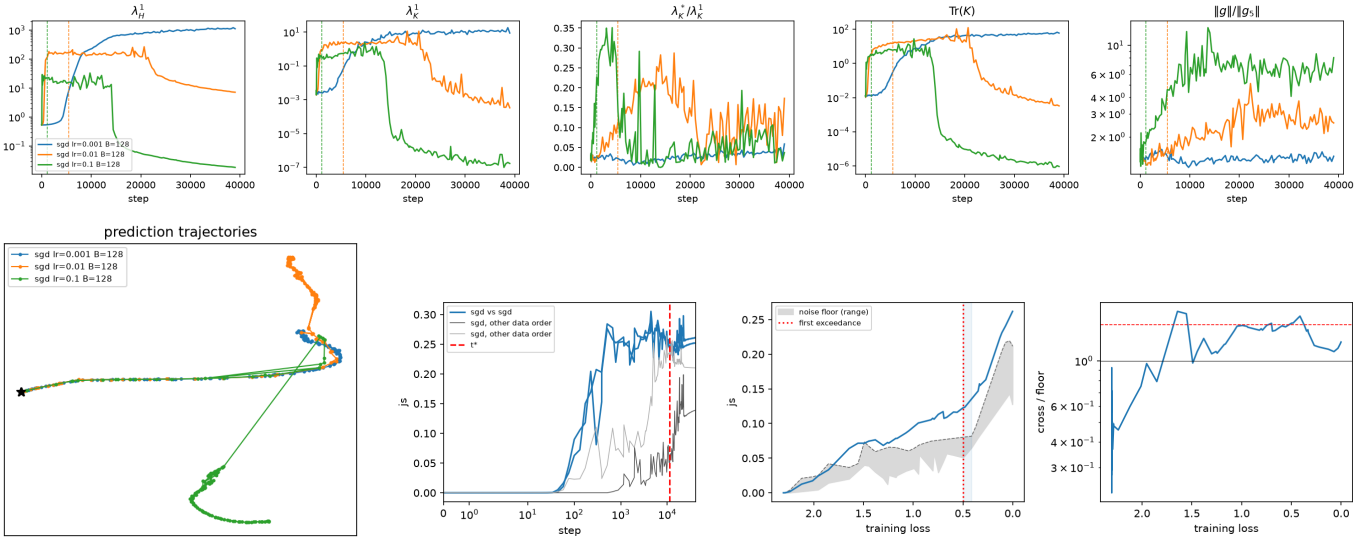


Figure 16: The break-even analysis of Jastrzebski et al. (2020) on CIFAR-10 with plain SGD at $\eta \in \{0.001, 0.01, 0.1\}$, batch 128. Top: λ_H^1, λ_K^1, λ_K^*/λ_K^1, $\operatorname{Tr} \hat{K}$ and $\|g\|/\|g_5\|$ against step ($d = 0$). Bottom left: prediction trajectories (PCA then UMAP), starting at the star. Bottom right: the divergence figure of Fig. 2 for $\eta = 0.01$ against $\eta = 0.1$, with two data orders. Shakespeare in Fig. 37.

1.04), with four short episodes and no persistent split. As for the optimizers, most of the separation in time is one run being faster.

Language models.† On Shakespeare, plain SGD at 0.1 and 0.3 without warmup blows up in the first ten or so steps, where the distance between any two runs, floor pairs included, approaches 1 bit; the runs recover but reach only training loss 3.09 (the lowest common loss). At equal loss the distance between the two rates is at most 1.17 times the floor and no split is found (Table 2). λ_K^*/λ_K^1 rises over training at both rates (Fig. 37), and λ_H^1 is lower at 0.3 apart from spikes. At $\eta = 0.03$ the runs do not blow up; against 0.1 the ratio exceeds the floor in one episode (loss 3.74–3.28, at most 5.2 times the floor, in 68% of resamples) and is 0.90 at the lowest common loss, 3.13. On TinyStories-10M the three rates reproduce the sharpness and noise effects: λ_H^1 levels off at about 125, 35 and 20 for $\eta = 0.03$, 0.1 and 0.3, and λ_K^1 and $\operatorname{Tr} \hat{K}$ decrease with the learning rate; but λ_K^*/λ_K^1 is, if anything, higher at the smallest rate, the opposite of CIFAR-10. In prediction space 0.03 against 0.1 shows the pattern of the optimizer pairs: up to 151 times the floor at loss 7.4, one episode down to loss 5.14, and 0.97 times the floor (median) below loss 4, down to the lowest common loss 2.92 (Fig. 38). The rate 0.3 recovers slowly from its initial blow-up, so 0.1 against 0.3 is compared only down to loss 4.80 (one episode, 7.78–6.24, ratio 1.07 at the end).

Three optimizers at a constant rate.† The main runs decay the rate after warmup, unlike those of Jastrzebski et al. (2020). At a constant rate (Section 4.9, Fig. 17) SGD reaches its threshold between steps 650 and 200, earlier at larger rates, but its paths at different rates separate before that, around steps 50–100. AdamW is already at its threshold when warmup ends, so its break-even step is 200 at every rate, and its paths stay together. Muon's $\lambda_{\max}(H)$ peaks within warmup and only falls after, so the rule returns steps 200–350, which mark nothing in particular; its paths fan out late. Reaching the threshold does not mark where the paths of different rates split.

Batch size.† On CIFAR-10 at $\eta = 0.05$, batch 30 against 300 also replicates the sharpness effect (the smaller batch levels off at a lower λ_H^1, about 12 against 50), and at equal loss the two batch sizes are 0.66–2.14 times the floor below loss 2 (median 1.22; at most 3.1 near the initialization), with no episode and no persistent split (Fig. 38).

5.10 Function metrics

The raw norms mostly measure how far each optimizer moves the weights (Table 4): Muon ends 5.0–6.6 initial norms away from the initialization, AdamW 2.7–2.8 and SGD 0.4–0.6, and $\|\theta\|_2$ and the largest spectral norm of the hidden matrices follow the same order. So SGD having the smallest ℓ_2 norm, and Muon the largest spectral norm, says little about implicit bias. The scale-free versions† match the predicted geometries in part. The ratio $\|\theta\|_\infty/\|\theta\|_2$, small when the entries are even, is 3–5 times smaller for AdamW than for SGD, as Adam's ℓ_∞ bias predicts (Zhang et al., 2024a); and the mean stable rank of the hidden matrices, large when their singular values are even, is highest for Muon (75 and

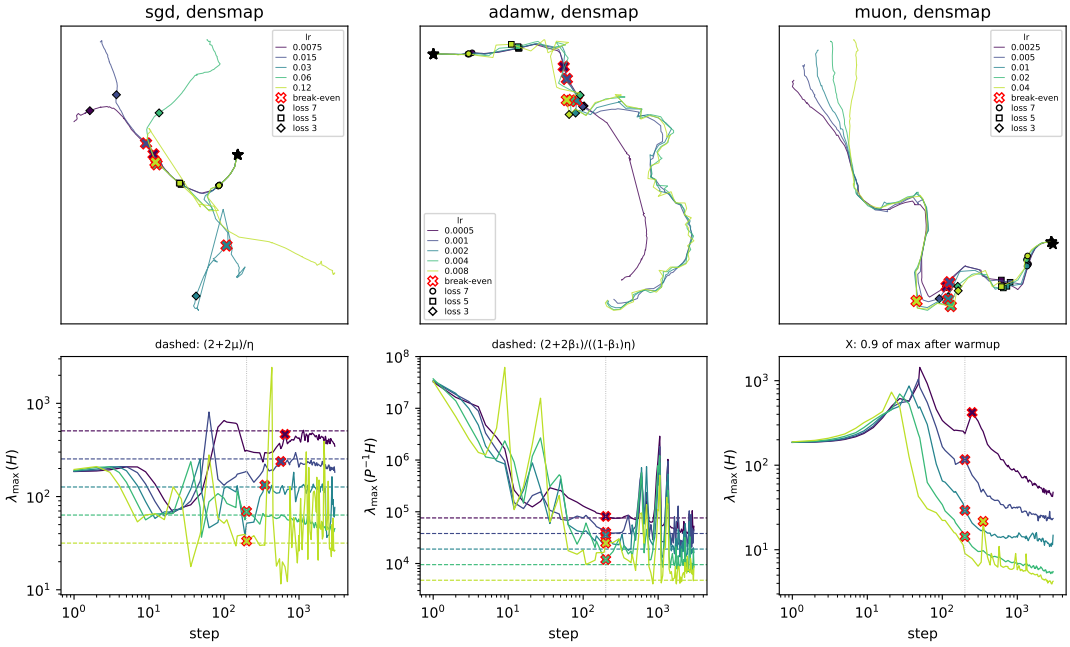


Figure 17: Break-even points of the three optimizers† on TinyStories-10M at a constant rate after 200 warmup steps, five rates each (data order 0, shared initialization). Top: densMAP of the $\sqrt{\text{JS}}$ distance between the logged steps of each optimizer's runs, with markers at training loss 7, 5 and 3 and the break-even step (red ×). Bottom: sharpness against step, with the thresholds (dashed: $(2+2\mu)/\eta$ for SGD, $(2+2\beta_1)/((1-\beta_1)\eta)$ for AdamW) and the end of warmup (dotted); Muon's mark is at 0.9 of its maximum after warmup.

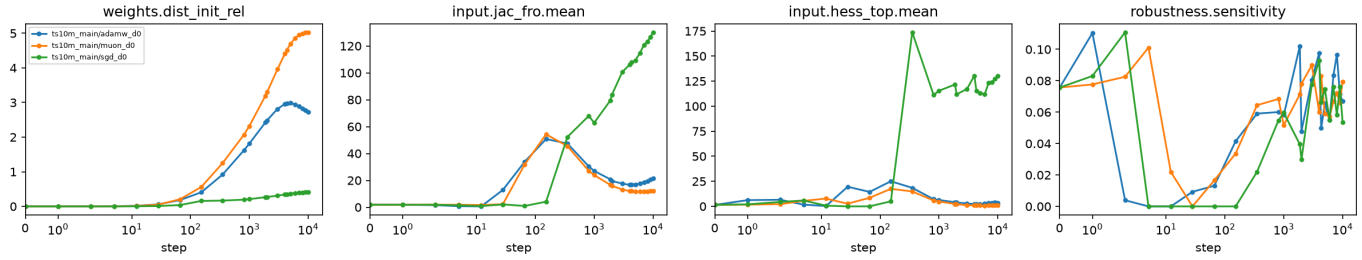


Figure 18: Function metrics over training for the $d = 0$ runs on TinyStories-10M: distance of the weights from the initialization relative to the initial norm, Frobenius norm of the input Jacobian of the softmax output, top eigenvalue of the input Hessian of the loss (inputs are the token embeddings), and one-token substitution sensitivity (Section B.1).

90 against 41–56), as its spectral bias predicts (Fan et al., 2025). Muon also has the smallest $\|\theta\|_\infty/\|\theta\|_2$. The weights therefore carry the geometry of their optimizer while the functions of AdamW and Muon are within the data-order floor of each other (Section 5.3), in line with the barriers of Section 5.6.

In input space SGD's function looks much sharper: its input Jacobian is 5–11 times and its top input-Hessian eigenvalue 15–110 times those of AdamW and Muon. The inputs are the token embeddings, which SGD moves least (Frobenius norm of the embedding matrix 27 for SGD against 97 for AdamW and 175 for Muon on TinyStories-10M), and the pre-LayerNorm transformer normalizes the residual stream, so derivatives with respect to the embeddings shrink as the embeddings grow; much of the difference is therefore one of scale. One-token substitution sensitivity, which does not depend on that scale, is 0.06–0.08 for all three; and the input Jacobian times the embedding norm is about the same for AdamW and Muon (2.1 and 2.2×10^3) and 1.7 times higher for SGD. Over training (Fig. 18) the input Jacobian of AdamW and Muon peaks at about step 200, the end of warmup, and then falls, while SGD's keeps rising.

On Shakespeare (Table 11) AdamW's learning rate of 3×10^{-4} moves the weights by only 0.25 initial norms, as little as SGD (0.36), and the two have the same $\|\theta\|_\infty/\|\theta\|_2$ (12.4 and 12.3×10^{-3}): the ℓ_∞ geometry only shows once the weights leave the initialization. Muon again has the largest stable rank (99 against 48 and 63), and, as it memorizes, twice to four times the substitution sensitivity of the others (0.19 against 0.10 and 0.05).

Table 4: Final function metrics on TinyStories, mean over the five data orders (standard deviations are small except for SGD's spectral norms and stable rank). "Hidden" are the matrices in the transformer blocks, the ones Muon updates; the stable rank is $\|W\|_F^2/\|W\|_2^2$.

	TinyStories-10M			TinyStories-30M		
	SGD	AdamW	Muon	SGD	AdamW	Muon
$\|\theta\|_2$	105	265	485	156	369	857
$\|\theta\|_\infty$	4.37	2.09	1.85	3.33	2.58	2.28
$\max \|W\|_2$ (hidden)	3.5	12.4	14.1	7.6	15.0	22.9
$\|\theta\|_\infty/\|\theta\|_2$ ($\times 10^3$)	41.6	7.9	3.8	21.3	7.0	2.7
stable rank (hidden, mean)	56	41	75	56	47	90
$\|\theta-\theta_0\|/\|\theta_0\|$	0.40	2.72	5.04	0.61	2.83	6.56
input Jacobian (Frobenius)	131.7	21.6	12.5	72.7	15.2	7.8
input Hessian $\lambda_{\max}$	123.6	3.5	1.1	37.2	2.4	0.5
substitution sensitivity	0.070	0.082	0.067	0.076	0.060	0.075
top-1 accuracy	0.576	0.651	0.657	0.637	0.677	0.680

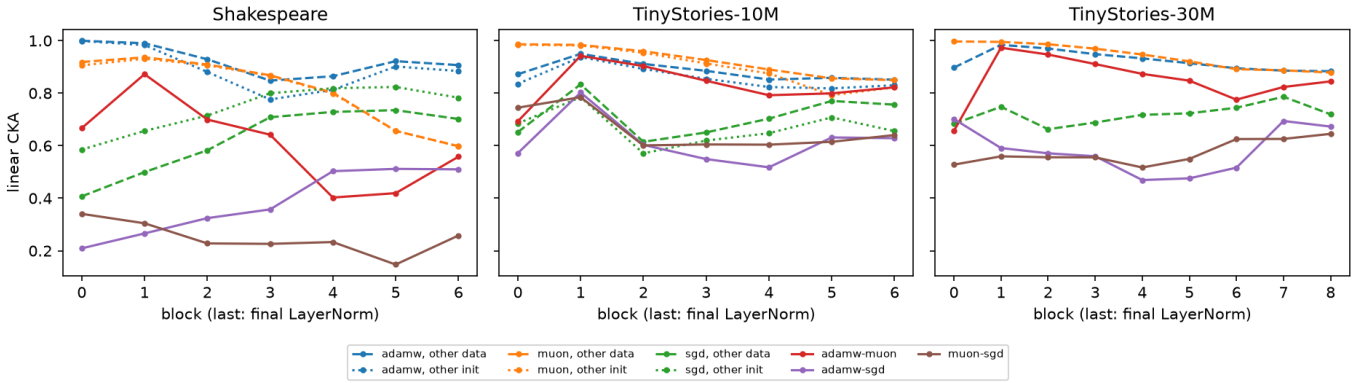


Figure 19: Linear CKA† between the final hidden states of two pure runs after each transformer block (the last point is the final LayerNorm), averaged over pairs. Solid: other optimizer, same initialization and data order; dashed: same optimizer, other data order; dotted: same optimizer, other initialization (not run on TinyStories-30M).

Representations†. Between weights and outputs sit the hidden states (Fig. 19). On TinyStories, AdamW and Muon trained on the same data have less similar representations than two data orders of either optimizer, but not by much outside the first block: linear CKA 0.69 and 0.66 after the first block against 0.87–1.0, and 0.78–0.97 against 0.85–0.98 after the others, 0.82–0.85 against 0.85–0.88 after the final LayerNorm. SGD's representations are not even reproducible across its own data orders (0.57–0.83), and pairs with SGD are lower still. On Shakespeare, where Muon memorizes, AdamW and Muon differ clearly (0.40–0.87). Another initialization moves the representations as much as another data order. So the optimizer's mark fades from the weights (their geometry, and barriers in the middle blocks) through the representations (a small CKA gap) to the outputs (none).

Off the data distribution†. The optimizers could agree on the data and differ elsewhere, where the data do not constrain the function and an implicit bias has room to act. We compared the final functions of the pure runs on the 32 validation sequences, on the same sequences with their tokens shuffled within each sequence, and on uniformly random tokens (Fig. 20). Off the distribution every pair of runs differs much more (on TinyStories, 0.20–0.25 bits on shuffled and 0.30–0.40 on random tokens, against 0.05–0.06 on validation), but the optimizer adds no more than the data order does: AdamW and Muon are 0.90–1.07 floors apart on every input type in all three settings. SGD, less trained at equal steps, is 1.2–2.7 floors from the adaptive optimizers on validation but only 1.05–1.8 off the distribution, so what separates it is how well it fits the data, not how it extrapolates.

Metrics under swaps. Figure 21 places each fork's final metrics (the scale-free ones added after the plan†) between those of pure A (0) and pure B (1). The weight metrics follow whichever optimizer moved the weights more, not the one that decides the function. Forks into SGD keep the old optimizer's $\|\theta\|_\infty/\|\theta\|_2$ and input Jacobian after switching at step 400 or later (shift ≤ 0.37), although at steps 400–1000 their function is within 1.0–1.4 floors of pure SGD (Fig. 9); Muon to SGD keeps Muon's stable rank at every fork step. Forks from SGD take the new optimizer's metrics, as they take its function. Between AdamW and Muon, whose functions cannot be told apart, the late forks are mixtures:

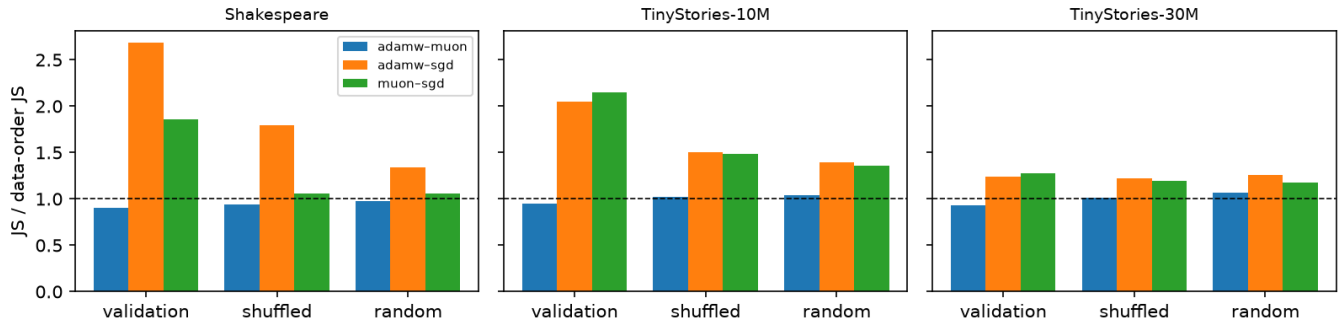


Figure 20: Distance D_{JS} off the data distribution† between the final functions of two optimizers (same initialization and data order), averaged over the five data orders, over the larger of the two optimizers' mean distance between data orders, on validation sequences, on the same sequences with shuffled tokens, and on uniformly random tokens.

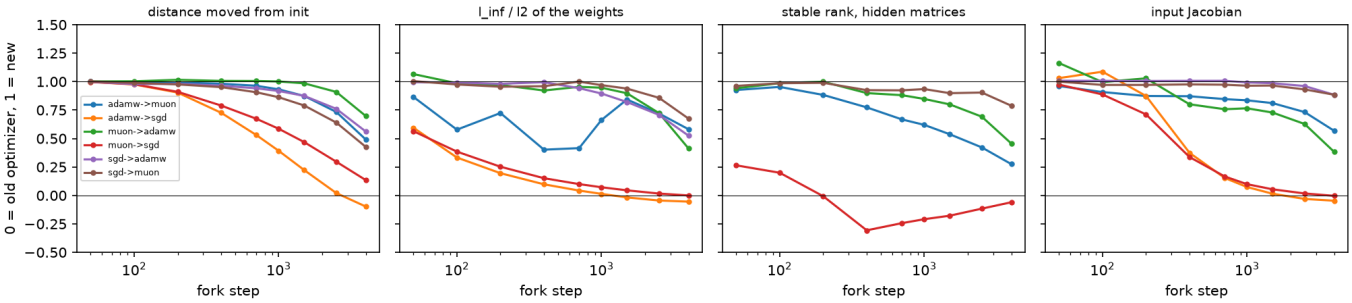


Figure 21: Final metrics of the forks on TinyStories-10M as a fraction of the gap between pure A (0) and pure B (1), against the fork step. Pairs whose pure gap is under three data-order standard deviations are left out (AdamW–SGD in stable rank).

switching at step 4000 goes only 0.28 (AdamW to Muon) and 0.45 (Muon to AdamW) of the way to the new optimizer's stable rank, and 0.4–0.6 of the way in $\|\theta\|_\infty/\|\theta\|_2$. The geometry of the endpoint records both optimizers and their order, while the function records only the faster of the two. On Shakespeare the same holds where the pure gap exceeds the noise, with Muon, which moves the weights furthest, setting the geometry (Fig. 44). On TinyStories-30M the picture is the same, from a later fork step for the swaps into SGD: switching at step 800 or later keeps the old optimizer's $\|\theta\|_\infty/\|\theta\|_2$ (shift ≤ 0.20), and at step 1400 or later its input Jacobian (≤ 0.19), while up to step 3000 the function of these forks is within 1.02–1.21 floors of pure SGD; AdamW and Muon swapped at step 8000 go 0.29 and 0.65 of the way in stable rank and 0.45 in $\|\theta\|_\infty/\|\theta\|_2$.

5.11 Further controls†

Second initialization. All other runs share one initialization, so the floor varies only the data order. We repeated the main runs of Shakespeare and TinyStories-10M from a second initialization (seed 1) with data orders 0 and 1, without curvature measurements. On Shakespeare (Fig. 22, top) the distance between the two initializations starts at 0.05 bits, falls to about 0.01 by step 10–100 and from about step 100 on equals the data-order floor for every optimizer; at the final step it is 0.068 against 0.063 (AdamW), 0.288 against 0.292 (Muon) and 0.053 against 0.067 (SGD). The split analysis at the second initialization, with two data orders, reproduces that at the first: AdamW–Muon has episodes at loss 4.18–2.86 and 2.41–2.02 (first initialization: 4.23–2.77 and 2.30–2.12) and no persistent split; AdamW–SGD 4.18–3.04 and 2.77–2.54 (first: 4.23–3.10), no persistent split; and Muon–SGD again has a persistent split, from loss 2.80 (first: 2.82). On TinyStories-10M (Fig. 40) the two distances also agree from about step 100 on, and at the final step the initialization, the data order and the change between AdamW and Muon all give the same distance: 0.054, 0.062 and 0.058 bits from AdamW, 0.055, 0.059 and 0.058 from Muon. The episodes are again close to those of the first initialization, AdamW–Muon 8.31–4.27 (first: 8.26–4.85), AdamW–SGD 8.31–6.21 (8.26–6.32) and Muon–SGD 8.43–6.78 (8.38–5.81), with no persistent split. Changing the initialization thus moves the function by as much as changing the data order, and the structure of the episodes does not depend on the initialization.

Rare and frequent tokens. Kunstner et al. (2024) attribute Adam's advantage over SGD on language models to the rare classes of a heavy-tailed vocabulary. We grouped the probe positions by the training frequency of their target

token (Fig. 22, middle, for TinyStories-10M). SGD's loss gap is concentrated on the rare tokens: on TinyStories-10M it is 1.15 nats above AdamW on the rarest fifth and 0.12 on the most frequent (TinyStories-30M: 0.48 and 0.07). At the final step its distance from AdamW and Muon is also largest on the rare tokens (1.8–2.1 times the floor, against 1.4 on the most frequent), but at the lowest training loss both runs reach it is 0.93–1.03 times the floor in every bin, for every pair on TinyStories. The gap on rare tokens is a difference in speed, not in the function learned at equal loss.

A positive control. Since the detector finds no persistent split between AdamW and Muon, we checked that it finds one where theory predicts it. Logistic regression on linearly separable data ($d = 50$, $n = 200$, labels from a sparse teacher on 3 coordinates, 2000 test points), from zero initialization with batch 20, trained by SGD with momentum ($\eta = 0.1$) and by Adam ($\eta = 0.01$) for 10^5 steps with five data orders: in the limit gradient descent converges in direction to the ℓ_2 max-margin classifier (Soudry et al., 2018) and Adam to the ℓ_∞ one (Zhang et al., 2024a). The same loss-matched JS rule, on the test predictions, finds the ratio below 1.5 down to training loss 10^{-2} (at most 1.47) and a persistent split from loss $7.5 \cdot 10^{-4}$, with the ratio up to 3.3 (Fig. 22, bottom). The detector can see a split, and in the linear case the split appears near interpolation, far below the losses our language models reach. Neither optimizer is at its limit after 10^5 steps: Adam's normalized margins exceed SGD's in both norms (0.221 against 0.202 in ℓ_2, 0.377 against 0.341 in ℓ_∞).

Training into memorization. If the language models split only near interpolation, as the linear model does, our runs may simply stop too early. On Shakespeare we therefore trained AdamW and Muon for 8000 steps instead of 1500 (same learning rates, cosine schedule over the longer run, five data orders each), so that both memorize: AdamW's training loss falls to 0.09 and its validation loss rises to 4.6. The loss-matched ratio (Fig. 41; Fig. 4, top left, red) has a single early episode (loss 4.23–2.64) and then falls steadily to 1.0 at the lowest common loss, 0.09; with top-1 disagreement the picture is the same (episode 4.23–2.85, no persistent split). At the final step the two optimizers on the same data order are as far apart as two data orders of either (0.359 bits against floors of 0.355 and 0.324). Memorizing the training set, AdamW and Muon reach functions that differ on held-out text by a lot, but by no more than two data orders do.

Other learning rates. The comparisons above use one tuned learning rate per optimizer. The sweep runs (data order 0, same initialization) give the comparison at every other rate for free: each against the tuned run of every optimizer on the same data, at equal training loss, over the data-order floor of the tuned runs, summarized by the median below the bigram loss (Fig. 23). On TinyStories every rate that trains stably, a 27-fold range for AdamW, 8-fold for Muon and 3- to 10-fold for SGD, is within 0.80–1.11 floors of all three tuned optimizers, with one exception: SGD at 0.1 is 1.35 floors from tuned SGD at 0.03 on TinyStories-10M, further than from tuned AdamW or Muon. Only rates at the edge of stability (Muon at 0.08, SGD at 0.3 on TinyStories-10M) are 1.5–2.3 floors away, and higher ones diverge. On Shakespeare, where SGD does not get below the bigram loss at most rates, AdamW at 3–13 times its tuned rate is 1.5–1.6 floors from tuned AdamW but 1.03–1.06 from tuned Muon, since the floor of a cross comparison is the larger of the two optimizers' floors and Muon's is large (it memorizes). The floor of the tuned runs may not be the floor at other rates, so this is a check, not a second analysis; but it shows that the absence of a split on TinyStories is not a property of the chosen rates.

Rates around the tuned one. On TinyStories-10M we also trained each optimizer at $\frac{1}{4}$, $\frac{1}{2}$, 2 and 4 times its tuned rate (SGD 0.0075–0.12, AdamW $5 \cdot 10^{-4}$–$8 \cdot 10^{-3}$, Muon 0.0025–0.04; data order 0, the schedule and length of the main runs) and compared each run with the run at the tuned rate. At equal loss every rate stays within 0.8–1.1 times the data-order floor of the tuned runs, from loss about 3–4 to the end, for all three optimizers. In a densMAP of the trajectories, one per optimizer (Fig. 24), the SGD runs at different rates take separate paths from about loss 3, while the AdamW runs share one path down to loss 3, and so do the Muon runs. The embedding keeps neighbourhoods, not distances: in the actual distances at equal loss the SGD rates do not split either.

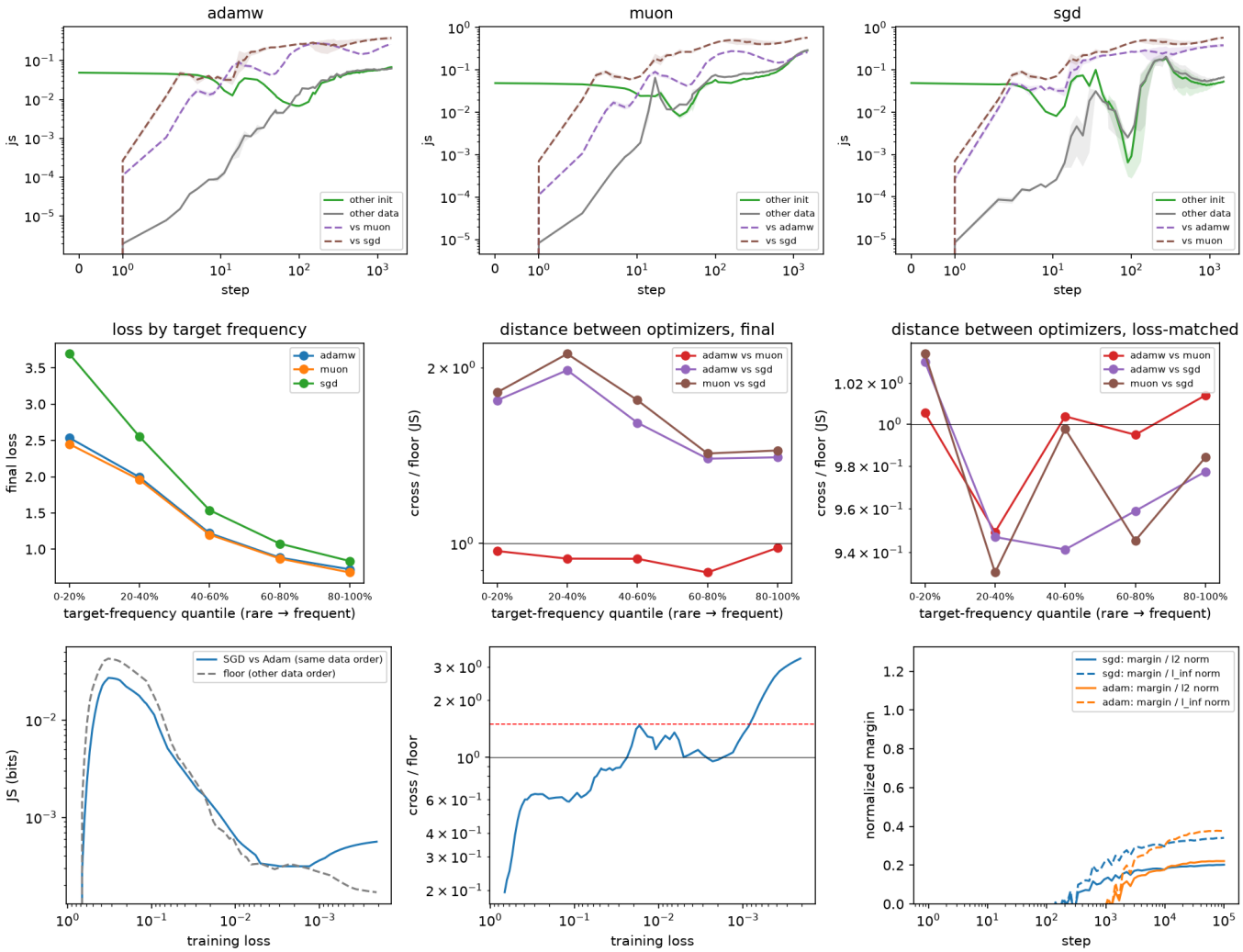


Figure 22: Controls added after the plan. Top: second initialization on Shakespeare; for each optimizer, at equal steps, the distance to the same optimizer from the other initialization (same data order), to the same optimizer with the other data order (same initialization), and to the other optimizers (same initialization and data order); median and range over the pairs. Middle: TinyStories-10M by target-token frequency (quintiles of the training frequency of the target, rare to frequent): final validation loss of each optimizer (mean over data orders), and the distance between optimizers over the floor at the final step and at the lowest training loss both runs reach. Bottom: the linear positive control; loss-matched JS distance between SGD and Adam on the test points against the data-order floor (left), their ratio with the margin $1 + m_\times = 1.5$ (middle), and the ℓ_2- and ℓ_∞-normalized margins over training (right).

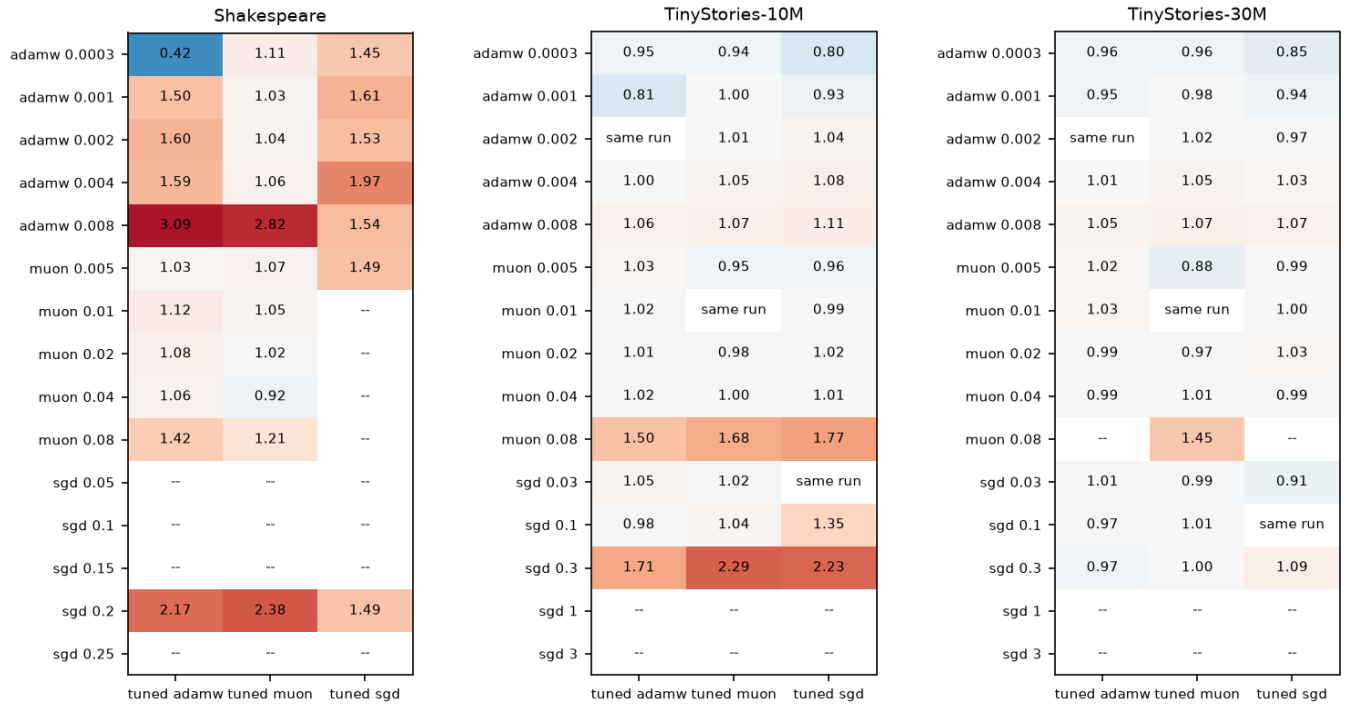


Figure 23: Each learning-rate sweep run (rows) against the tuned run of each optimizer (columns) on the same data order: median loss-matched JS over the data-order floor of the tuned runs, below the bigram loss (2.47 on Shakespeare, 3.8 on TinyStories). Blue below 1, red above; "–": diverged or never below that loss.

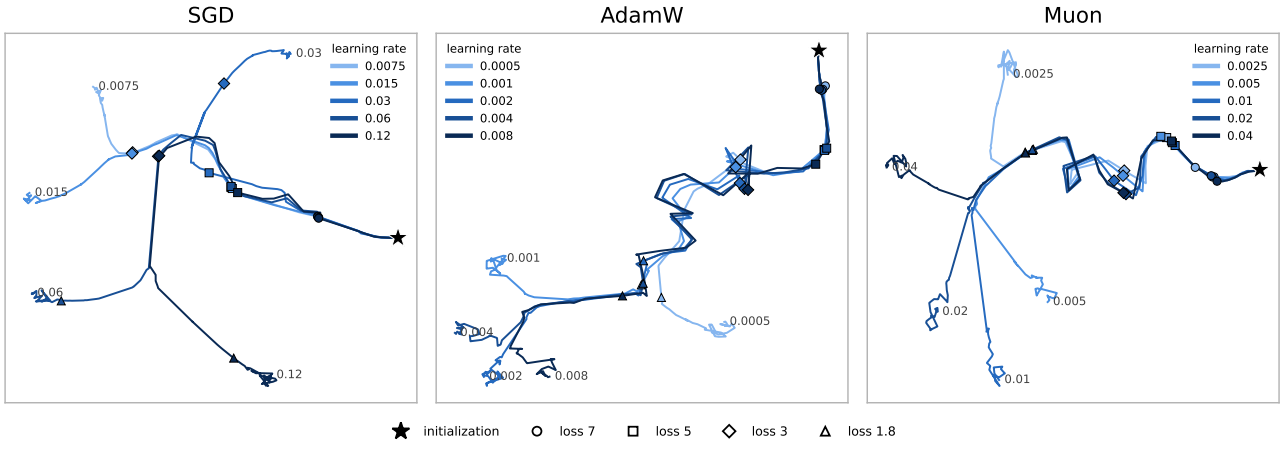


Figure 24: Runs at $\frac{1}{4}$ to 4 times each optimizer's rate on TinyStories-10M (data order 0, shared initialization; the 1× run is the main run): densMAP of the $\sqrt{\text{JS}}$ distance between their logged steps, one embedding per optimizer. Markers: the initialization and the steps at which each run reaches training loss 7, 5, 3 and 1.8. densMAP keeps neighbourhoods, not distances.

6 A model of the results†

The results share one pattern: the optimizers differ in how they get somewhere (speed, route, weights, curvature), but after warmup not in which function they compute at a given training loss. Here we give a simple model of this pattern and of the role of the break-even point, with proofs or derivations, a toy model in which every quantity is exact (Fig. 25), and checks against the measurements.

Setup. Contexts $c \sim \mathcal{D}$ have next tokens $y \sim q(\cdot \mid c)$, and a predictor π has loss $L(\pi) = \mathbb{E}_c\, \mathbb{E}_{y\sim q(\cdot|c)}[-\log \pi(y \mid c)]$. For a statistic s of the context (nothing, the last token, the last k tokens), let $\mathcal{F}_s$ be the predictors that depend on c only through $s(c)$: the unigram, bigram, … predictors. $\mathcal{F}_s$ is convex; its best element is the n-gram model of the data, $q_s(\cdot \mid s) = \mathbb{E}[q(\cdot \mid c) \mid s(c) = s]$, with loss the n-gram entropy H_s, and by the tower property every $\pi \in \mathcal{F}_s$ satisfies

$$L(\pi) = H_s + \mathbb{E}_s\, \mathrm{KL}\big(q_s(\cdot \mid s) \,\|\, \pi(\cdot \mid s)\big). \tag{10}$$

Proposition 1 (funnel). *If $\pi_A, \pi_B \in \mathcal{F}_s$ have excess losses $\Delta L_A = L(\pi_A) - H_s$ and ΔL_B (in nats), then, with* JS *in bits,*

$$\mathbb{E}_s \mathrm{JS}(\pi_A, \pi_B) \le \sqrt{2}\big(\sqrt{\Delta L_A} + \sqrt{\Delta L_B}\big), \qquad \text{and} \qquad \mathbb{E}_s \mathrm{JS}(\pi_A, \pi_B) \le \frac{\Delta L_A + \Delta L_B}{2 \ln 2}\,(1 + o(1)) \ \text{ as } \Delta L_A, \Delta L_B \to 0. \tag{11}$$

Proof. $\sqrt{\mathrm{JS}}$ is a metric (Endres and Schindelin, 2003). For one outcome with probabilities p, q the divergence (in nats) is $\frac{p+q}{2}(\ln 2 - h(t))$ with $t = p/(p+q)$ and h the binary entropy, and $\ln 2 - h(t) \le \ln 2\,|2t - 1|$ because the left side is convex and the two agree at $t \in \{0, \frac{1}{2}, 1\}$; summing, JS in bits is at most the total variation TV. With Pinsker's inequality $\mathrm{TV} \le \sqrt{\mathrm{KL}/2}$, $\mathrm{JS}(\pi_A, \pi_B) \le (\sqrt{\mathrm{JS}(\pi_A, q_s)} + \sqrt{\mathrm{JS}(q_s, \pi_B)})^2 \le 2\,\mathrm{TV}(\pi_A, q_s) + 2\,\mathrm{TV}(q_s, \pi_B) \le \sqrt{2}(\sqrt{\mathrm{KL}(q_s \| \pi_A)} + \sqrt{\mathrm{KL}(q_s \| \pi_B)})$; take the mean over s, use Jensen's inequality and (10). For the second bound, $\mathrm{JS}(q, \pi)$ in nats and $\mathrm{KL}(q \| \pi)$ are $\frac{1}{8}$ and $\frac{1}{2}$ of $\sum_y (\pi(y) - q(y))^2 / q(y)$ to leading order, and $(a+b)^2 \le 2a^2 + 2b^2$. □

Within a class, all predictors with a given loss thus lie in a shell around q_s that shrinks with the excess loss, whichever optimizer produced them. Two optimizers can be far apart at equal loss only when the loss is far above the optimum of the class their functions lie in, or when the class has no best element: on separable data the loss tends to zero along directions chosen by each optimizer's implicit bias (Soudry et al., 2018; Zhang et al., 2024a), which is where the linear control splits (Section 5.11). Networks learn the statistics of their data roughly in order of complexity (Belrose et al., 2024), and ours pass close to the bottom of the bigram funnel: on Shakespeare all three optimizers come within 0.036–0.041 bits of q_2 at losses 2.46–2.48, 0.01–0.03 nats above the bigram entropy of the training split ($H_2 = 2.452$), where the AdamW–Muon ratio is 0.91 (Section 5.3). In the toy below, where the model can only represent bigram predictors, (11) holds exactly, and the loss-matched distance between optimizers falls with the excess loss as the second bound allows (Fig. 25, bottom left).

The toy model. Logits $AB\, e_c + b$ for the previous character c, trained on the bigram pairs of Shakespeare (Fig. 25). The function is the 65×65 table of next-character distributions, all losses are exact, and the class of reachable predictors is $\mathcal{F}_{\text{bigram}}$, so its only funnel is the bigram one. The bias plays the part of directions that are trained from the first step, the product AB that of directions that must first leave the saddle at zero. It reproduces the main observations: a loss-matched distance up to 40 times the floor (18 for AdamW–Muon) that falls below 1.5 at loss 2.62–2.69 and ends below 1 at H_2, with no persistent split; the order in which the routes pass the unigram model; forks that end on the optimizer they switch to; and barriers between optimizers well above those between data orders.

Routes before the funnel. Near the initialization the gradient of the loss with respect to the product is $G_0 = -\sum_a p(a)(q(\cdot \mid a) - u)e_a^\top$, with u uniform and p the frequency of the context; its top singular direction is close to the unigram direction $(q_1 - u)p^\top$. With SGD, b learns the unigram at once and, for gradient flow from a small balanced initialization, each singular value of AB grows as $\dot{s}_k \approx 2\sigma_k s_k$, with σ_k the singular values of G_0 (Saxe et al., 2014): exponentially, at rates proportional to σ_k, so the modes are picked up one at a time, strongest first, and the route passes through the unigram model before adding the bigram modes. Muon updates A by $-\eta\,\mathrm{msign}(GB^\top)$ and B by $-\eta\,\mathrm{msign}(A^\top G)$, which move every singular direction in the range of the gradient at the same speed whatever σ_k: all modes of AB grow together, the route heads for the spectrally flattened target and stays away from the unigram model. AdamW normalizes each coordinate, which makes it free of scale like Muon but in the coordinate basis: b and AB move at the same speed, and the route passes near the unigram model, less closely than SGD's. In the toy the closest approach to the unigram model is 0.036 bits for SGD, 0.064 for AdamW and 0.18 for Muon, the order measured on the Shakespeare transformers (0.001, 0.054, 0.111; Section 5.3). At this stage the mean gradient dominates the mini-batch noise. Two data orders of one optimizer differ by a random walk, of squared size $\propto \eta^2 t \sigma^2 / B$ after t steps, while two optimizers drift apart linearly in t, so the ratio of their squared distances grows like t times the squared signal-to-noise ratio of the gradient: the ratio to the floor is largest before the funnel, as in Fig. 4.

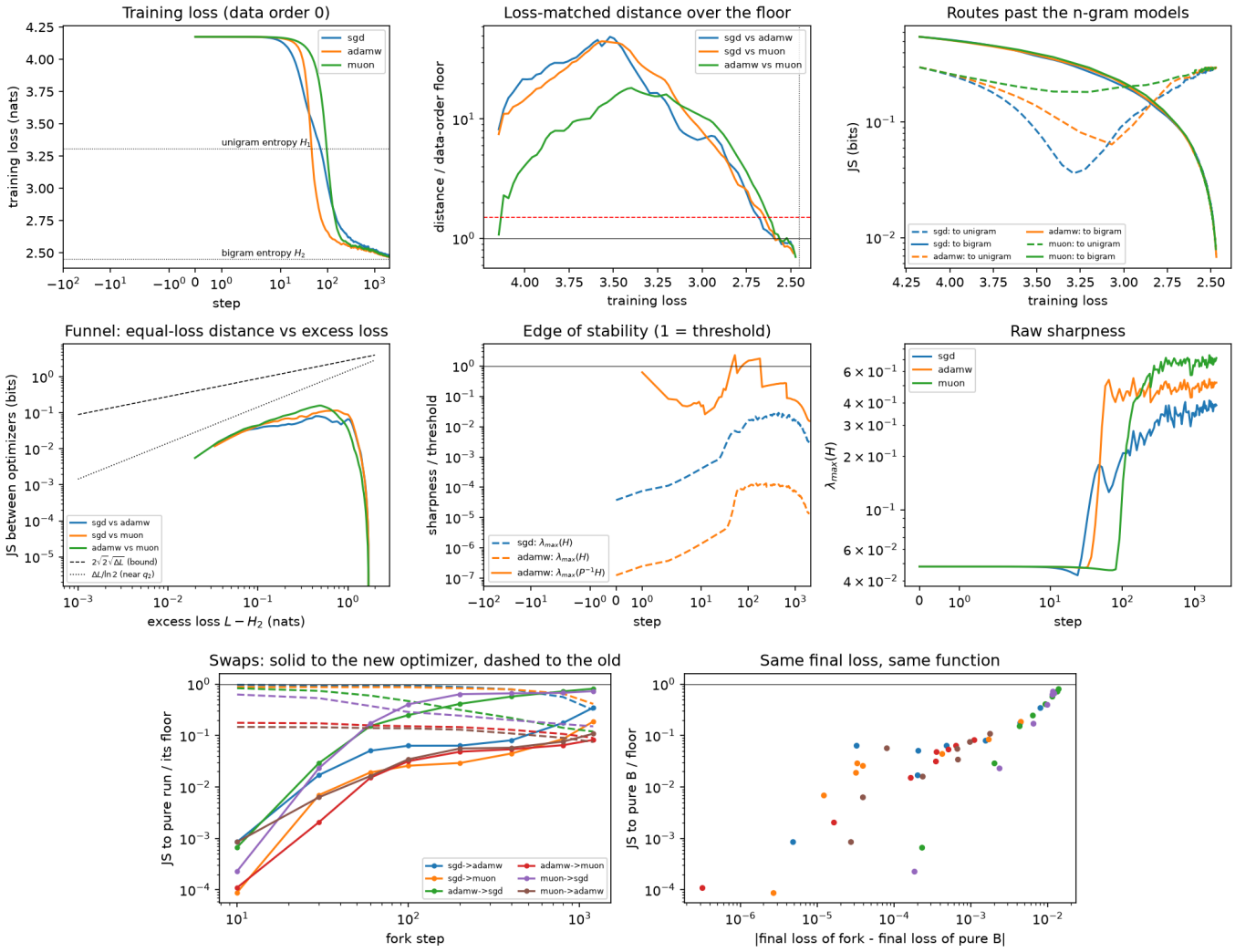


Figure 25: Toy model†: next-character logits $AB\,e_c + b$ for the previous character c on Shakespeare ($V = 65$, inner width 128, initialization $\mathcal{N}(0, 0.02^2)$ for A, B and $b = 0$), batch 128, 2000 steps with the schedule of the language models, learning rates from a sweep (SGD 0.3, AdamW 0.01, Muon 0.03 on A, B with AdamW at 0.01 on b), five data orders. Top: training loss with the unigram and bigram entropies; loss-matched distance over the floor (red: 1.5; dotted: H_2); JS of the $d = 0$ runs from the unigram (dashed) and bigram (solid) models. Bottom: loss-matched JS between optimizers against the excess loss over H_2, with the two bounds of (11); sharpness over its threshold; raw sharpness. Last row: swaps from $d = 0$ at eight fork steps, distance to pure B (solid) and A (dashed) in floors, and distance to pure B against the gap between the final losses.

Proposition 2 (the floor near the funnel). Near q_s write each run's function as $\pi = q_s + \mu_X + \xi_X^d$, with μ_X common to all data orders of optimizer X and ξ_X^d a zero-mean part due to data order d. To second order $\mathrm{JS}(\pi, \pi') = \frac{1}{8\ln 2}\|\pi - \pi'\|_F^2$ in bits, with $\|\delta\|_F^2 = \mathbb{E}_s \sum_y \delta(y \mid s)^2/q_s(y \mid s)$, so

$$F_X = \tfrac{1}{4\ln 2}\mathbb{E}\|\xi_X\|_F^2, \quad C_{\text{other}} = \tfrac{F_A+F_B}{2} + \tfrac{1}{8\ln 2}\|\mu_A - \mu_B\|_F^2, \quad C_{\text{same}} = C_{\text{other}} - \tfrac{1}{4\ln 2}\mathbb{E}\langle \xi_A^d, \xi_B^d\rangle_F, \tag{12}$$

for the floor F_X and the cross distances on other and on the same data orders. The cross distance over the larger floor is therefore at most about one plus the optimizer's own term $\frac{1}{8\ln 2}\|\mu_A - \mu_B\|_F^2$ over the floor, and below one when two optimizers respond alike to the same batches. The measurements at equal steps (Section 5.3) fit this: for AdamW–Muon on TinyStories-30M at step 20,000, $C_{\text{other}} = 0.0556$ bits exceeds the mean of the floors (0.0547, 0.0512) by 0.0027, 5% of the floor, and $C_{\text{same}} = 0.0511$; at step 10,000 the excess is 0.0039 (5%), and on TinyStories-10M at step 10,000 it is 0.0037 (6%) with $C_{\text{same}} = 0.0574$ against 0.0632. In the toy the excess is 6% of the floor. Over training the excess shrinks as the funnel predicts: on TinyStories it is 8–10 times the mean floor at step 30, 0.3–0.4 times at step 120 and about 0.1 by step 1000, and 4–7% from there on (at equal steps; on Shakespeare, where Muon memorizes faster, it stays near 50%). Most of the distance between two optimizers is data-order noise, and part of it is shared when they see the same batches.

Swaps. A fork from A to B is a run of B from another starting point. In the regime of Proposition 2 it differs from pure B on the same data order through its mean part, which the loss bounds, so it lands on B once it reaches B's loss, that is whenever B is fast enough to get there in the remaining steps, and otherwise keeps a loss, and with it a function, set by the parent's progress. In the toy every optimizer reaches H_2, all forks end within 0.81 floors of pure B, and the distance to pure B grows with the gap between the final losses (Fig. 25, last row). In the transformers SGD is the one optimizer too slow to undo its parent's progress, and the forks into it are the ones that stay near their parent (Section 5.6).

Weight space. The function $AB + b$ is unchanged by $(A, B) \mapsto (AR, R^{-1}B)$ for invertible R, and by any change of A and B that leaves their product unchanged: each function is a high-dimensional fiber of weights. Gradient flow conserves $A^\top A - BB^\top$ (Du et al., 2018), so SGD from a small initialization stays near the balanced point of the fiber, which minimizes $\|A\|_F^2 + \|B\|_F^2$ for its product ($\|A\|_F^2 + \|B\|_F^2 \geq 2\|AB\|_*$, with equality for balanced factors). Normalized updates do not conserve it and end elsewhere on the fiber. Two points of one fiber compute the same function, but the segment between them leaves it, since $(\alpha A_1 + (1-\alpha)A_2)(\alpha B_1 + (1-\alpha)B_2) \neq A_1 B_1$ in general, so a loss barrier between optimizers is compatible with equal functions. In the toy the imbalance $\|A^\top A - BB^\top\|_F / \|A^\top A\|_F$ is 0.39 (SGD), 0.91 (AdamW) and 2.1 (Muon), SGD has the smallest ℓ_2 norm (27, against 43 for AdamW and 36 for Muon), and the barriers between optimizers on one data order (0.013–0.024 nats) are 20–50 times those between data orders of SGD or AdamW, at a function distance below the floor. This is the parameter-space implicit bias of the asymptotic theory acting along the fiber rather than on the function, and it accounts for the weights carrying each optimizer's geometry and for the barriers between AdamW and Muon while their functions agree (Sections 5.6 and 5.10).

Break-even points by optimizer. Jastrzebski et al. (2020) model the early phase by one quadratic direction whose curvature grows until the step becomes unstable along it. Where this happens depends on how the update scales with the gradient. (i) SGD with momentum is linear in the gradient; along an eigenvector of H with eigenvalue λ it is stable if and only if $\eta\lambda < 2 + 2\mu$, so its break-even sharpness is $(2 + 2\mu)/\eta$, lower for larger rates (their first conjecture). (ii) AdamW is linear in the gradient only for a frozen preconditioner, and P changes over about $1/(1-\beta_2)$ steps. The threshold applies to $\lambda_{\max}(P^{-1}H)$ (Cohen et al., 2022), so the raw sharpness at break-even is about $38\,p/\eta$, with p the preconditioner along the sharp direction, of the size of the gradient's root mean square. On TinyStories-30M $\lambda_{\max} \approx 5$ at $\eta = 1.5 \times 10^{-3}$ gives $p \approx 2 \times 10^{-4}$, and AdamW sits at the lowest raw sharpness of the three (5–10, against 12–20 for Muon and 20–30 for SGD; Fig. 32). A growing oscillation also raises v and with it P, so AdamW stabilizes itself through its preconditioner as well as through the curvature. The test is whether, at the steps at which AdamW is at its threshold after warmup, $\lambda_{\max}\eta$ grows with the gradient: at the tuned rate, over those steps of the five data orders, a log-log fit of $\lambda_{\max}\eta$ against the gradient norm has slope 1.15 on TinyStories-30M (77 steps, $r = 0.82$) and 1.61 on TinyStories-10M (69 steps, $r = 0.89$), and on Shakespeare the relation is weak (31 steps, $r = 0.52$). These are slopes against the gradient norm at one rate, not against η (for which see Section 5.4). The dependence on the gradient holds, with a slope above the predicted 1: the global gradient norm is only a proxy for p, which is set where the direction lies, on TinyStories mostly in the embeddings (Fig. 12). (iii) Muon's update $\mathrm{msign}(M)$ does not change when the gradient is scaled, so it has no linear regime, and its threshold is not on $\lambda_{\max}(H)$. On a quadratic $\frac{\lambda}{2}\|X\|_F^2$ in a matrix X, without momentum, $X \leftarrow X - \eta\,\mathrm{msign}(X)$ maps each singular value s to $|s - \eta|$: the iterates enter $\{\|X\|_2 \leq \eta\}$ and stay there for every $\lambda > 0$. No curvature makes Muon diverge; the curvature sets only the excess loss of the oscillation, at most $\frac{\lambda}{2}\eta^2 \operatorname{rank} X$. The iterates still settle at an edge of stability, but one set by the gradient. For a step $-\eta U$ with $\|U\|_2 = 1$ and $\langle G, U\rangle = \|G\|_*$, the nuclear norm, as $U = \mathrm{msign}(G)$ gives, the loss changes to second order by $-\eta\|G\|_* + \frac{1}{2}\eta^2\langle U, H[U]\rangle$, which is negative only while the curvature along the update is below $2\|G\|_*/\eta$. Islamov et al. (2026) show that the stability condition of such scale-invariant updates bounds a sharpness measured in the spectral norm by $2\|\nabla L\|_*/\eta$, and that full-batch spectral gradient descent sits at this bound. Muon's break-even point is thus set by this spectral-norm sharpness rather than by $\lambda_{\max}(H)$; on the part left to its auxiliary AdamW it is AdamW's (Section 4.4). In our runs the bound binds only around the end of warmup (ratio 0.80–0.93) and S falls well below it after (Section 5.4): a stochastic step is less aligned with the full gradient than the polar factor U the bound assumes. Across learning rates, Muon's $\lambda_{\max}(H)$ falls about as $1/\eta$, like the thresholds of the other two (Section 5.4).

Why the break-even point does not decide the function. Stability caps the curvature along the few sharpest directions of each optimizer's operator. This moves the run along the fiber of weights and sets how fast the loss falls, but it does not choose between functions with the same loss: by Proposition 1, a function in a stage class is pinned by its loss up to the excess loss. An oscillation of amplitude a along the sharpest direction costs $\frac{1}{2}\lambda a^2$ of loss, so it moves the function no more than that excess allows, and equally in floor pairs and in cross pairs. This is why the edge-of-stability events (AdamW at its threshold for much of training, SGD only early) do not line up with changes in the loss-matched distance (Section 5.4), and why on CIFAR-10 the break-even effects on sharpness and gradient noise replicate while two learning rates stay within 1.75 floors at equal loss (Section 5.9).

Predictions. A persistent split needs a class without a best element, as when training into interpolation; AdamW and Muon memorize Shakespeare without splitting, so either their implicit biases agree on held-out text or the runs are not yet in that regime. Across learning rates the sharpness of SGD and AdamW should scale as $1/\eta$; in the sweeps it does, and so does Muon's $\lambda_{\max}(H)$ (Section 5.4). Muon should sit at its spectral-norm threshold in full-batch training; with mini-batches it comes near it only around the end of warmup (Section 5.4).

7 Discussion and limitations

No split after warmup. The hypothesis of the title, that optimizers first follow one path and separate later, is not what we observe. After warmup, at equal training loss, two optimizers on the same data order are about as far apart as one optimizer on two data orders, down to the lowest loss we reach. The exceptions involve SGD: on TinyStories-10M it is up to about twice as far from AdamW or Muon at losses 4–2.5 and within 1.1 times below, and on Shakespeare, where it barely trains, it stays above the floor, with the only persistent split, against Muon, a fragile one. Within warmup the ratios to the floor are large, but there the runs still share their initialization and the floor is near zero, so these ratios are inflated; in this stage the optimizers pick up simple statistics in different orders (on Shakespeare SGD and AdamW pass through the unigram model and Muon does not). After it the function is set mostly by the data, and the optimizer matters mainly through speed (Section 6 makes this precise). This rests on comparing at equal loss: at equal steps all pairs look different for most of training, Muon being faster than AdamW throughout, and on Shakespeare Muon's faster memorization dominates the comparison.

What the floor means. Being within the floor does not mean computing the same function: at the end of the TinyStories runs two data orders of AdamW or Muon differ by 0.051–0.062 bits. The finding is that changing the optimizer perturbs the learned function no more than changing the order of the data or, where we tested it, the initialization (Section 5.11), not that it does not perturb it. The asymptotic implicit-bias results (Gunasekar et al., 2018; Zhang et al., 2024a; Fan et al., 2025) describe the limit of long training of linear models on separable data. Our TinyStories models see at most 1.2 epochs, and by then the loss-matched difference between optimizers is within the data-order noise; in the linear control the split appears only near interpolation. Nor do the swaps show an early decision: a fork ends at the function of the optimizer it switches to whenever that optimizer is fast enough to get there in the remaining steps (on TinyStories, whenever it is the faster one, at every fork step we tried), and keeps the mark of its parent only when the new optimizer is too slow to move away from it.

Function space and weight space. The mode-connectivity barriers tell a different story from the function distances. After a change of optimizer a fork needs a later fork step to stay linearly connected to its parent than after a change of data order, except for Muon on Shakespeare, which is itself unstable to the data order while it memorizes. Between pure runs the size of the barrier does not single out the optimizer (AdamW and Muon are further apart than two data orders only on TinyStories-10M), but its location does: the barrier between AdamW and Muon extends into the middle blocks, while that between two data orders lies in the first block, although their functions are as close. Different optimizers thus reach functions that are equivalent up to data-order noise through weights that differ beyond the first block, whereas data orders differ mainly in the first. The weights also carry the geometry each optimizer is biased towards: more even entries for AdamW than for SGD, and more even singular values for Muon (Section 5.10), and after a swap that geometry is a mixture weighted by how far each optimizer moved the weights, even when the function is entirely the second optimizer's. The hidden states sit in between: on TinyStories AdamW and Muon differ after the first block and little after the others. Part of the barriers may be scale: the optimizers move the weights by very different amounts (on TinyStories Muon's end 5–6.6 initial norms from the initialization, AdamW's 2.7–2.8 and SGD's 0.4–0.6), and interpolating between weights of different scales can raise the loss without any change in function.

Break-even point. On CIFAR-10 we reproduce the sharpness and gradient-noise effects of Jastrzebski et al. (2020), but the separation of trajectories between learning rates is mostly a difference in speed: at equal loss, 0.01 and 0.1 stay within 1.75 times the floor. Comparing at equal steps overstates how differently two training procedures shape the function.

Edge of stability. On TinyStories AdamW is at its adaptive edge of stability through much of training whether or not the optimizers differ, and SGD meets its threshold only early in training; neither explains the start or the end of the early episode, which ends within warmup in all pairs but one and in the same way for AdamW and Muon. With the cosine schedule the threshold acts as a ceiling: once the learning rate decays, the sharpness levels off instead of following the threshold up, and the runs leave the edge of stability. Across learning rates the top Hessian eigenvalue falls about as $1/\eta$ for all three optimizers, Muon included, whose threshold is on a spectral-norm sharpness (Islamov et al., 2026). That sharpness comes near its bound only around the end of warmup, and at a constant rate the break-even step does

not mark where the paths of different rates separate, for any of the three optimizers. Along the step they actually take, AdamW and Muon sit at an edge of stability of the population loss, not of their batch loss (Section 5.5). The one edge-of-stability event that lines up with a change in function, Muon's auxiliary AdamW overshooting its threshold just before Muon memorizes on Shakespeare, is not what drives the memorization, which persists in the matched control (where the overshoot comes later and is smaller). The anatomy adds that the sharp direction is optimizer-specific, lies at the input end of the network, and, except in AdamW's embeddings, mostly comes from curvature between layers, but its changes do not line up with changes in function distance either.

Future work. Per-layer learning rates cannot be allocated by stability, since each layer's curvature along its step adapts to its own rate (Section 5.8); they could instead be set from loss hypergradients, the derivative of the loss K steps later with respect to $\log m_a$. Beyond that: whether the sharp direction moves to the new optimizer's layers after a swap.

Limitations.

- The noise floor varies only the data order; a second initialization (Section 5.11) gives the same distance as a new data order, but only two of each were run, on Shakespeare and TinyStories-10M.
- Learning rates come from five-point sweeps on one data order, and the Shakespeare AdamW rate is the smallest of its grid. The other rates of the sweeps agree with the tuned ones on TinyStories (Fig. 23), but only against the tuned runs' floor. SGD trains poorly at the stable rates on Shakespeare, so comparisons with SGD there stop at loss 2.61, and its large data-order noise on TinyStories makes "no split" against SGD a weak statement.
- The first exceedance is degenerate (Section 4.2); episodes and the persistent split depend on the margin $m_\times$, the run length k and the matching tolerance τ; the conclusions hold over the variants we tried (Section 5.2), except the Shakespeare Muon–SGD split.
- The loss grid ends at the lowest loss that all runs of a pair reach, so a split below it would be missed.
- The thresholds are those of linear stability on a quadratic with a frozen preconditioner; Muon's spectral-norm sharpness is measured only on TinyStories-10M, without momentum and as a Frank–Wolfe lower bound (Section 4.4), and the AdamW ratio at the first steps reflects a preconditioner built from few gradients.
- The sweeps with sharpness use one data order and three or four rates per optimizer.
- Sharpness is measured on the training loss of a fixed batch of 32 training sequences with 20 Lanczos iterations, in float32 with TF32 matrix products, while training uses bfloat16 autocast and the predictions are compared on held-out validation contexts.
- Loss matching and the mode-connectivity barrier use a training probe of 64 sequences.
- The landing coordinate is a ratio of distances, not a projection onto the segment between the pure runs.
- The models are small (up to 30M parameters) and trained for at most 655M tokens.

8 Conclusion

From the same initialization and data order, SGD with momentum, AdamW and Muon do not first share a path and then split. After warmup, at equal training loss, two optimizers are about as far apart as two data orders of one optimizer. This holds in all three language-model settings, with SGD up to about twice as far from AdamW or Muon at losses 4–2.5 on TinyStories-10M, every pair within 1.1 times on TinyStories-30M, and one short AdamW–Muon episode on Shakespeare; the exceptions are the pairs with SGD on Shakespeare, where SGD barely trains, with the only persistent split, Muon–SGD, a fragile one. The large ratios within warmup are inflated by a floor near zero. Muon is faster than AdamW, and at equal steps the two are at most about a fifth closer on the same data order than two data orders of one optimizer, because they share their recent batches; at the end of TinyStories-30M, replacing AdamW by Muon changes the function no more than reshuffling the data. Neither the edge of stability nor the end of warmup explains the changes in distance; at a constant rate, reaching the threshold does not mark where runs at different rates separate, and Muon's spectral-norm sharpness comes near its bound only around the end of warmup. Switching optimizers during training does not reveal an early decision either: the run ends at the function of the optimizer it switches to whenever that one is fast enough to get there, on TinyStories whenever it is the faster one, even late in training. The optimizer does leave a mark in weight space: after a change of optimizer a fork usually needs a later fork step to stay linearly connected to its parent, the barrier between AdamW and Muon extends beyond the first block, where that between two data orders lies, and the weights carry each optimizer's geometry, mixed after a swap according to how far each moved them. On TinyStories the hidden representations differ after the first block and little after the others, and the functions no more than with another data order even off the data distribution. As a check, on TinyStories every rate of the sweeps that trains stably is within 0.8–1.1 floors of the tuned runs, with one exception at 1.35, and on TinyStories-10M rates from a quarter to four times each optimizer's rate stay within 0.8–1.1 times the

floor of the tuned runs at equal loss. A linear model on separable data, where the limits of training are known to differ, shows that our test can detect a split, and places it near interpolation; but AdamW and Muon trained on Shakespeare until they memorize the training set do not split either, so for these models the absence of a split is not only a matter of training too briefly. A simple model accounts for this pattern (Section 6): at equal loss, functions of one n-gram class are pinned by their excess loss whatever the optimizer, while the optimizer chooses the route, the speed, the weights among those computing a function, and a cap on the curvature, which for Muon applies to a spectral-norm sharpness. Per-layer rates that even out the curvature help only SGD, as a self-tuned rate: for AdamW and Muon each layer's curvature along its step adapts to its own rate, so stability cannot allocate the rate across layers, and at equal loss the balanced runs compute the function of the baseline up to data-order noise.

References

Adilova, L., M. Andriushchenko, M. Kamp, et al. (2024). "Layer-wise Linear Mode Connectivity". In: *ICLR*.

Andreyev, A., A. Ananthkumar, M. Walden, et al. (2026). *Momentum Further Constrains Sharpness at the Edge of Stochastic Stability*. Preprint.

Andreyev, A. and P. Beneventano (2024). *Edge of Stochastic Stability: Revisiting the Edge of Stability for SGD*. Preprint.

Bartlett, P. L., D. J. Foster, and M. Telgarsky (2017). "Spectrally-normalized Margin Bounds for Neural Networks". In: *NeurIPS*.

Belrose, N., Q. Pope, L. Quirke, et al. (2024). "Neural Networks Learn Statistics of Increasing Complexity". In: *ICML*.

Cattaneo, M. D., J. M. Klusowski, and B. Shigida (2024). "On the Implicit Bias of Adam". In: *ICML*.

Clark, K., M.-T. Luong, Q. V. Le, and C. D. Manning (2020). "ELECTRA: Pre-training Text Encoders as Discriminators Rather Than Generators". In: *ICLR*.

Cohen, J. M., A. Damian, A. Talwalkar, et al. (2025). "Understanding Optimization in Deep Learning with Central Flows". In: *ICLR*.

Cohen, J. M., B. Ghorbani, S. Krishnan, et al. (2022). *Adaptive Gradient Methods at the Edge of Stability*. Preprint.

Cohen, J. M., S. Kaur, Y. Li, et al. (2021). "Gradient Descent on Neural Networks Typically Occurs at the Edge of Stability". In: *ICLR*.

Cortesi, F. V., G. Iannone, G. Crippa, et al. (2026). *Same Error, Different Function: The Optimizer as an Implicit Prior in Financial Time Series*. Preprint.

Damian, A., E. Nichani, and J. D. Lee (2023). "Self-Stabilization: The Implicit Bias of Gradient Descent at the Edge of Stability". In: *ICLR*.

Demmel, J. (2023). "Nearly Optimal Block-Jacobi Preconditioning". In: *SIAM Journal on Matrix Analysis and Applications* 44.1, pp. 408–413.

Du, S. S., W. Hu, and J. D. Lee (2018). "Algorithmic Regularization in Learning Deep Homogeneous Models: Layers are Automatically Balanced". In: *NeurIPS*.

Eldan, R. and Y. Li (2023). *TinyStories: How Small Can Language Models Be and Still Speak Coherent English?*

Endres, D. M. and J. E. Schindelin (2003). "A New Metric for Probability Distributions". In: *IEEE Transactions on Information Theory* 49.7, pp. 1858–1860.

Fan, C., M. Schmidt, and C. Thrampoulidis (2025). "Implicit Bias of Spectral Descent and Muon on Multiclass Separable Data". In: *NeurIPS*.

Ford, N., J. Gilmer, N. Carlini, and E. D. Cubuk (2019). "Adversarial Examples Are a Natural Consequence of Test Error in Noise". In: *ICML*.

Frankle, J., G. K. Dziugaite, D. M. Roy, and M. Carbin (2020). "Linear Mode Connectivity and the Lottery Ticket Hypothesis". In: *ICML*.

Gunasekar, S., J. Lee, D. Soudry, and N. Srebro (2018). "Characterizing Implicit Bias in Terms of Optimization Geometry". In: *ICML*.

Hahn, M., D. Jurafsky, and R. Futrell (2021). "Sensitivity as a Complexity Measure for Sequence Classification Tasks". In: *Transactions of the Association for Computational Linguistics*.

Howard, J. and S. Ruder (2018). "Universal Language Model Fine-tuning for Text Classification". In: *ACL*.

Islamov, R., M. Crawshaw, J. Cohen, and R. Gower (2026). *Non-Euclidean Gradient Descent Operates at the Edge of Stability*. Preprint.

Jastrzebski, S., M. Szymczak, S. Fort, et al. (2020). "The Break-Even Point on Optimization Trajectories of Deep Neural Networks". In: *ICLR*.

Jiang, Y., B. Neyshabur, H. Mobahi, et al. (2020). "Fantastic Generalization Measures and Where to Find Them". In: *ICLR*.

Jordan, K., Y. Jin, V. Boza, et al. (2024). *Muon: An Optimizer for Hidden Layers in Neural Networks*. Blog post. URL: https://kellerjordan.github.io/posts/muon/.

Kalra, D. S. and M. Barkeshli (2024). "Why Warmup the Learning Rate? Underlying Mechanisms and Improvements". In: *NeurIPS*.
Karpathy, A. (2022). *nanoGPT*. Software. URL: https://github.com/karpathy/nanoGPT.
Keskar, N. S. and R. Socher (2017). *Improving Generalization Performance by Switching from Adam to SGD*. Preprint.
Kornblith, S., M. Norouzi, H. Lee, and G. Hinton (2019). "Similarity of Neural Network Representations Revisited". In: *ICML*.
Krizhevsky, A. (2009). *Learning Multiple Layers of Features from Tiny Images*. Tech. rep. University of Toronto.
Kunstner, F., A. Milligan, R. Yadav, et al. (2024). "Heavy-Tailed Class Imbalance and Why Adam Outperforms Gradient Descent on Language Models". In: *Advances in Neural Information Processing Systems*.
Lee, J. and K. M. Lee (2026). *The Road Taken: The Role of Optimizers at the Edge of Stability*. Preprint.
Liu, J., J. Su, X. Yao, et al. (2025). *Muon is Scalable for LLM Training*. Preprint.
Moosavi-Dezfooli, S.-M., A. Fawzi, J. Uesato, and P. Frossard (2019). "Robustness via Curvature Regularization, and Vice Versa". In: *CVPR*.
Nagarajan, V. and J. Z. Kolter (2019). *Generalization in Deep Networks: The Role of Distance from Initialization*. NeurIPS 2017 Workshop on Deep Learning: Bridging Theory and Practice.
Neyshabur, B., R. Tomioka, and N. Srebro (2015). "Norm-Based Capacity Control in Neural Networks". In: *COLT*.
Novak, R., Y. Bahri, D. A. Abolafia, et al. (2018). "Sensitivity and Generalization in Neural Networks: an Empirical Study". In: *ICLR*.
Sanyal, A., P. H. S. Torr, and P. K. Dokania (2020). "Stable Rank Normalization for Improved Generalization in Neural Networks and GANs". In: *ICLR*.
Saxe, A. M., J. L. McClelland, and S. Ganguli (2014). "Exact Solutions to the Nonlinear Dynamics of Learning in Deep Linear Neural Networks". In: *ICLR*.
Sluis, A. van der (1969). "Condition Numbers and Equilibration of Matrices". In: *Numerische Mathematik* 14.1, pp. 14–23.
Smith, S. L., B. Dherin, D. G. T. Barrett, and S. De (2021). "On the Origin of Implicit Regularization in Stochastic Gradient Descent". In: *ICLR*.
Soudry, D., E. Hoffer, M. S. Nacson, et al. (2018). "The Implicit Bias of Gradient Descent on Separable Data". In: *JMLR*.
Srinivas, S., K. Matoba, H. Lakkaraju, and F. Fleuret (2022). "Efficient Training of Low-Curvature Neural Networks". In: *NeurIPS*.
Vasudeva, B., J. W. Lee, V. Sharan, and M. Soltanolkotabi (2025). "The Rich and the Simple: On the Implicit Bias of Adam and SGD". In: *NeurIPS*.
Wang, J., M. Wang, Z. Zhou, et al. (2025). "The Sharpness Disparity Principle in Transformers for Accelerating Language Model Pre-Training". In: *ICML*.
Wen, K., D. Hall, T. Ma, and P. Liang (2025). *Fantastic Pretraining Optimizers and Where to Find Them*. Preprint.
Wilson, A. C., R. Roelofs, M. Stern, et al. (2017). "The Marginal Value of Adaptive Gradient Methods in Machine Learning". In: *NeurIPS*.
Xie, S. and Z. Li (2024). "Implicit Bias of AdamW: ℓ_∞ Norm Constrained Optimization". In: *ICML*.
Yang, G., E. J. Hu, I. Babuschkin, et al. (2021). "Tensor Programs V: Tuning Large Neural Networks via Zero-Shot Hyperparameter Transfer". In: *NeurIPS*.
You, Y., I. Gitman, and B. Ginsburg (2017). *Large Batch Training of Convolutional Networks*. Preprint.
You, Y., J. Li, S. Reddi, et al. (2020). "Large Batch Optimization for Deep Learning: Training BERT in 76 minutes". In: *ICLR*.
Zeng, A., X. Liu, Z. Du, et al. (2023). "GLM-130B: An Open Bilingual Pre-trained Model". In: *ICLR*.
Zhang, C., D. Zou, and Y. Cao (2024a). "The Implicit Bias of Adam on Separable Data". In: *NeurIPS*.
Zhang, Y., C. Chen, T. Ding, et al. (2024b). "Why Transformers Need Adam: A Hessian Perspective". In: *NeurIPS*.
Zhang, Y., C. Chen, Z. Li, et al. (2025). "Adam-mini: Use Fewer Learning Rates To Gain More". In: *ICLR*.
Zhao, R., D. Morwani, D. Brandfonbrener, et al. (2025). "Deconstructing What Makes a Good Optimizer for Language Models". In: *ICLR*.

A Hyperparameters

Table 5: Optimizer hyperparameters, shared by all language-model settings unless noted. The Shakespeare control matches Muon's auxiliary AdamW to AdamW: $\beta = (0.9, 0.99)$, weight decay 0.1, learning rate $3 \cdot 10^{-4}$.

	SGD	AdamW	Muon
Update	heavy ball, $\mu = 0.9$, not Nesterov	$\beta = (0.9, 0.95)$; Shakespeare $(0.9, 0.99)$; $\epsilon = 10^{-8}$	Nesterov momentum 0.95; 5 Newton–Schulz steps (bfloat16); scale $\sqrt{\max(1, n_{\text{out}}/n_{\text{in}})}$
Parameters	all	all	matrices in transformer blocks; auxiliary AdamW ($\beta = (0.9, 0.95)$, $\epsilon = 10^{-8}$) on embeddings, head, gains
Weight decay	none	0.1 on tensors with ≥ 2 dims, none on gains	none (Muon and auxiliary AdamW)
Clipping	none	none	none

Table 6: Learning-rate sweeps (data order 0, five rates per optimizer). Rates used in the main runs in bold. Selection: final validation loss for TinyStories, minimum validation loss over training for Shakespeare. The auxiliary AdamW of Muon keeps the learning rate of Table 1 throughout the sweep. SGD (†: best on data order 0) uses a lower rate, since its best rate diverged or blew up on other data orders: 0.2 on Shakespeare (orders 1 and 2), 0.1 on TinyStories-10M (orders 1–3) and 0.3 on TinyStories-30M (orders 2–4). On TinyStories-10M the divergence analysis uses SGD re-tuned to 0.06 (Fig. 27).

Setting	SGD	AdamW	Muon
Shakespeare-char	**0.05**, 0.1, 0.15, 0.2†, 0.25	$\mathbf{3 \cdot 10^{-4}}$, 10^{-3}, $2 \cdot 10^{-3}$, $4 \cdot 10^{-3}$, $8 \cdot 10^{-3}$	0.005, 0.01, 0.02, **0.04**, 0.08
TinyStories-10M	**0.03**, 0.1†, 0.3, 1, 3	$3 \cdot 10^{-4}$, 10^{-3}, $\mathbf{2 \cdot 10^{-3}}$, $4 \cdot 10^{-3}$, $8 \cdot 10^{-3}$	0.005, **0.01**, 0.02, 0.04, 0.08
TinyStories-30M	0.03, **0.1**, 0.3†, 1, 3	$3 \cdot 10^{-4}$, 10^{-3}, $\mathbf{2 \cdot 10^{-3}}$, $4 \cdot 10^{-3}$, $8 \cdot 10^{-3}$	0.005, **0.01**, 0.02, 0.04, 0.08

Table 7: Fork steps of the swaps (all six ordered pairs, plus each optimizer into itself with data order 1).

Setting	Fork steps
Shakespeare-char	25, 50, 100, 200, 300, 450, 600, 800, 1000
TinyStories-10M	50, 100, 200, 400, 700, 1000, 1500, 2500, 4000
TinyStories-30M	100, 200, 400, 800, 1400, 2000, 3000, 5000, 8000

Break-even runs: plain SGD (momentum 0), constant learning rate, no warmup. CIFAR-10: 100 epochs, i.e. 39,000 steps at batch 128, 166,600 at batch 30 and 16,600 at batch 300; data orders 0 and 1. Shakespeare and TinyStories-10M: 1,500 and 10,000 steps at their training batch size, learning rates 0.03, 0.1 and 0.3; data orders 0, 1 and 2.

B Measurement details

A grid "every $N + K$ log" is $\{0, N, 2N, \dots\} \cup \{\text{round}(T^u) : u \in K \text{ evenly spaced values in } [0, 1]\} \cup \{T\}$, i.e. the union of a linear grid and K log-spaced steps between 1 and T.

Sharpness. Lanczos runs for 20 iterations with full reorthogonalization (two Gram–Schmidt passes per iteration) from a fixed Gaussian start, and returns the top Ritz pair. Hessian-vector products use double backpropagation through one gradient graph, with the exact (math) attention kernel, in float32 without autocast but with TF32 matrix products, on the model before the update of the measured step.

Split. $\tau = 0.02$, $m_\times = 0.5$, $m_+ = 0$, $k = 3$, 1000 bootstrap resamples. The training loss of a run is its loss on the training probe at its prediction steps.

Mode connectivity. Barrier on the 64 sequences of the training probe, at 11 evenly spaced values of α, between final checkpoints.

Table 8: Logging grids, probe sizes and measurement settings of the main language-model runs (forks inherit their parent's) and of the CIFAR-10 runs at batch 128. Fixed sets of examples are drawn once, from random streams separate from the training batches, and are the same for all runs of a setting.

	Shakespeare-char	TinyStories-10M	TinyStories-30M	CIFAR-10 (batch 128)
Prediction steps $\mathcal{S}$	every 50 + 40 log	every 250 + 30 log	every 500 + 30 log	every 390 + 40 log
Checkpoints	every 250 + 12 log	every 1000 + 12 log	every 2000 + 12 log	every 3900 + 12 log
Measurement steps $\mathcal{M}$	every 50 + 30 log	every 500 + 30 log	every 1000 + 30 log	every 390 + 30 log
Validation probe (N)	$64 \times 256 = 16{,}384$	$128 \times 16 = 2{,}048$	$128 \times 16 = 2{,}048$	10,000 images
Training probe	64 seq. (16,384 tok.)	64 seq. (16,384 tok.)	64 seq. (16,384 tok.)	2,000 images
Hessian batch $\mathcal{H}$	32 seq. (8,192 tok.)	32 seq. (8,192 tok.)	32 seq. (8,192 tok.)	2,500 images
Lanczos iterations	20	20	20	20
Batch sharpness	8×64 seq.	8×128 seq.	8×128 seq.	8×128 images
$\hat{K}$ (break-even runs)	25×64 seq.	25×128 seq.	–	25×128 images

Prediction trajectories. Break-even runs: 128 evenly spaced probe positions; top 50 principal components of the centered probability vectors, then UMAP. Figures 1 and 24: densMAP of the $\sqrt{\text{JS}}$ distance between all pairs of logged steps (Section 4.1).

B.1 Function metrics

The metrics are computed on 1024 fixed validation examples (sequences or test images), drawn separately from the probe and shared by all runs; the input-derivative metrics use the first 32.

Input derivatives. Let $f(x)$ be the loss at the true labels as a function of the input x. For the language models $x \in \mathbb{R}^{64\times d}$ is the token-embedding matrix of the first 64 tokens of a sequence (position embeddings are added inside the model) and f is the mean next-token cross-entropy over these 64 positions; for the CNN x is the normalized image and f its cross-entropy. With $g = \nabla f(x)$, $H = \nabla^2 f(x)$ and $T[u,u,\cdot] = \nabla_x\big(u^\top H(x)u\big)$, and $z_1, \dots, z_8$ standard Gaussian probes:

- gradient norm $\|g\|$;
- Jacobian norm $\|J\|_F$ of the softmax output with respect to x (Novak et al., 2018): exact for the CNN (one vector-Jacobian product per class), and estimated as $\big(\frac{1}{8}\sum_j \|J^\top z_j\|^2\big)^{1/2}$ over the $64 \times V$ outputs for the language models;
- Hessian norm $\|H\|_F \approx \big(\frac{1}{8}\sum_j \|Hz_j\|^2\big)^{1/2}$;
- top input-Hessian eigenvalue $\lambda_x = v^\top H v$, with v from 20 power iterations (the eigenvalue of largest magnitude) (Moosavi-Dezfooli et al., 2019);
- level-set curvature $\big(\frac{1}{8}\sum_j \|PHPz_j\|^2\big)^{1/2}/\|g\| \approx \|PHP\|_F/\|g\|$, with $P = I - \hat{g}\hat{g}^\top$ the projection orthogonal to the gradient, a projected variant of the normalized curvature of Srinivas et al. (2022);
- third order: the norm $\|T\|_\sigma = \max_{\|u\|=1} T[u,u,u]$, by shifted symmetric power iteration $u \leftarrow (T[u,u,\cdot] + \alpha u)/\|T[u,u,\cdot] + \alpha u\|$ with $\alpha = \|T[u_0,u_0,\cdot]\|$, 20 iterations, best of 3 random starts (as $T[u,u,u]$ is odd in u, this is also $\max|T[u,u,u]|$); the third derivative along the sharpest direction $T[v,v,v]$, with the sign of v fixed by $v^\top g \geq 0$; and the curvature length $|\lambda_x|/\|T[v,v,\cdot]\|$, the distance over which, to first order, λ_x changes by its own size (for a simple eigenvalue $\nabla_x \lambda_x = T[v,v,\cdot]$).

Each is reported as mean and standard deviation over the 32 points.

Weight norms. Over all parameters θ: ℓ_0 (entries with $|\theta_i| > 10^{-8}$), ℓ_1, ℓ_2, ℓ_3, ℓ_∞. For each weight tensor with at least two dimensions, reshaped to a matrix W (output $\times$ input): the spectral norm $\|W\|_2$, nuclear norm $\|W\|_*$ and stable rank $\|W\|_F^2/\|W\|_2^2$ (Sanyal et al., 2020). Distance from initialization $\|\theta - \theta_0\|_2$, also relative to $\|\theta_0\|_2$ (Nagarajan and Kolter, 2019). The log spectral complexity $\log\big(\prod_i \|W_i\|_2 \cdot (\sum_i (\|(W_i - W_i^0)^\top\|_{2,1}/\|W_i\|_2)^{2/3})^{3/2}\big)$ of Bartlett et al. (2017), with the initialization as reference matrices. For the CNN only, the path norm as computed by Jiang et al. (2020) (after Neyshabur et al., 2015): the sum of the outputs on an all-ones input of the network with every parameter squared and identity activations.

Accuracy and robustness. Accuracy on the 1024 validation examples (for the language models, next- token accuracy over all positions). For the CNN, accuracy under Gaussian input noise of standard deviation 0.05, 0.1 and 0.2 (in normalized pixel units) (Ford et al., 2019). For the language models, a one-token substitution sensitivity after Hahn et al. (2021): for 32 validation contexts of 64 tokens, we replace each position in turn by a uniformly random different

token and count how many of the 64 substitutions change the argmax prediction at the last position; we report this count as a fraction of 64, averaged over contexts. All derivatives are taken by automatic differentiation in float32, with the exact (math) attention kernel.

C Compute

All runs used one NVIDIA L40S GPU each on the MIT ORCD cluster. The project took 103 GPU-hours in all, 60% of them for the forks (Table 9), and 89 CPU core-hours for the offline analyses that run on CPU (mode connectivity, representations, off-distribution distances, swap landing and the split statistics). Twelve smoke-test runs (0.2 GPU-hours) are not counted. The GPU-hours include runs that were later replaced and are not in the run counts (SGD at its best sweep rate on data orders 1–4, which diverged), and the Shakespeare break-even count includes the three runs at rate 1.0, which diverged. Not in the table are the learning-rate runs of Fig. 24 (12 runs on TinyStories-10M), the sweeps with sharpness measurements (10 runs on TinyStories-10M, 15 on Shakespeare), the break-even runs of the three optimizers at a constant rate (15 runs on TinyStories-10M), the spectral-norm sharpness of Muon, the stochastic edge of stability, and the balanced and fixed-multiplier runs of Section 5.8.

Table 9: Runs per suite and compute. The matched-Muon control has one more run, on data order 0, that saves the fork checkpoints for the sharpness of the auxiliary block (Section 4.4).

Suite	Runs	GPU-hours
Learning-rate sweeps (3 settings)	3×15	9.0
Main runs (3 settings)	3×15	16.7
Forks (3 settings)	3×81	61.1
Break-even, Shakespeare and TinyStories-10M	$12 + 9$	2.6
Break-even, CIFAR-10	10	1.1
Shakespeare control (matched Muon)†	$5 + 1$	0.5
Second initialization, Shakespeare and TinyStories-10M†	2×6	1.0
Shakespeare trained into memorization†	10	0.5
Offline function metrics	–	10.7
Total	392	103.2

D Additional figures and tables

Table 10: Edge of stability in the language models: end of warmup w; measured steps at the edge of stability ($\mathcal{Q} \geq 0.9\,\Theta$; $\lambda_{\max}$ for SGD, preconditioned sharpness for AdamW) of the $d = 0$ run, from first to last; onset s_{EoS} over the 5 data orders; fraction of measured steps at the edge of stability over the 5 data orders; first exceedance t^*_X of the $d = 0$ run against the other two optimizers (in the order SGD, AdamW, Muon); and its persistent split, if any.

Setting	Run	w	at EoS ($d = 0$)	s_{EoS} (all d)	fraction of $\mathcal{M}$	t^*_X	$t^*_{\mathrm{p},X}$
Shakespeare	SGD	100	21–120	21	2–14%	1 / 1	191 (vs Muon)
	AdamW	100	1–500	1	11–16%	1 / 1	–
	Muon	100	–	–	–	1 / 2	14 (vs SGD)
TinyStories-10M	SGD	200	62–1500	45–500	6–12%	3 / 1	–
	AdamW	200	1–4500	1	50–65%	1 / 1	–
	Muon	200	–	–	–	1 / 7	–
TinyStories-30M	SGD	400	60–658	60–467	10–16%	2 / 1	–
	AdamW	400	1–14000	1	40–69%	1 / 1	–
	Muon	400	–	–	–	1 / 8	–

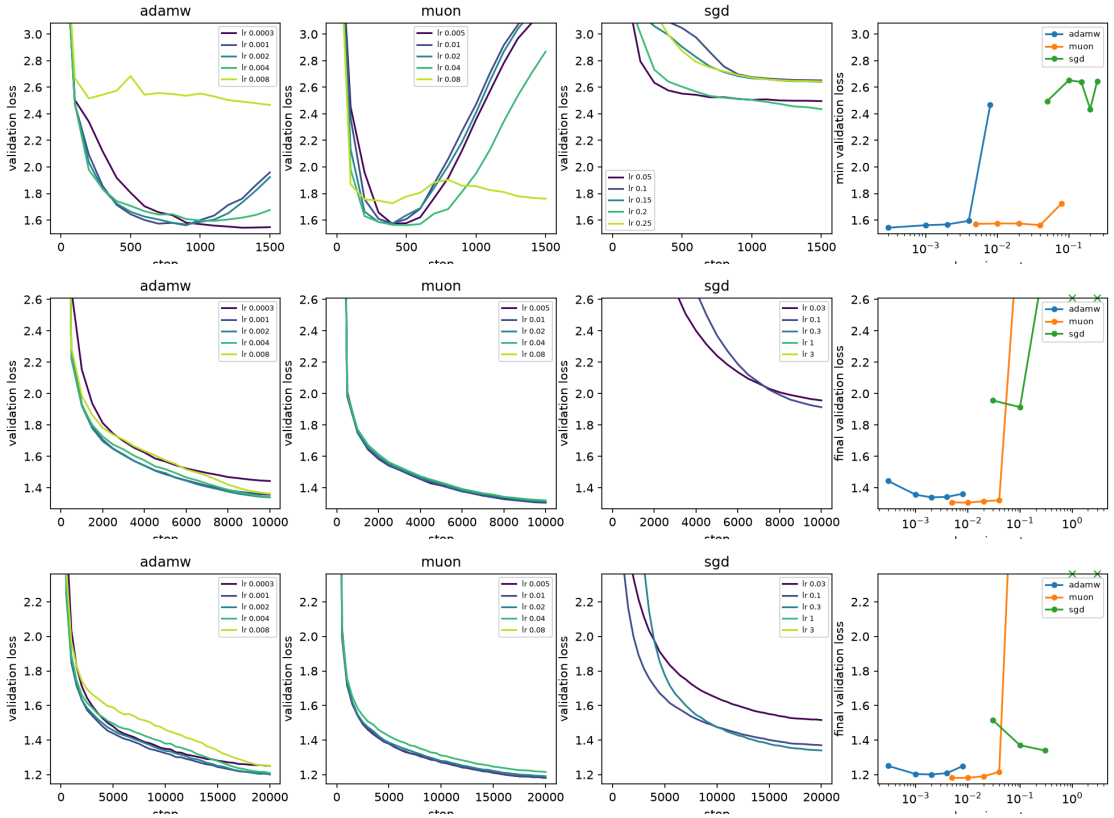


Figure 26: Learning-rate sweeps on Shakespeare (top), TinyStories-10M and TinyStories-30M (bottom), data order 0. One panel per optimizer with the validation loss of each rate against step, and a last panel with the minimum (Shakespeare) or final (TinyStories) validation loss against the learning rate; × marks runs that diverged. Curves above the plotted range are cut.

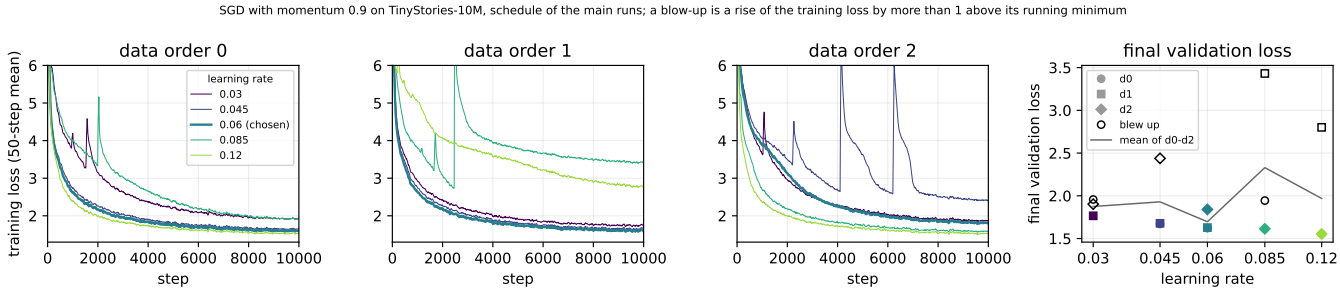


Figure 27: Re-tuning SGD on TinyStories-10M, data orders 0–2, schedule of the main runs. Left three panels: training loss (50-step mean) of each rate on one data order. Right: final validation loss against the rate; open markers: runs that blew up (a rise of the training loss by more than 1 nat above its running minimum). Only 0.06 never blows up; it has the best mean, 1.70 (1.64 in the rerun with the main logging, where order 2 skipped the early spike of its tuning run), against 1.88 at 0.03.

Table 11: Final function metrics on Shakespeare, mean over the five data orders, as in Table 4.

	SGD	AdamW	Muon
$\|\theta\|_2$	97	92	654
$\|\theta\|_\infty/\|\theta\|_2$ ($\times 10^3$)	12.3	12.4	2.4
stable rank (hidden, mean)	63	48	99
$\|\theta-\theta_0\|/\|\theta_0\|$	0.36	0.25	7.39
input Jacobian (Frobenius)	46.8	120.1	162.9
substitution sensitivity	0.048	0.097	0.191
top-1 accuracy	0.323	0.554	0.533

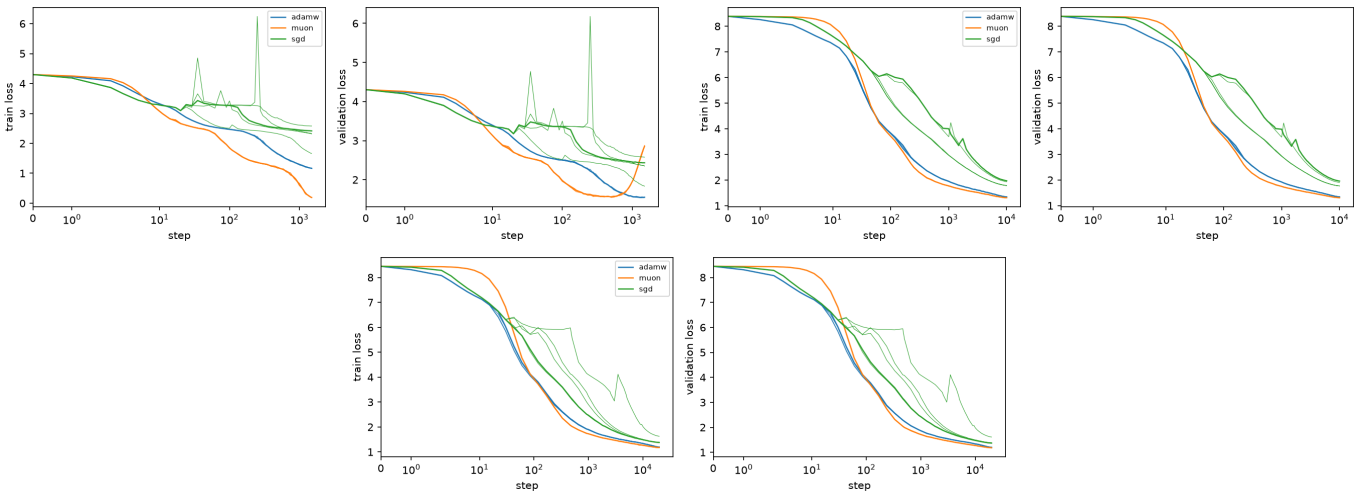


Figure 28: Training-probe and validation loss of the main runs against step: Shakespeare (top left), TinyStories-10M (top right), TinyStories-30M (bottom). Data order 0 thick, orders 1–4 thin.

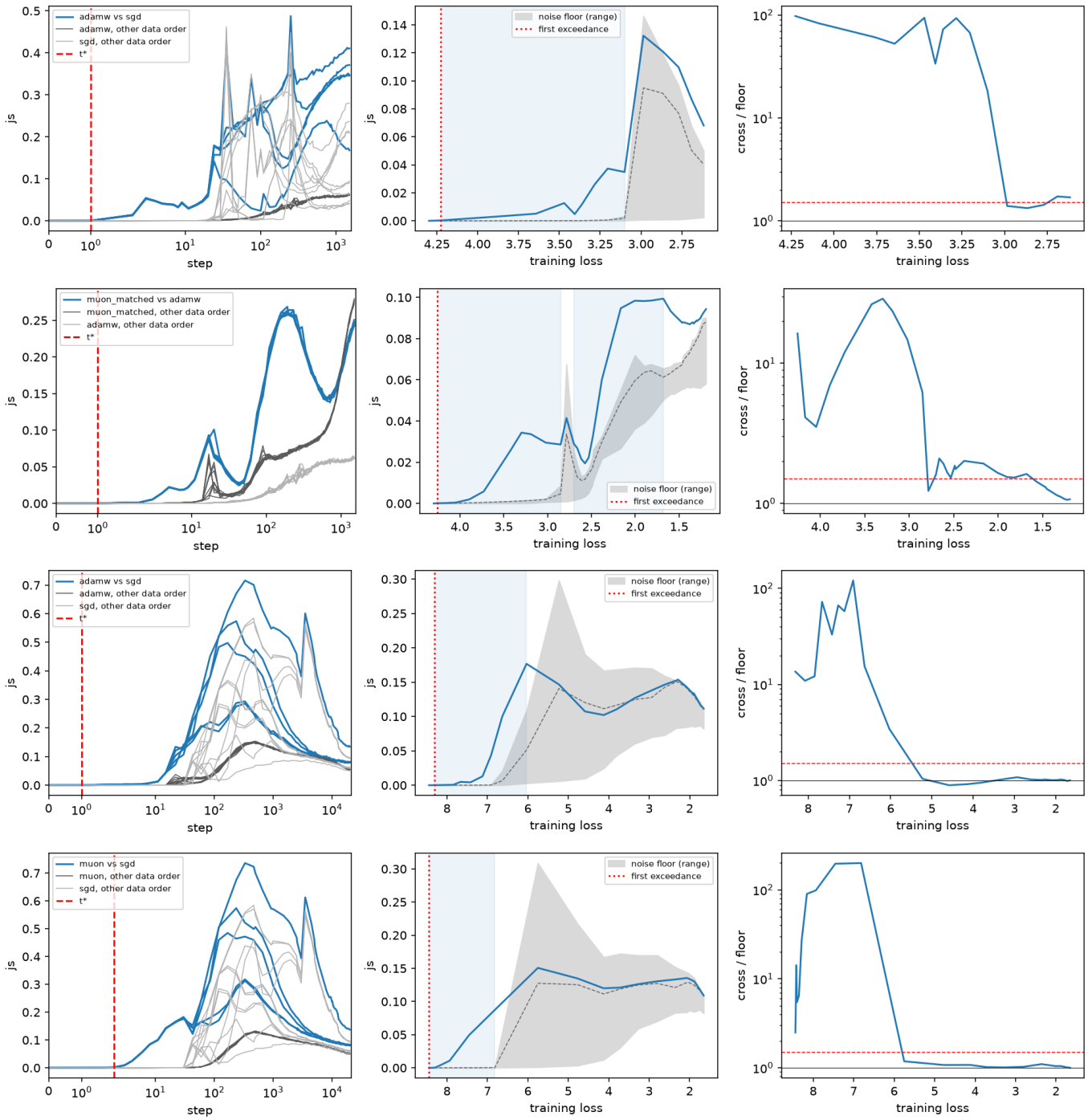


Figure 29: Divergence figures as in Fig. 2. Rows: Shakespeare AdamW–SGD; Shakespeare matched Muon–AdamW (control); TinyStories-30M AdamW–SGD; TinyStories-30M Muon–SGD.

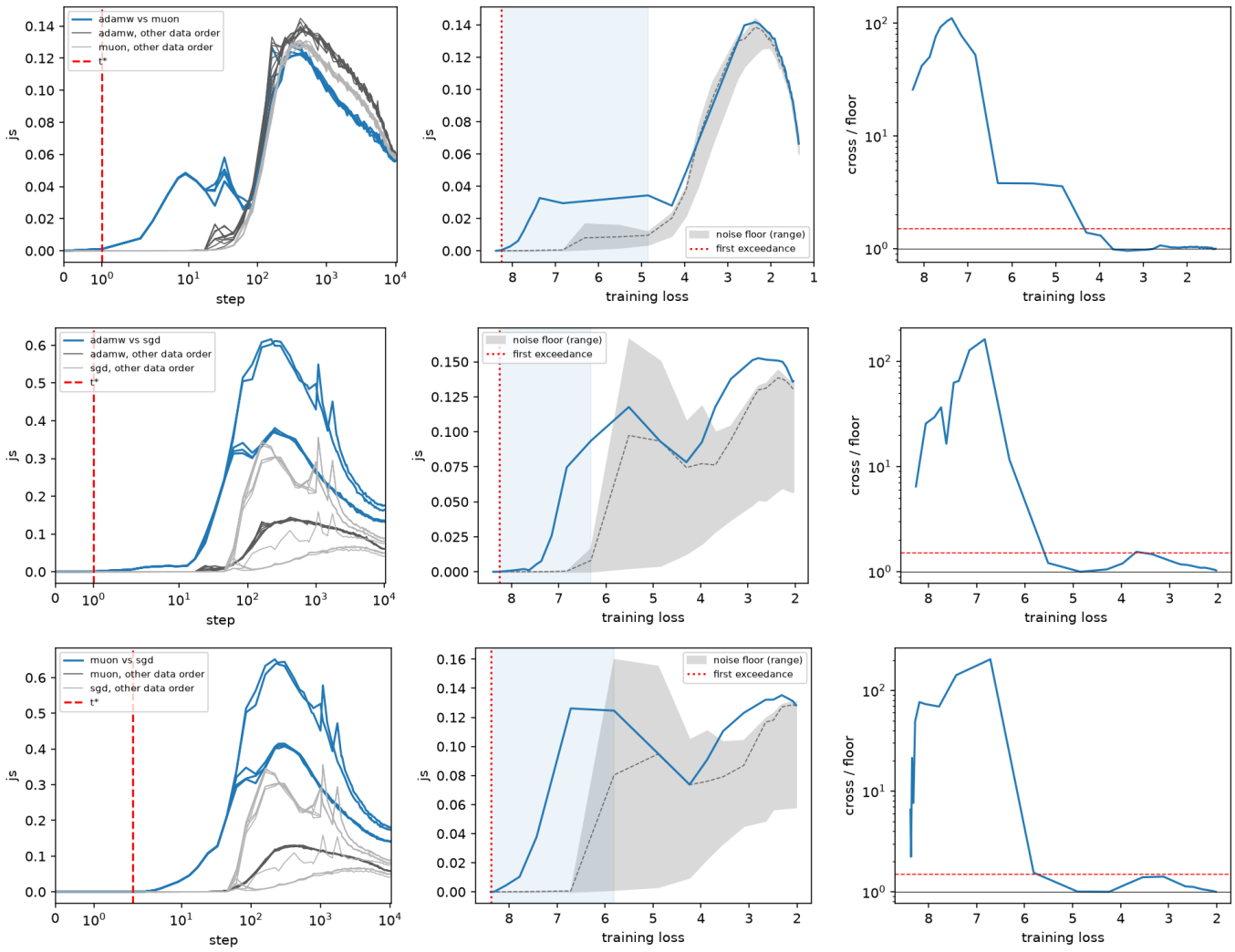


Figure 30: Divergence figures as in Fig. 2 for TinyStories-10M. Rows: AdamW–Muon, AdamW–SGD, Muon–SGD.

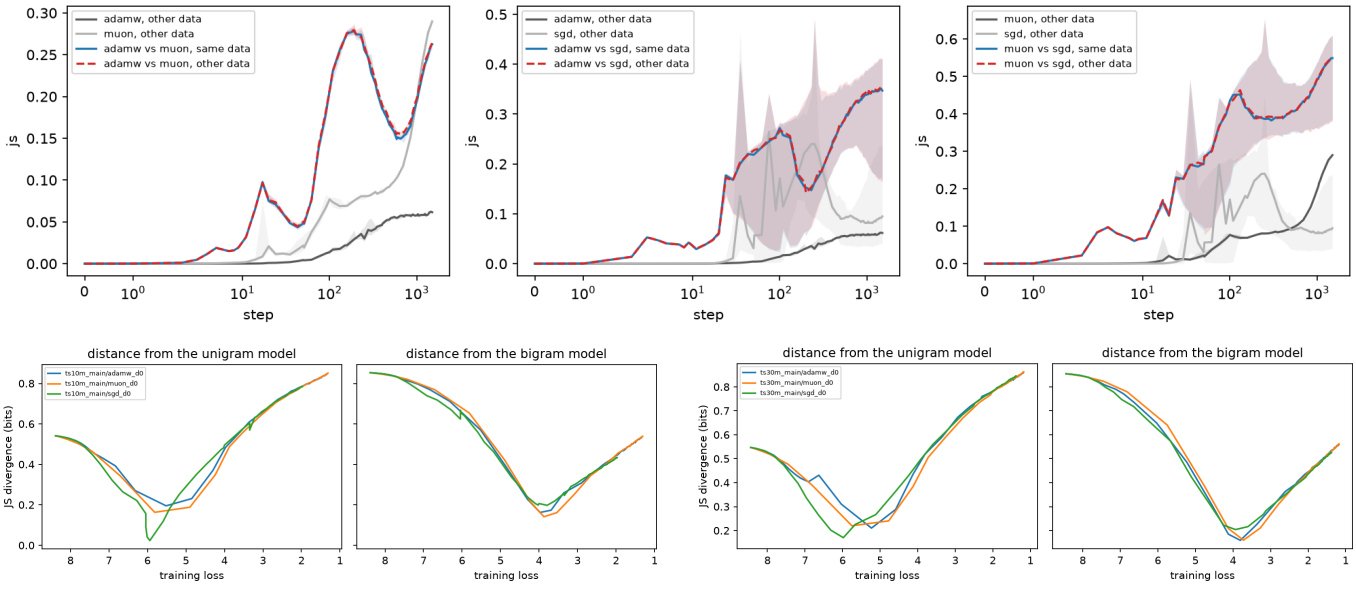


Figure 31: Top: optimizer against data order at equal steps on Shakespeare, as in Fig. 5 (top), for the three pairs. Bottom: distance of the $d = 0$ runs from the unigram and bigram models on TinyStories-10M (left) and TinyStories-30M (right), as in Fig. 5 (bottom).

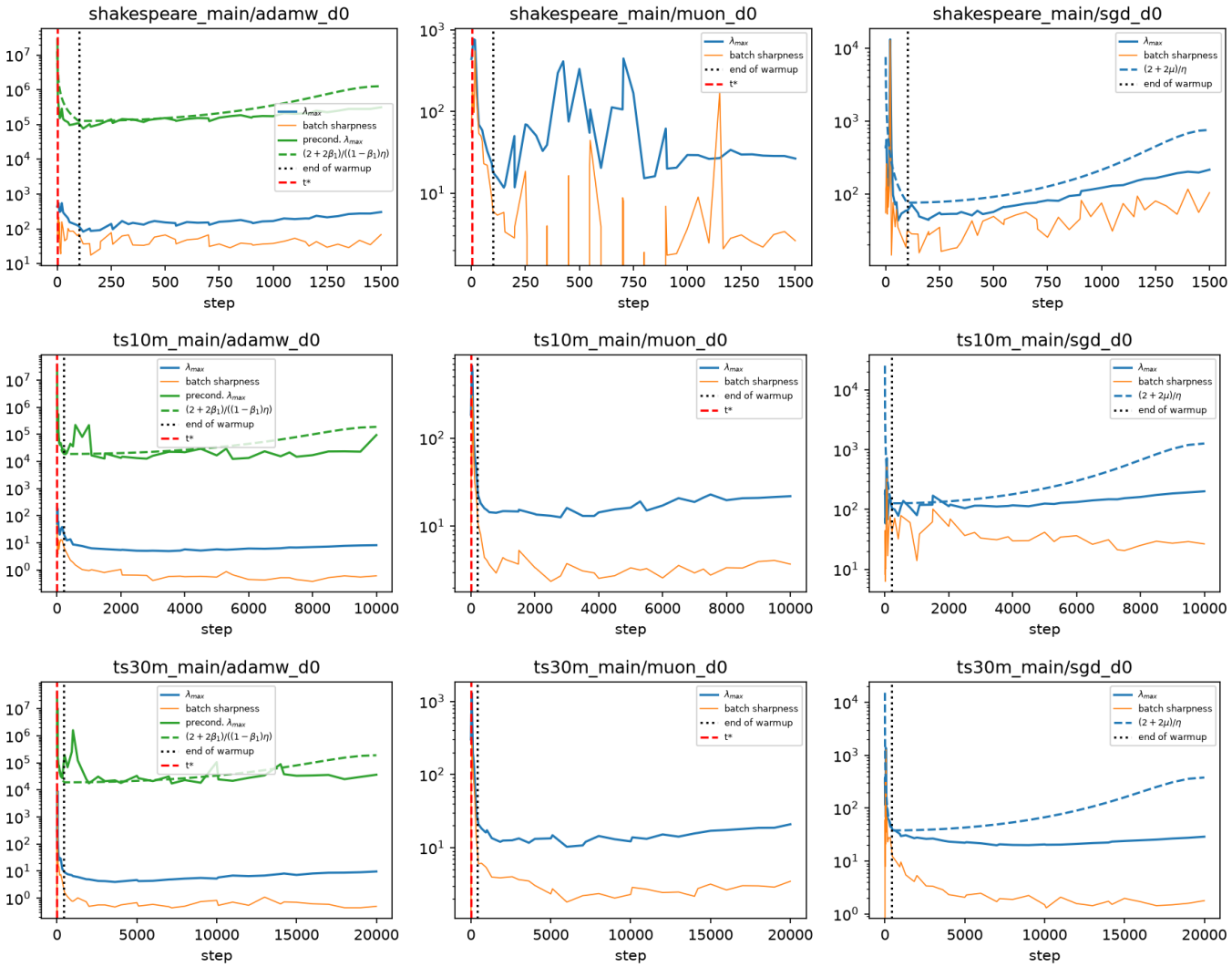


Figure 32: Sharpness of the $d = 0$ runs on Shakespeare (top), TinyStories-10M and TinyStories-30M (bottom), log scale: $\lambda_{\max}$, batch sharpness, for AdamW the preconditioned sharpness, and the thresholds $(2 + 2\mu)/\eta_s$ (SGD) and $(2 + 2\beta_1)/((1 - \beta_1)\eta_s)$ (AdamW, dashed); end of warmup dotted, first exceedance t^*_X of the AdamW–Muon pair red dashed.

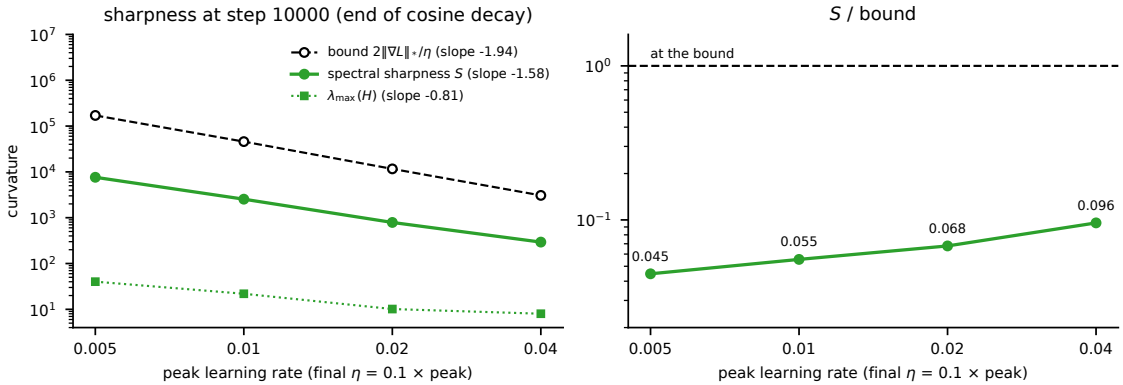


Figure 33: Muon's spectral-norm sharpness† S against its bound $2\sum_l c_l\|\nabla_l L\|_*/\eta$ at the last step (10,000, where η is 10% of the peak) of the TinyStories-10M sweep with sharpness measurements, against the peak learning rate; left also $\lambda_{\max}(H)$, with the log-log slopes.

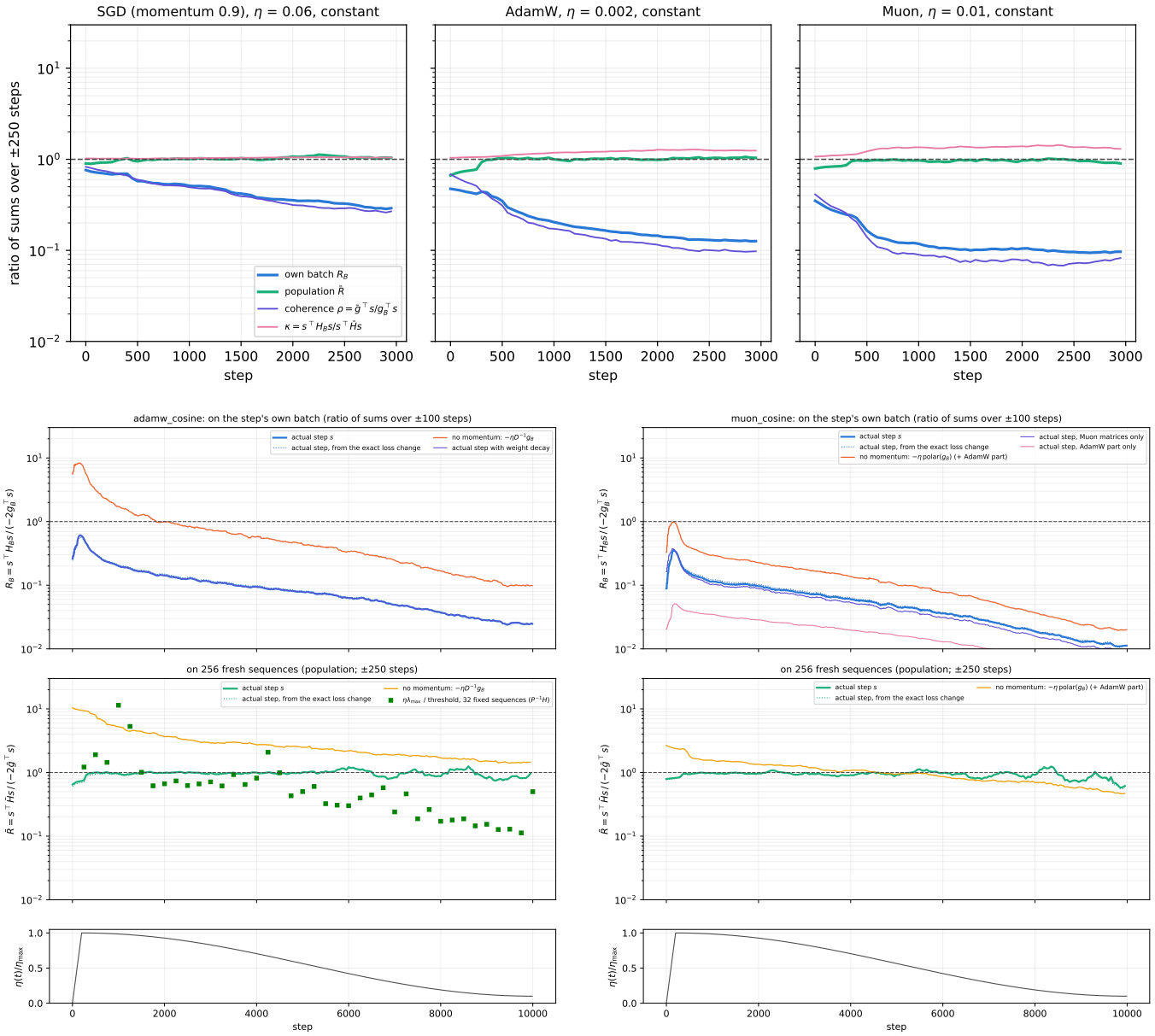


Figure 34: Stochastic edge of stability† on TinyStories-10M (Section 5.5). Top: at the tuned rate and a constant schedule, the own-batch ratio R_B, the population ratio $\bar{R}$ and the two factors between them, $R_B = \bar{R}\,\kappa\,\rho$ (ratio of sums over ±250 steps). Bottom: AdamW (left) and Muon (right) with warmup and cosine decay over 10,000 steps; ratios on the step's own batch (top panel) and on 256 fresh sequences (middle), $\eta(t)$ (bottom). Hessian-vector products in float32 on the step's batch every 10 steps (+ 60 log-spaced steps) and on fresh sequences every 50; $\eta\lambda_{\max}$ on 32 fixed sequences every 500. Per-run pages for all 14 runs, Muon's two parts and the step with weight decay are in `figures/stochastic_eos.pdf`.

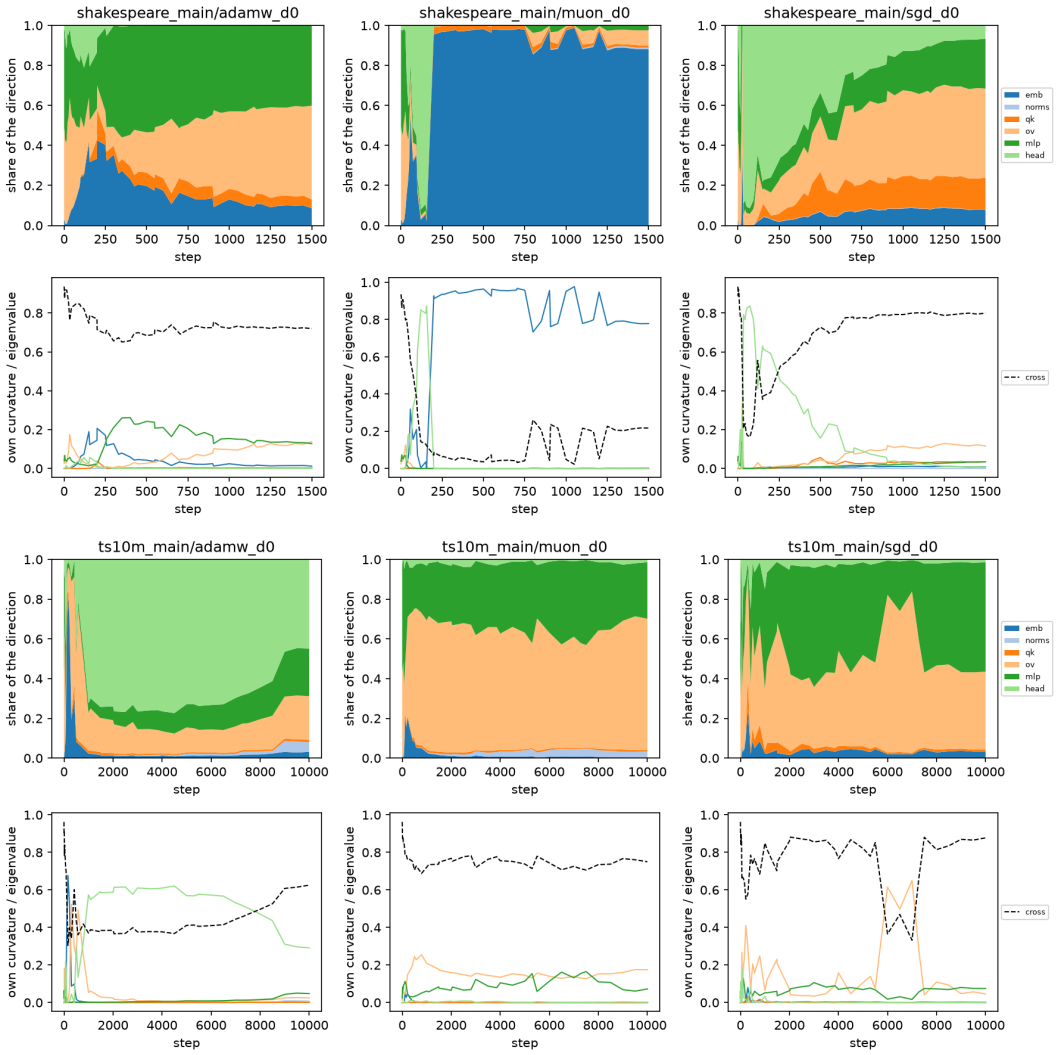


Figure 35: Anatomy of the top Hessian eigenvector as in Fig. 11, on Shakespeare (top two rows) and TinyStories-10M (bottom two rows).

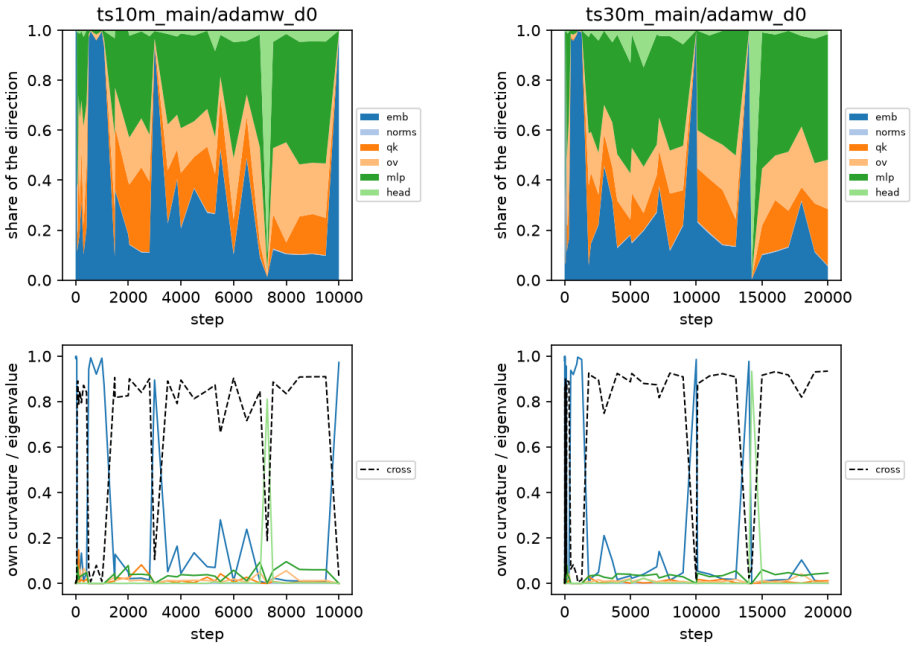


Figure 36: Anatomy of the top eigenvector of AdamW's preconditioned operator $P^{-1/2}HP^{-1/2}$ on TinyStories-10M (left) and TinyStories-30M (right), as in Fig. 11.

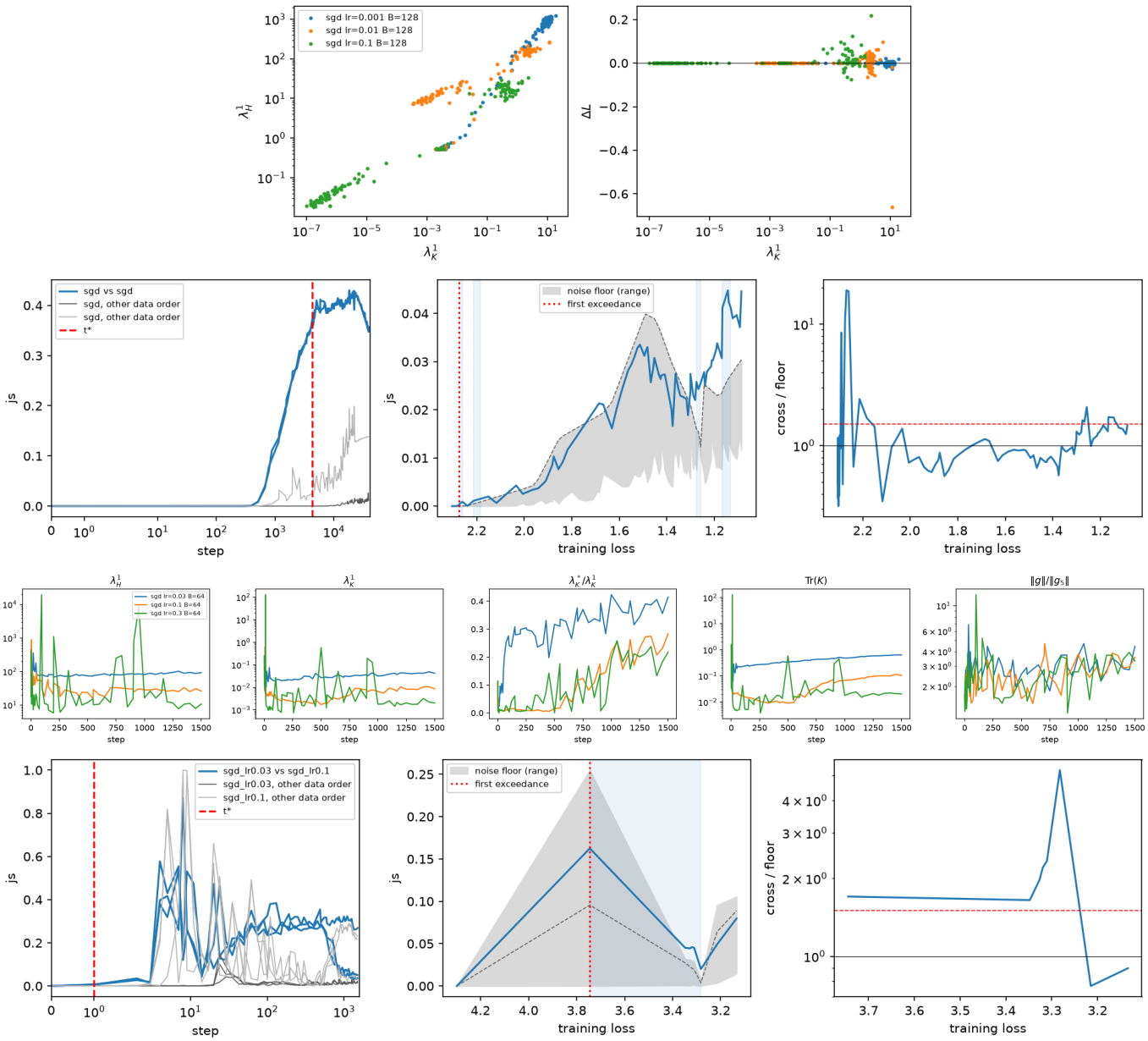


Figure 37: Break-even analysis, continued. First row: CIFAR-10, λ_H^1 (left) and ΔL (right) against λ_K^1. Second row: CIFAR-10 divergence figure for $\eta = 0.001$ against 0.01. Third and fourth rows: Shakespeare with plain SGD at $\eta \in \{0.03, 0.1, 0.3\}$, the quantities of Fig. 16 (top) and the divergence figure for 0.03 against 0.1.

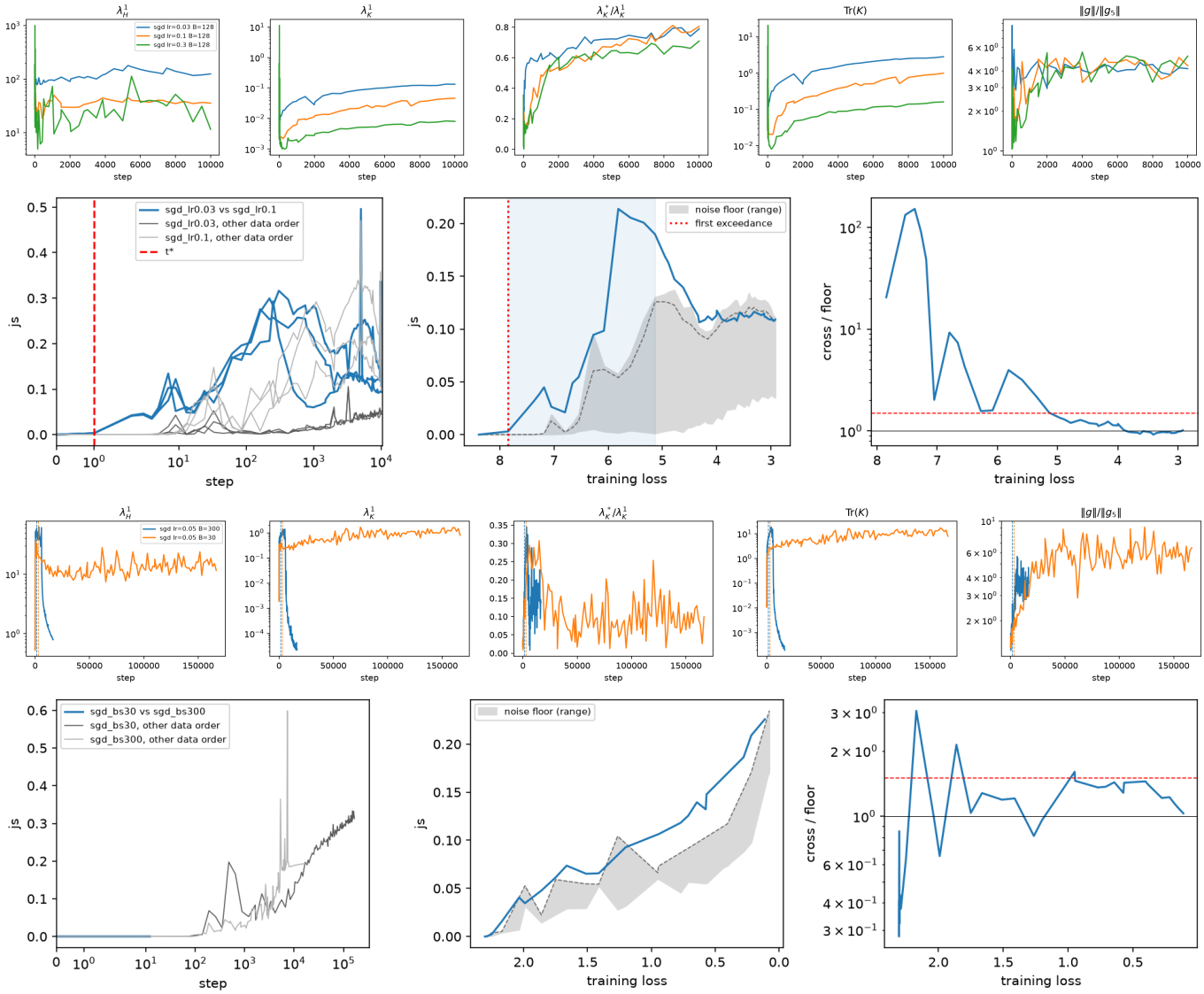


Figure 38: Break-even analysis, continued. First two rows: TinyStories-10M with plain SGD at $\eta \in \{0.03, 0.1, 0.3\}$, the quantities of Fig. 16 and the divergence figure for 0.03 against 0.1. Last two rows: CIFAR-10 at $\eta = 0.05$ with batch sizes 30 and 300.

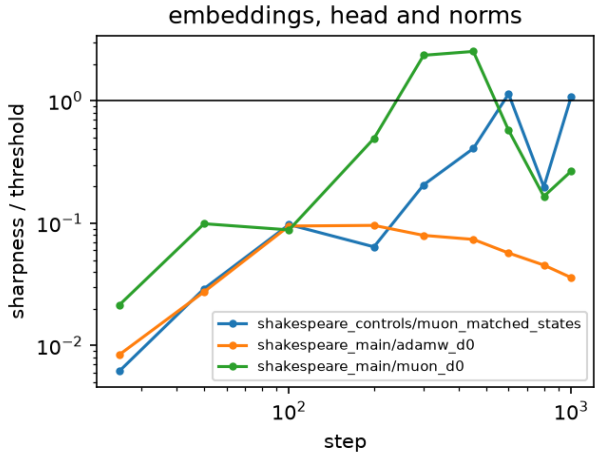


Figure 39: Sharpness of the embeddings, head and gains over their stability threshold (Section 4.4) on Shakespeare, at the fork checkpoints of AdamW, Muon, and Muon with its auxiliary AdamW matched to AdamW (learning rate $3 \cdot 10^{-4}$, betas 0.9/0.99, weight decay 0.1).

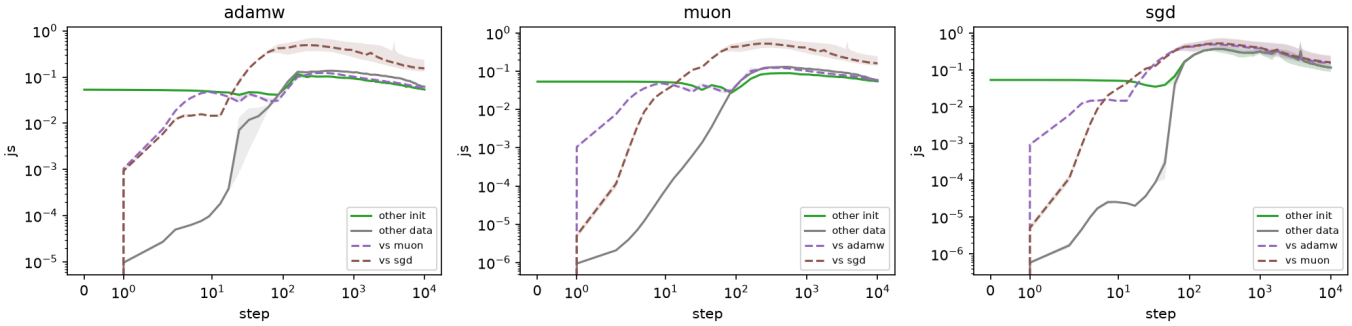


Figure 40: Second initialization on TinyStories-10M, as in Fig. 22 (top).

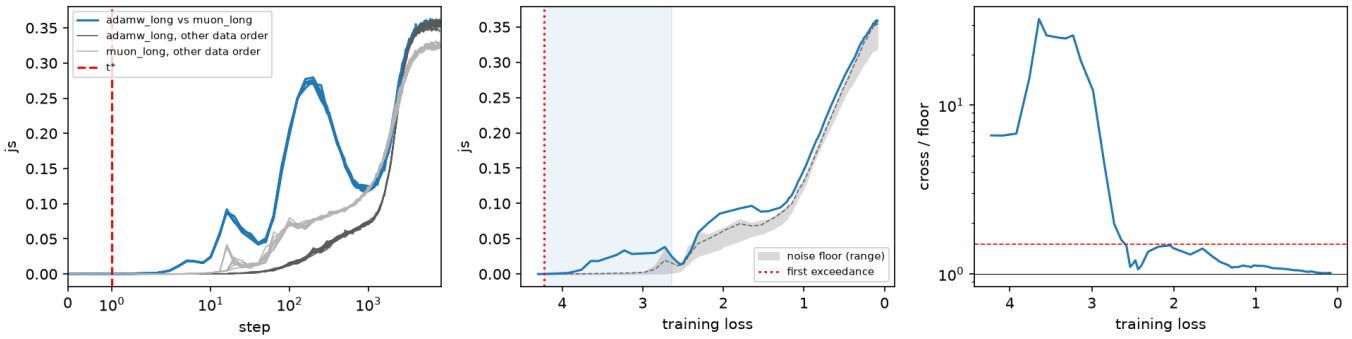


Figure 41: AdamW against Muon on Shakespeare trained for 8000 steps into memorization, as in Fig. 2.

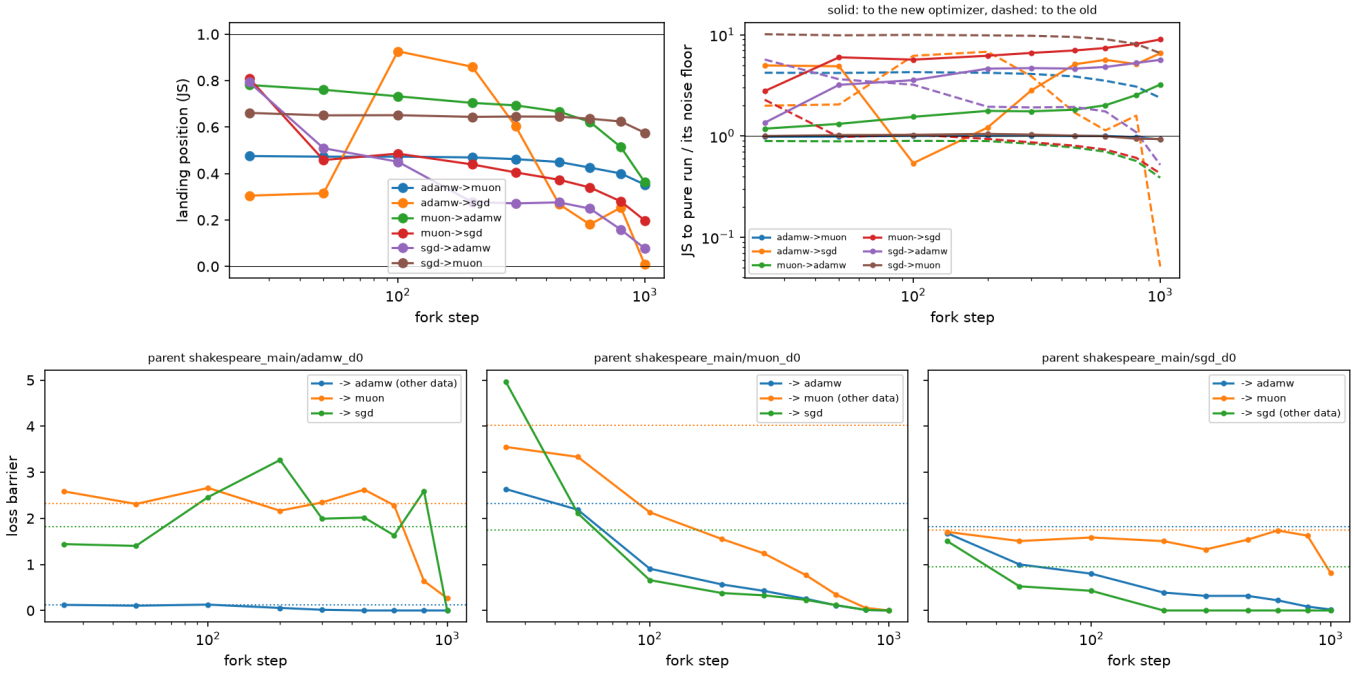


Figure 42: Swaps (top) and mode connectivity (bottom) on Shakespeare, as in Fig. 9.

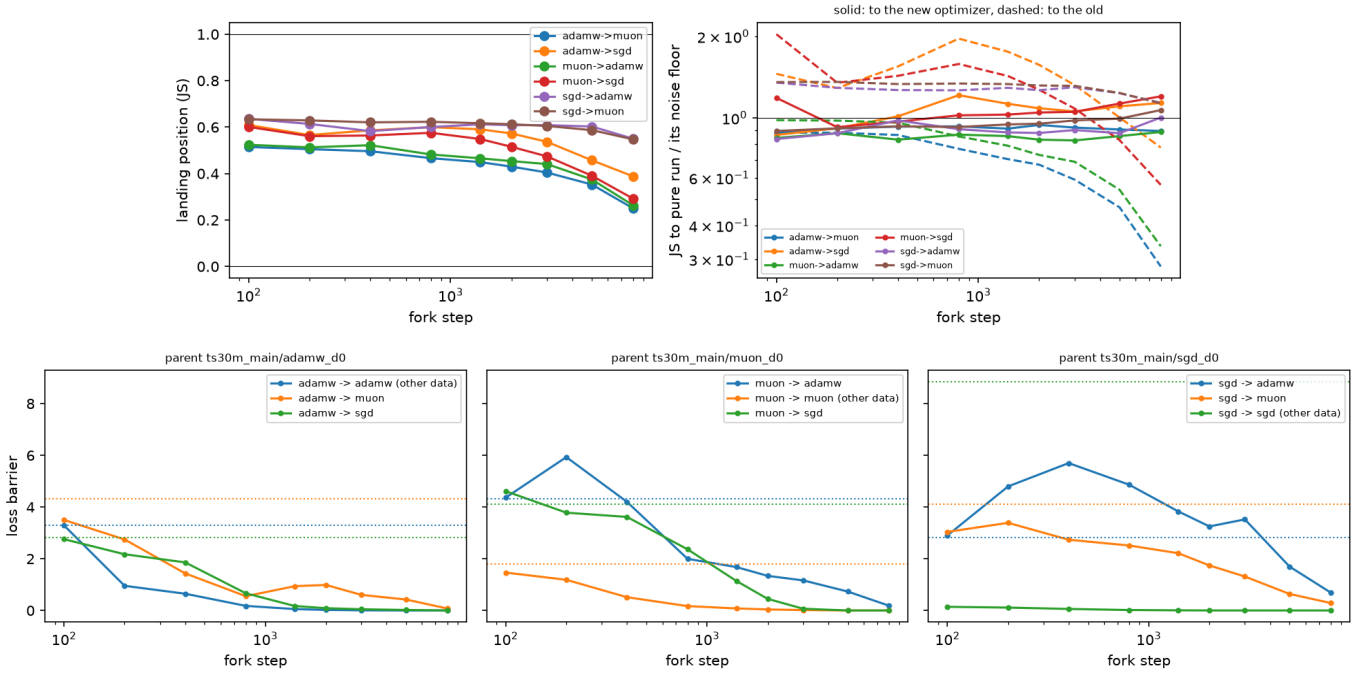


Figure 43: Swaps (top) and mode connectivity (bottom) on TinyStories-30M, as in Fig. 9.

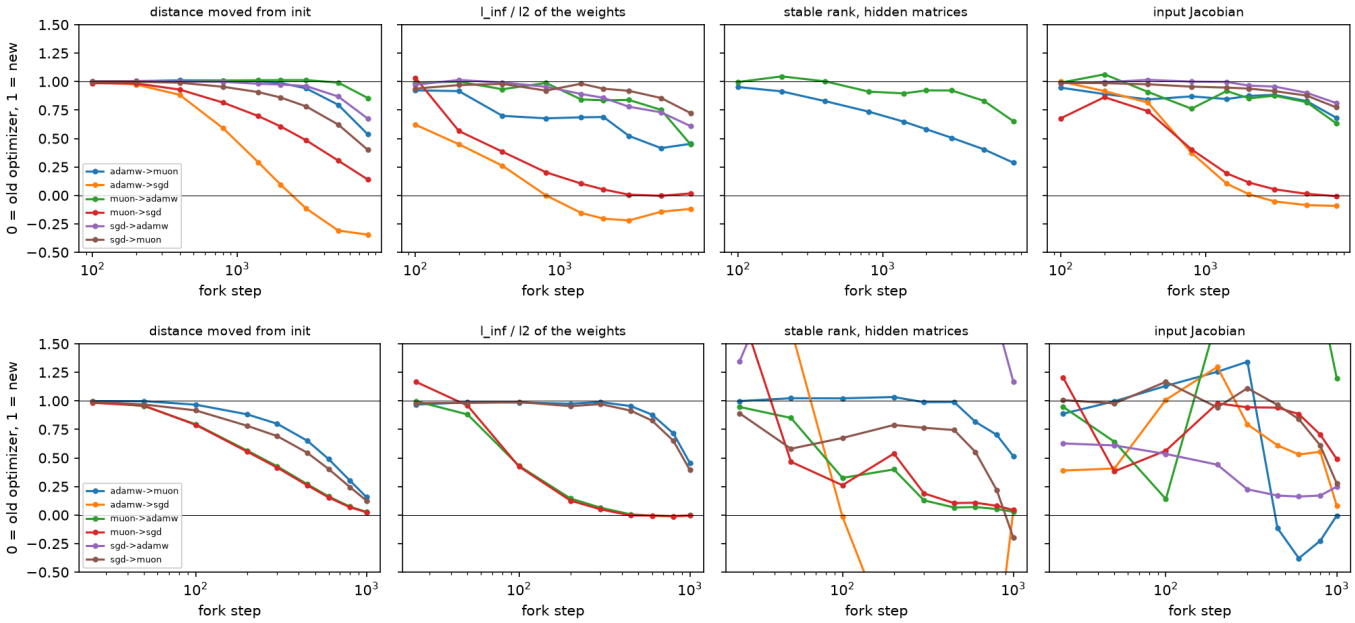


Figure 44: Final metrics of the TinyStories-30M (top) and Shakespeare (bottom) forks between pure A (0) and pure B (1), as in Fig. 21. The stable rank of SGD varies too much over data orders for the pairs with SGD to show on TinyStories-30M; on Shakespeare SGD's metrics vary widely in general, so most pairs with SGD drop out or are noisy.

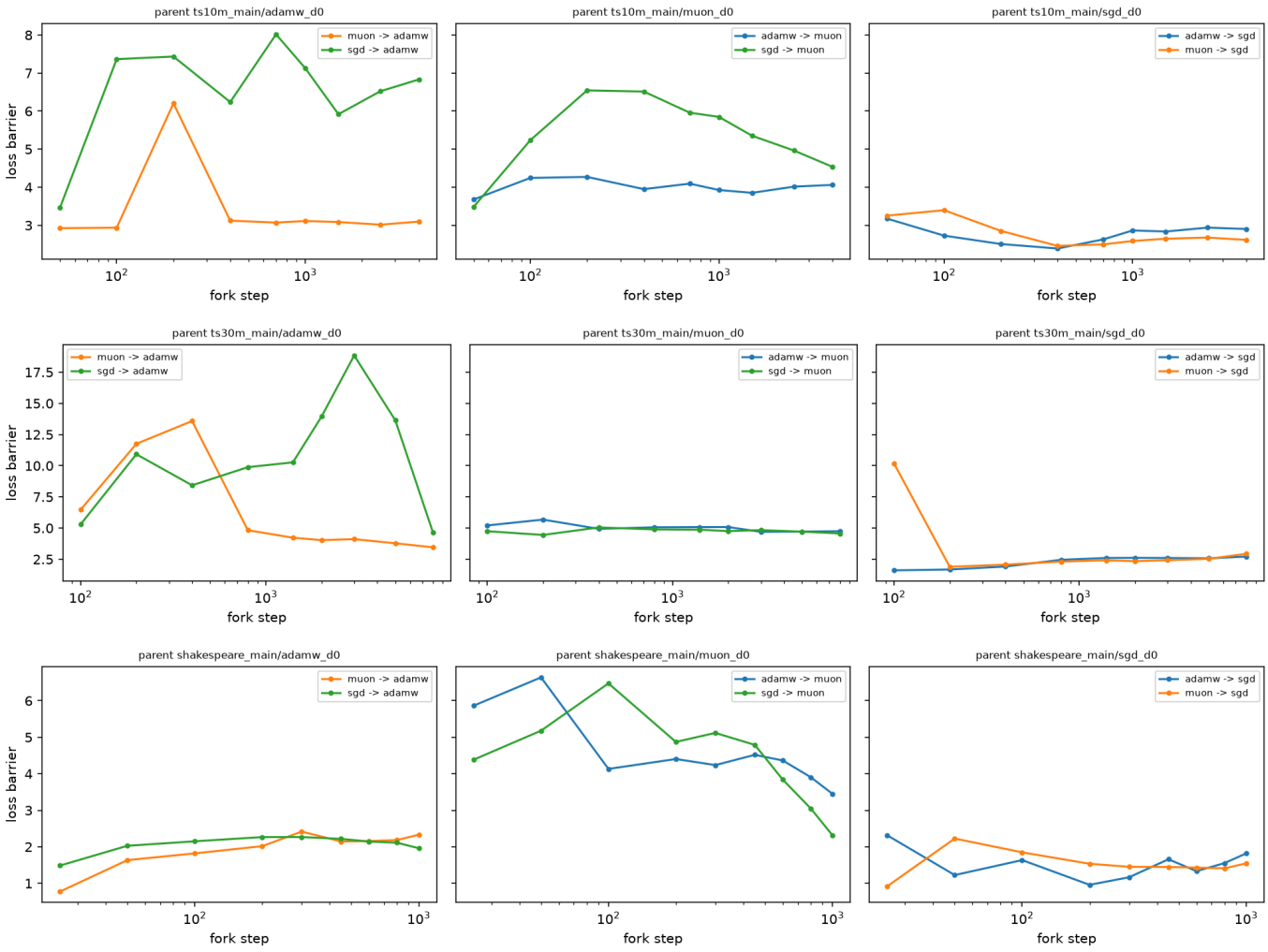


Figure 45: Barrier between the final weights of each pure $d = 0$ run and of every fork into its optimizer, against the fork step, on TinyStories-10M (top), TinyStories-30M (middle) and Shakespeare (bottom).

E Curvature-balanced learning rates†

Shares. Let $\hat{H} = \eta_t M^{1/2} A M^{1/2}$ as in Section 5.8, with top eigenpair (λ, v), $\|v\| = 1$, and E_a the projection on the coordinates of layer a. Since $\partial M^{1/2}/\partial \log m_a = \frac{1}{2} E_a M^{1/2}$, first-order perturbation gives $\partial\lambda/\partial \log m_a = \frac{1}{2} v^\top (E_a \hat{H} + \hat{H} E_a) v = \lambda \|v_a\|^2$: the share $\omega_a = \|v_a\|^2$ is the elasticity of λ in m_a, and $\sum_a \omega_a = 1$. Since $\hat{H} v = \lambda v$, $\omega_a = \sum_b v_a^\top \hat{H} v_b / \lambda$, the row sum of layer a in the heatmap of $\hat{H}$, which by symmetry is its diagonal cell plus half of the off-diagonal cells in its row and column. Maximizing $\sum_a \log m_a$ at fixed $\log \lambda$ gives, with a Lagrange multiplier ν, $1 = \nu\, \omega_a$ for every balanced layer: equal shares. Scaling each layer so that its own diagonal block is equally curved instead is block-Jacobi scaling, within a factor depending only on the number of blocks of the best block-diagonal scaling (Sluis, 1969; Demmel, 2023); the own-block variant of v2 (below) is its analogue along the step.

Updates. At every refresh (every 20 steps) the shares are smoothed by an exponential moving average with weight 0.2, $\bar{\omega}_a$. After warmup, $\log m_a \leftarrow \log m_a - 0.3\,(\bar{\omega}_a - \mathrm{mean}_b\, \bar{\omega}_b)$ over the balanced layers, then $\log m$ is recentred to mean 0 and clipped to $[\log 0.1, \log 10]$. Variant B also sets $\log \kappa \leftarrow \log \kappa - 0.2 \log\big(\lambda_{\max}(\hat{H})/(\tau\Theta)\big)$, with Θ the threshold and τ the target (B0.9: $\tau = 0.9$), and every κm_a is clipped to $[0.1, 10]$. The top eigenpairs of $\hat{H}$ are tracked by a warm-started block iteration: 3 vectors (AdamW A-fe: 4) and 2 (AdamW A-fe: 3) Rayleigh–Ritz steps on the vectors and their residuals per refresh; the shares are averaged over the Ritz vectors within 5% of the top, and a 20-step Lanczos check of $\lambda_{\max}(\hat{H})$ runs every 500 steps. For AdamW, P is frozen from the current second moments. "-fe" keeps the token and position embeddings at the schedule's rate while they stay in the operator, and the other layers equalize their shares among themselves: the 32 sequences of the Hessian batch contain only part of the 4096-token vocabulary, so the curvature of the embedding rows of the other tokens is not seen, and SGD B0.9 blew up at step 3020 with its embeddings at 10 times the rate. "-fresh" computes $\hat{H}$ on 32 fresh training sequences at every refresh instead of the fixed batch. Batch sharpness is not measured in these runs.

Muon. Muon's hidden matrices are balanced on the spectral-norm sharpness of Section 4.4, with c_l replaced by $m_a c_l$ for the matrices of block a. The stability ratio is $\rho = \eta_t S/(2\sum_l c_l \|\nabla_l L\|_*)$, 1 at the bound, and the share of block a is $\partial \log \rho/\partial \log m_a = 2\langle D_a, H[D]_a\rangle/\langle D, H[D]\rangle$ minus block a's part of $\sum_l c_l \|\nabla_l L\|_*$ (the shares sum to 1), with D the maximizer, tracked by 5 warm-started Frank–Wolfe steps per refresh and checked from scratch every 500 steps. The embeddings and the head, trained by the auxiliary AdamW, keep the schedule's rate.

v2. At every refresh, with s the update just taken without the weight decay, and g and H the gradient and Hessian before it on 256 fresh training sequences, layer a has $R_a = s_a^\top (Hs)_a/(-2 g_a^\top s_a)$; summed over the layers, numerators and denominators give the R of the whole step (Section 5.5). Numerators and denominators are smoothed by a moving average (weight 0.2), and $\bar{R}_a$ is their ratio. After warmup, $\log m_a \leftarrow \log m_a - 0.3\,(\log \bar{R}_a - \mathrm{mean}_b \log \bar{R}_b)$, with the deviation clipped to $[\log 0.1, \log 10]$ (a layer whose step does not lower the loss to first order counts as too fast); the geometric mean of m stays 1, every m_a lies in $[0.1, 10]$, and there is no global scale. Every layer is balanced, including those of Muon's auxiliary AdamW. The own-block variant uses $s_a^\top H_{aa} s_a$ as numerator.

Runs. All runs start from the initialization and use the rates and data orders of the main runs, which are the baselines. On TinyStories-10M every variant ran on data order 0 and the best ones also on orders 1 and 2; TinyStories-30M and Shakespeare repeat the settings that did best on TinyStories-10M. The speedup is the step of the baseline's best validation loss over the first evaluated step at which the balanced run reaches it, on the same data order; the best validation loss is the final one except on Shakespeare. Muon A on TinyStories-30M was cancelled at step 6500. Figures 46 to 50 show the runs, and Figs. 51 to 56 the losses and the table of every group of runs, and the balancer trace of the SGD runs on TinyStories-10M.

Failures. Several B runs blew up: on TinyStories-10M B0.9 at step 3020, B0.9-fe at step 5470 and B1-fe, and on TinyStories-30M B1 on data order 1 at step 7510. In each, $\lambda_{\max}(\hat{H})$ on the 32 fixed sequences sat just under the target, so κ kept rising until an instability those sequences did not show. Computing $\hat{H}$ on fresh sequences fixes target 0.9 (B0.9-fe-fresh) but not 1 (B1-fe-fresh blew up at step 1240). With targets 0.5 and 0.7, κ collapses to its floor as the sharpness rises (B0.5 ends at 3.91 on TinyStories-10M). For AdamW, B pushed κ down to 0.1 below the target and up to 5–10 times above it, both worse than the baseline.

SGD at a higher rate. Plain SGD at 0.12, four times the baseline rate, reaches 1.552, 2.801 and 1.554 on data orders 0–2 of TinyStories-10M (on order 1 it spikes in warmup and never catches up), and A-fe at 0.12 reaches 1.567, 2.516 and 1.569. Where the higher rate is stable it does as well as B1 (1.682, 1.553, 1.577) or better, and redistributing it across layers adds nothing (Fig. 46).

Shares after balancing. Figure 47 shows the heatmaps of Fig. 13 for the balanced runs, with their logged multipliers, over steps 5000–10,000. In $\hat{H}$ the shares are even (SGD B1: 12–15% per block and the head, 3% in the embeddings; Muon A: 9–24% per block and the head, 0% in the embeddings). In the plain Hessian H of the same runs the top eigenvector lies in block 1, the block with the lowest rate (91% for SGD B1, 74% for Muon A): balancing lowers the rate where the curvature is, rather than spreading the curvature. AdamW A-fe kept no optimizer state for P; with its final multipliers fixed (data order 2), $(MP^{-1})^{1/2}H(MP^{-1})^{1/2}$ has 8–19% per block, 7% in the embeddings and 3% in the head.

Fixed multipliers. Trained from the start with the final multipliers of the A and A-fe runs of data order 0 and no balancing (Fig. 48), AdamW reaches 1.370, 1.373 and 1.384 on data orders 0–2 against 1.339, 1.346 and 1.345 for the baseline, and Muon 1.353, 1.358 and 1.360 against 1.305, 1.307 and 1.314. SGD diverges at steps 180, 460 and 260; with the multipliers ramped in from 1 between steps 200 and 1200, after warmup, it diverges at step 1270 on order 0 and reaches 1.688 and 1.954 on orders 1 and 2 (baseline 1.766, 1.903).

v2. R_a settles at 0.8–1.2 in every layer from step 100 on, whatever the layer's rate (Fig. 50, right): AdamW's rates drift 100 times apart, into the caps, and its loss falls behind (Table 3). With the own block, R_a settles at 0.25–0.3, so the curvature between layers is about 70% of the numerator.

Related work. Layer-wise rates are common: LARS and LAMB scale each layer's step by the ratio of its weight and update norms (You et al., 2017; You et al., 2020); fine-tuning lowers the rate towards the input (Howard and Ruder, 2018; Clark et al., 2020); μP sets per-layer rates by width (Yang et al., 2021); Adam-mini shares one second moment per Hessian block (Zhang et al., 2025), motivated by the block structure of transformer Hessians (Zhang et al., 2024b); Wang et al. (2025) set blockwise rates from the sharpness of each block type; GLM-130B shrinks the gradient of the embeddings for stability (Zeng et al., 2023); and Kalra and Barkeshli (2024) relate warmup to the sharpness. Our rule differs in balancing the top eigenvector of the preconditioned operator, or the curvature along the actual step, layer by layer and online.

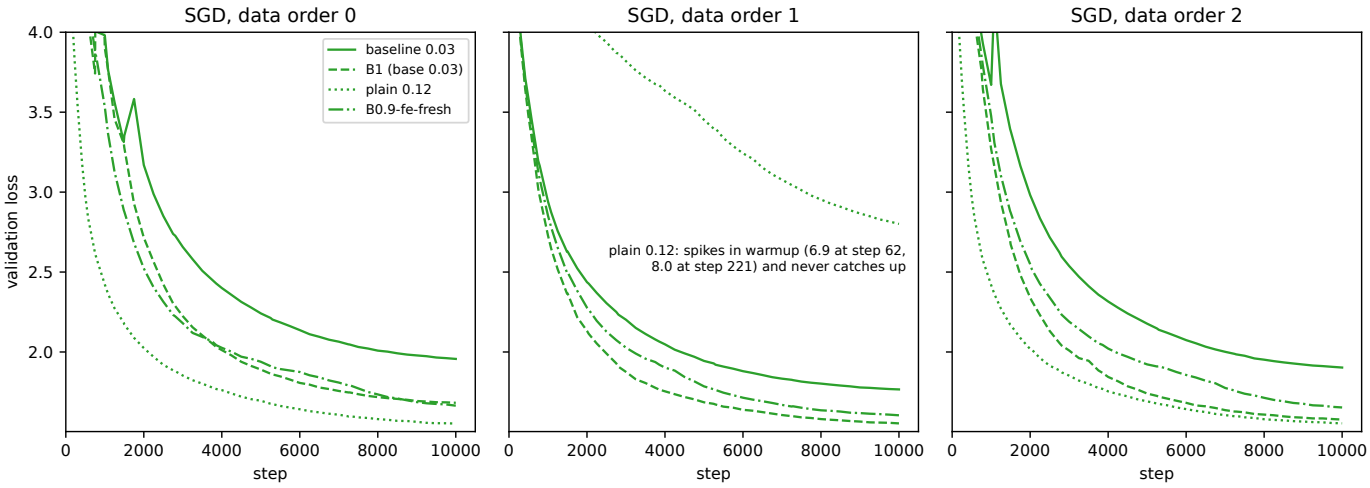


Figure 46: SGD on TinyStories-10M, one panel per data order: the baseline at 0.03, B1, B0.9-fe-fresh and plain SGD at 0.12. On order 1 plain SGD at 0.12 spikes in warmup (to 6.9 at step 62 and 8.0 at step 221) and never catches up.

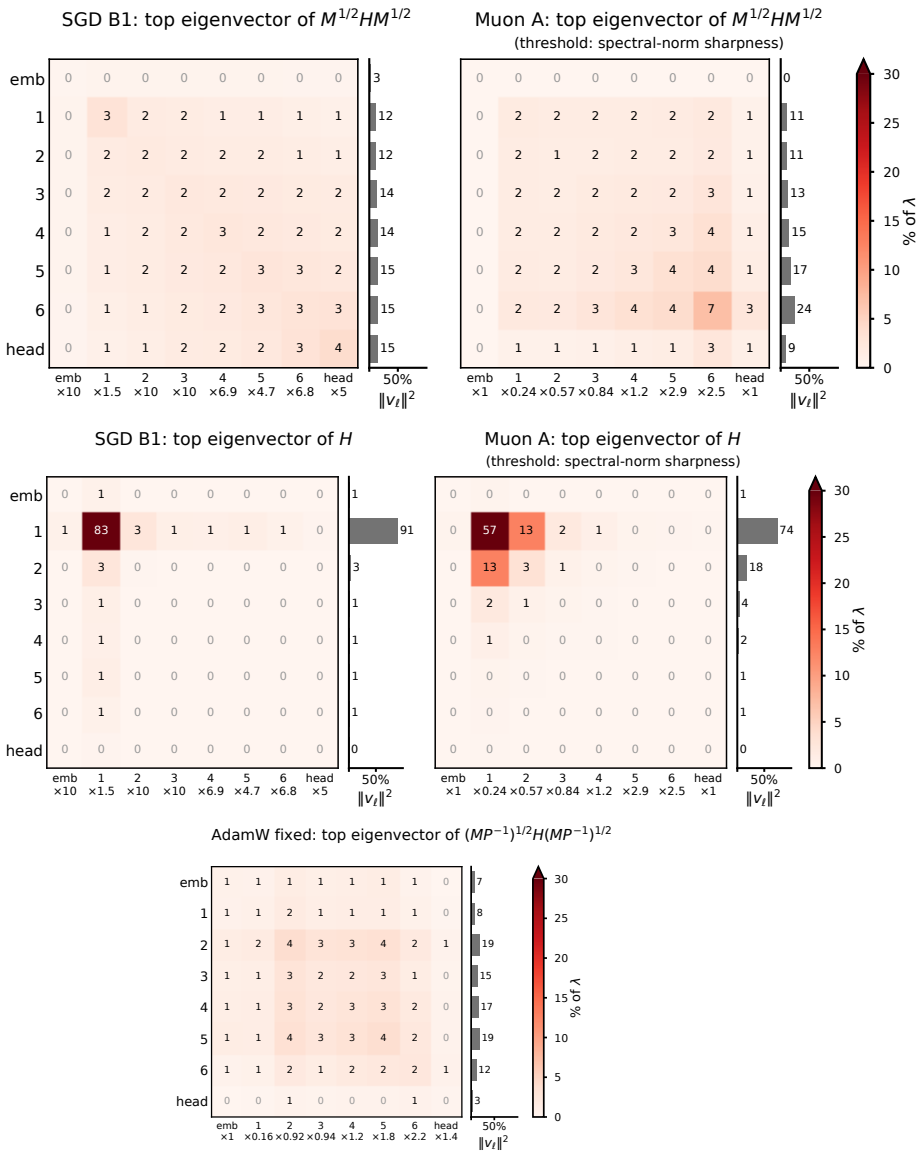
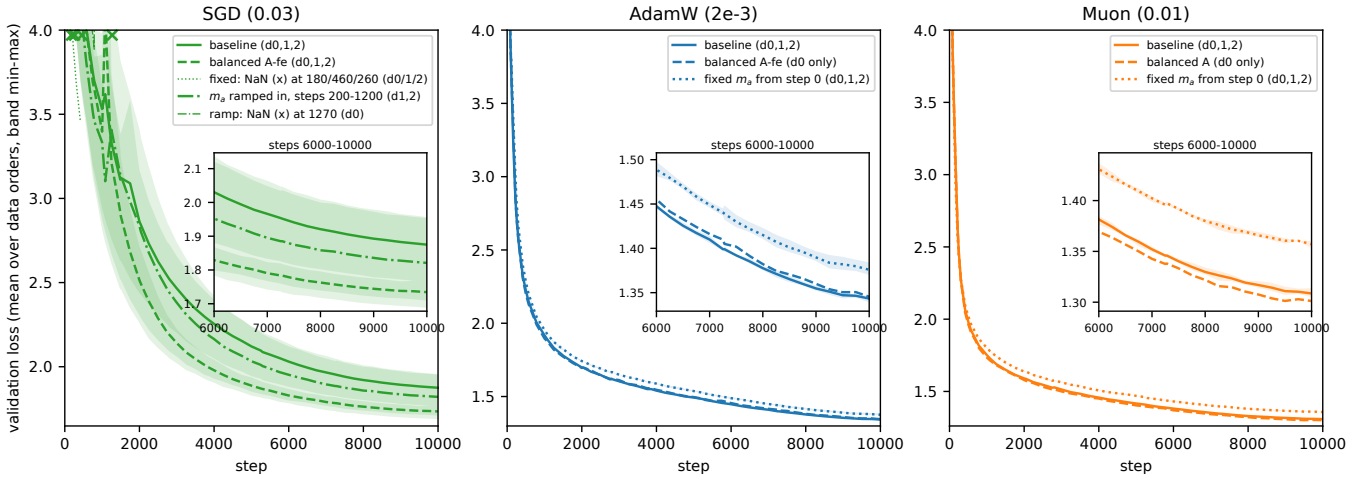


Figure 47: Layer heatmaps as in Fig. 13 for the balanced runs on TinyStories-10M, mean over the checkpoints at steps 5000–10,000, with each layer's rate multiplier under its label. Top: of the balanced operator $M^{1/2}HM^{1/2}$ for SGD B1 and Muon A. Middle: of H for the same runs. Bottom: AdamW trained with fixed multipliers (the final ones of A-fe), of $(MP^{-1})^{1/2}H(MP^{-1})^{1/2}$ at steps 5000, 7500 and 10,000 (data order 2).

TinyStories-10M: fixed final multipliers (dotted; no balancing, m_a of the balanced run on d0, from step 0; SGD also ramped in after warmup, dash-dot) against the balanced run (dashed) and the baseline (solid)



	emb	1	2	3	4	5	6	head
m_a	1*	0.13	0.92	0.84	0.58	0.95	2.71	6.51

	emb	1	2	3	4	5	6	head
m_a	1*	0.16	0.92	0.94	1.24	1.84	2.25	1.42

	emb	1	2	3	4	5	6	head
m_a	1*	0.25	0.56	0.84	1.20	2.92	2.47	1*

* kept at the schedule's rate: the embedding as in the -fe runs; for Muon the embedding, head and norms (its auxiliary AdamW). SGD ramp: m_a^f, f = 0 to step 200, linear to 1 at step 1200.

final validation loss per data order; speedup = 10000 / first evaluated step at which the run reaches the baseline's final loss (same order)

order	baseline	A-fe	speedup	fixed	speedup	ramp	speedup
d0	1.957	1.766	2.11x	NaN @180	-	NaN @1270	-
d1	1.766	1.709	1.38x	NaN @460	-	1.688	1.54x
d2	1.903	1.727	2.11x	NaN @260	-	1.954	never

order	baseline	A-fe	speedup	fixed	speedup
d0	1.339	1.345	never	1.370	never
d1	1.346	-	-	1.373	never
d2	1.345	-	-	1.384	never

order	baseline	A	speedup	fixed	speedup
d0	1.305	1.301	1.08x	1.353	never
d1	1.307	-	-	1.358	never
d2	1.314	-	-	1.360	never

Figure 48: TinyStories-10M trained with the final multipliers of the balanced runs fixed from step 0 (dotted; for SGD also ramped in after warmup, dash-dot), against the balanced runs (dashed) and the baseline (solid); insets: steps 6000–10,000. Tables: the multipliers and the final validation loss per data order.

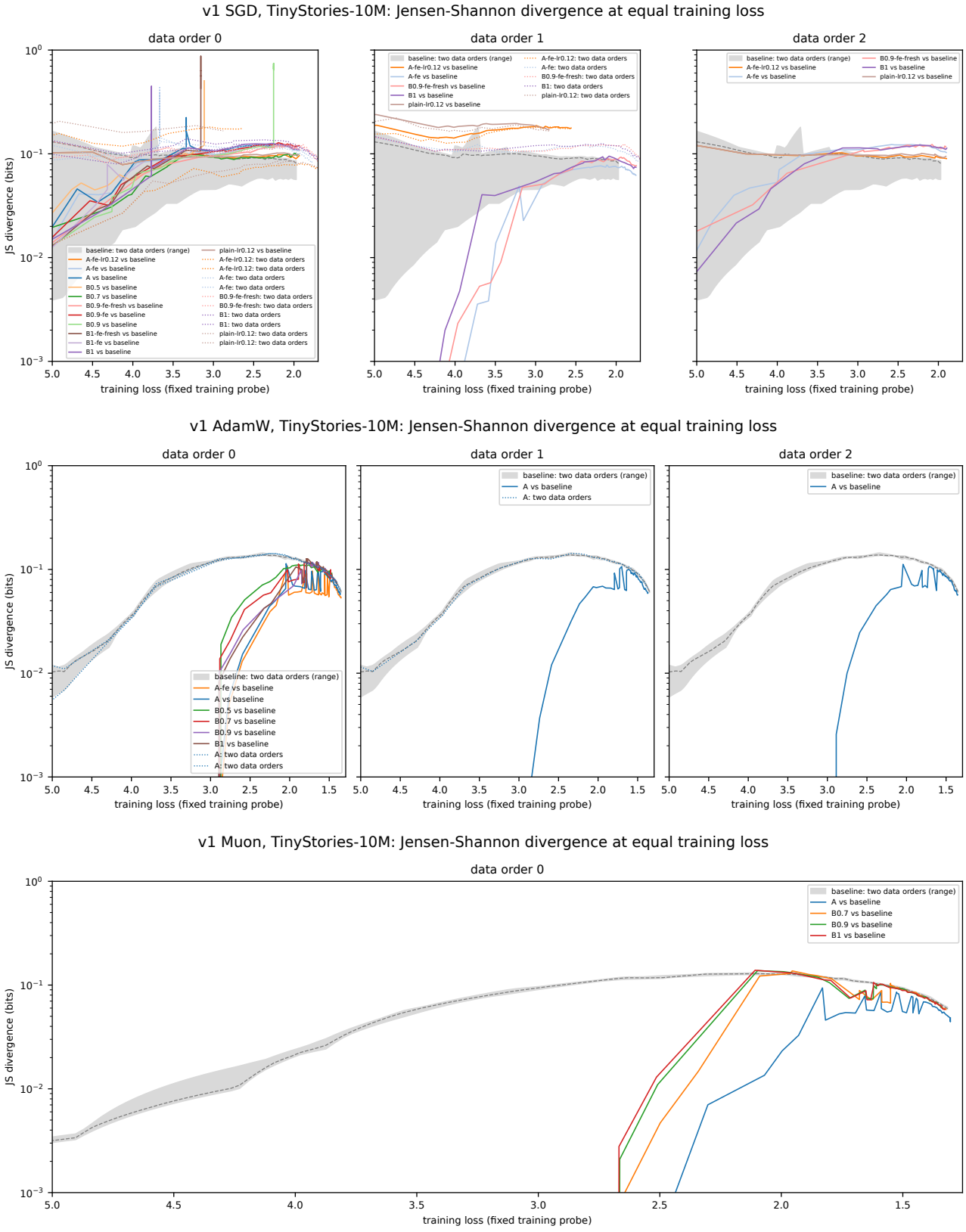


Figure 49: Jensen–Shannon divergence at equal training loss between each balanced run and the baseline on the same data order, against the baseline's data-order floor (grey band), on TinyStories-10M: SGD (top), AdamW and Muon (bottom).

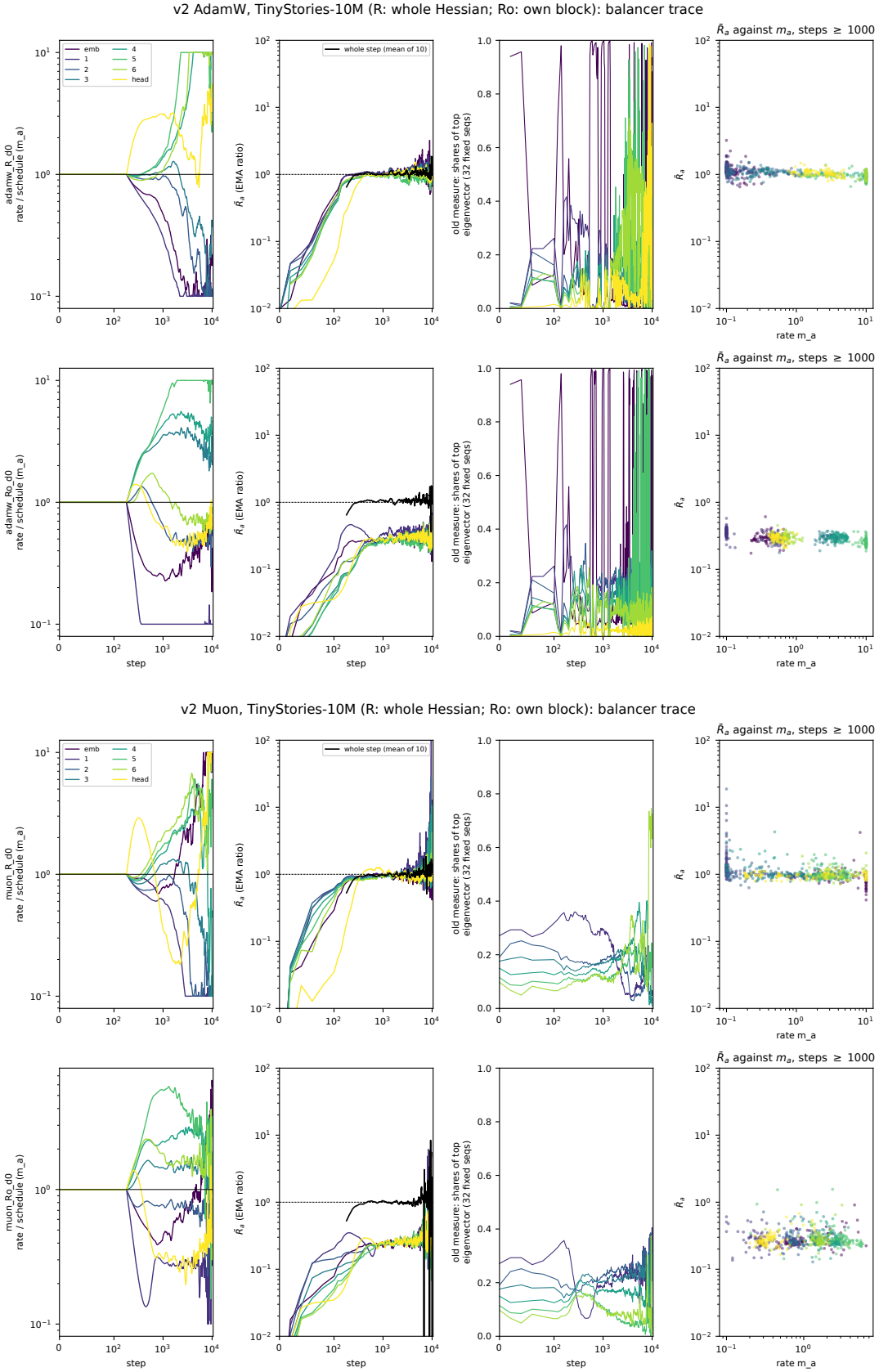
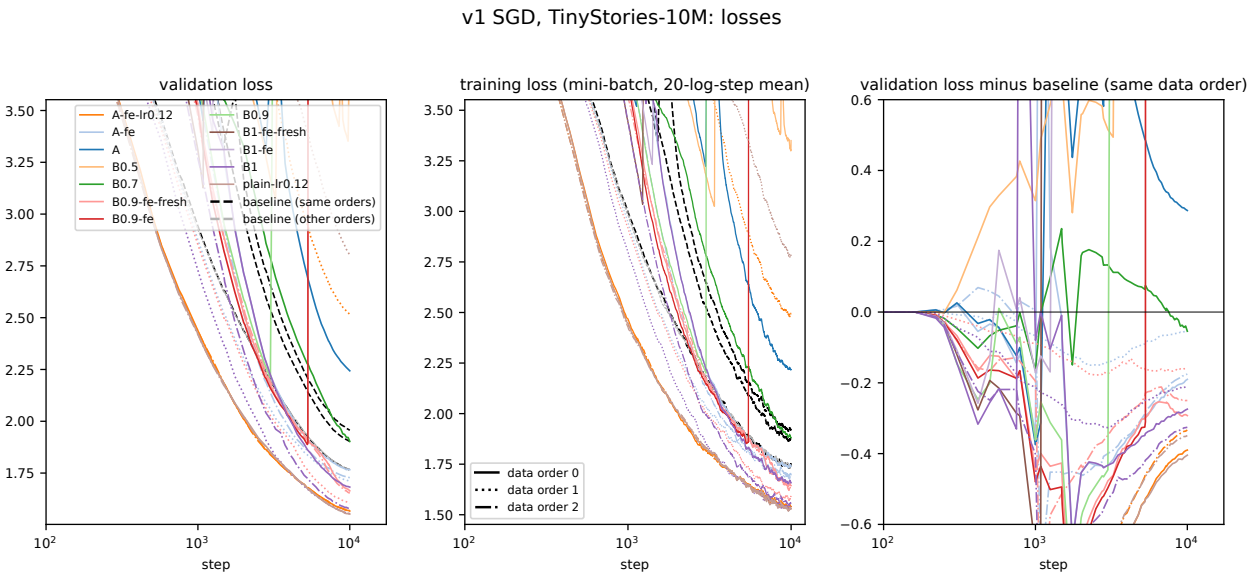


Figure 50: v2 on TinyStories-10M (data order 0), AdamW (top two rows) and Muon (bottom two rows), each with R_a (first row) and with the own block (second row): rate multipliers, smoothed R_a, the shares of v1 logged alongside, and R_a against the layer's multiplier after step 1000.



v1 SGD, TinyStories-10M: final and best validation loss

run	last step	final val	minus baseline	best val	minus baseline	reaches baseline best	speedup
sgd_A-fe-lr0.12_d0	10000	1.5666	-0.3900	1.5666	-0.3900	2250	4.44x
sgd_A-fe_d0	10000	1.7664	-0.1902	1.7663	-0.1903	4750	2.11x
sgd_A_d0	10000	2.2435	+0.2869	2.2435	+0.2869	-	-
sgd_B0.5_d0	10000	3.9069	+1.9503	3.0746	+1.1180	-	-
sgd_B0.7_d0	10000	1.9024	-0.0542	1.9024	-0.0542	9000	1.11x
sgd_B0.9-fe-fresh_d0	10000	1.6641	-0.2925	1.6641	-0.2925	5000	2.00x
sgd_B0.9-fe_d0	5500	38.2695	+36.3129	1.8878	-0.0688	4500	2.22x
sgd_B0.9_d0	10000	5.1156	+3.1590	2.2110	+0.2544	-	-
sgd_B1-fe-fresh_d0	10000	4.8535	+2.8969	3.1267	+1.1701	-	-
sgd_B1-fe_d0	10000	5.8562	+3.8996	3.3927	+1.4361	-	-
sgd_B1_d0	10000	1.6822	-0.2744	1.6822	-0.2744	4500	2.22x
sgd_plain-lr0.12_d0	10000	1.5522	-0.4044	1.5522	-0.4044	2500	4.00x
sgd_A-fe-lr0.12_d1	10000	2.5155	+0.7497	2.5155	+0.7497	-	-
sgd_A-fe_d1	10000	1.7094	-0.0565	1.7094	-0.0565	7250	1.38x
sgd_B0.9-fe-fresh_d1	10000	1.6040	-0.1619	1.6040	-0.1619	5298	1.89x
sgd_B1_d1	10000	1.5533	-0.2125	1.5533	-0.2125	3857	2.59x
sgd_plain-lr0.12_d1	10000	2.8009	+1.0351	2.8009	+1.0351	-	-
sgd_A-fe-lr0.12_d2	10000	1.5693	-0.3334	1.5693	-0.3334	2750	3.64x
sgd_A-fe_d2	10000	1.7274	-0.1753	1.7274	-0.1753	4750	2.11x
sgd_B0.9-fe-fresh_d2	10000	1.6530	-0.2497	1.6530	-0.2497	5500	1.82x
sgd_B1_d2	10000	1.5772	-0.3255	1.5772	-0.3255	3750	2.67x
sgd_plain-lr0.12_d2	10000	1.5537	-0.3490	1.5537	-0.3490	2750	3.64x

Figure 51: SGD on TinyStories-10M, all variants: validation loss, mini-batch training loss and validation loss minus the baseline's (top), and final and best validation loss of every run (bottom).

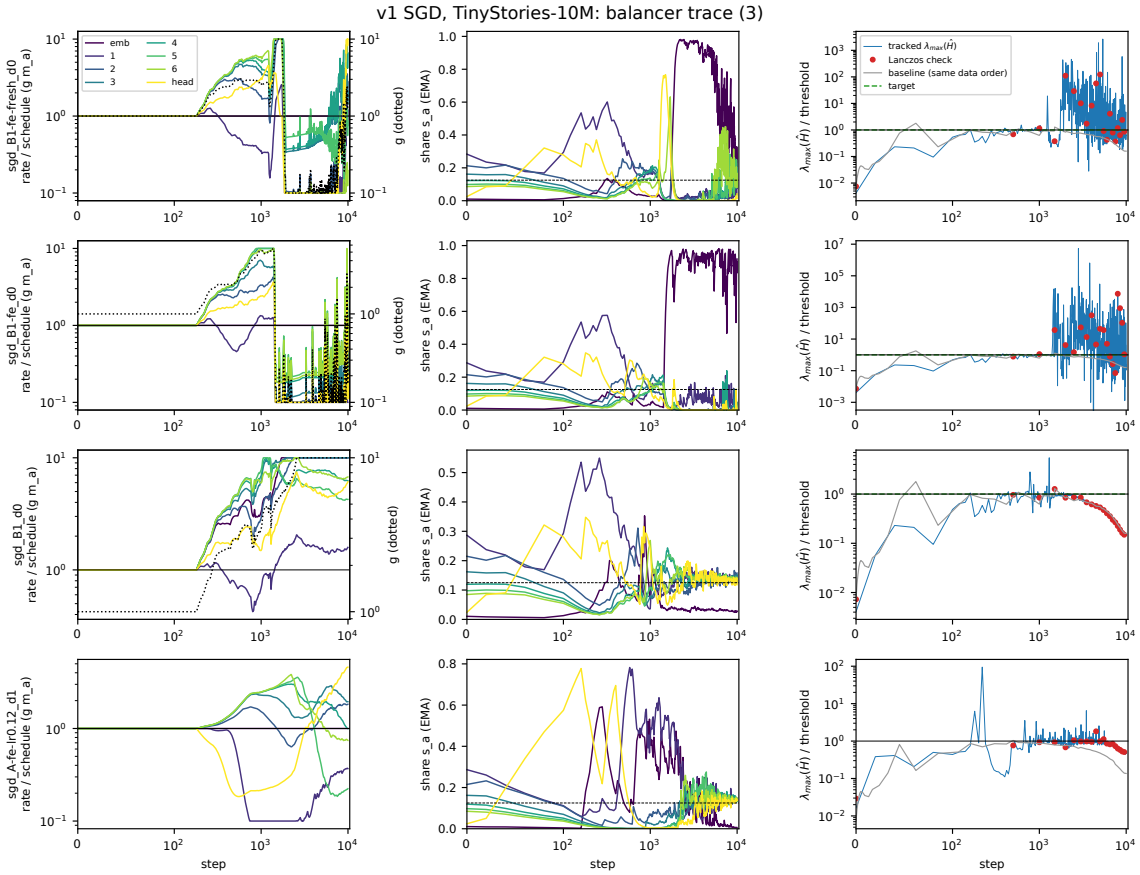
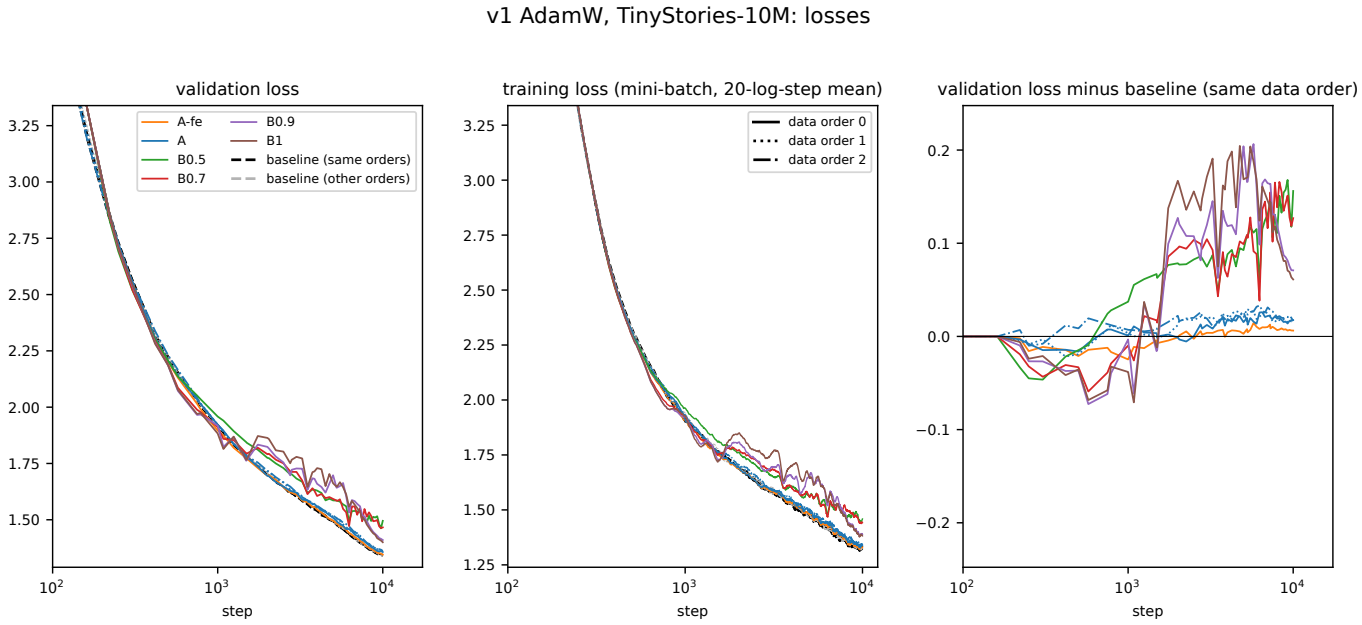
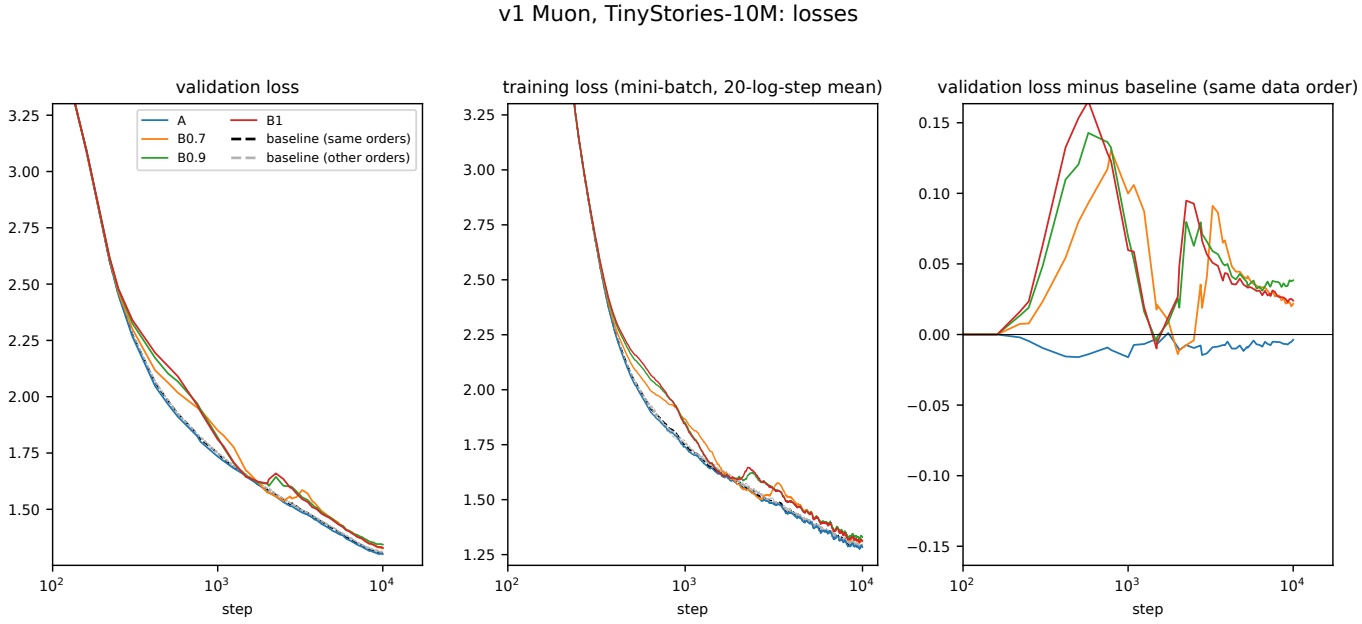


Figure 52: Balancer trace of SGD B1-fe-fresh, B1-fe and B1 (data order 0) and A-fe at 0.12 (order 1) on TinyStories-10M: rate multipliers κm_a with κ dotted, smoothed shares, and $\lambda_{\max}(\hat{H})$ over the threshold, tracked and checked by Lanczos, with the target dashed.

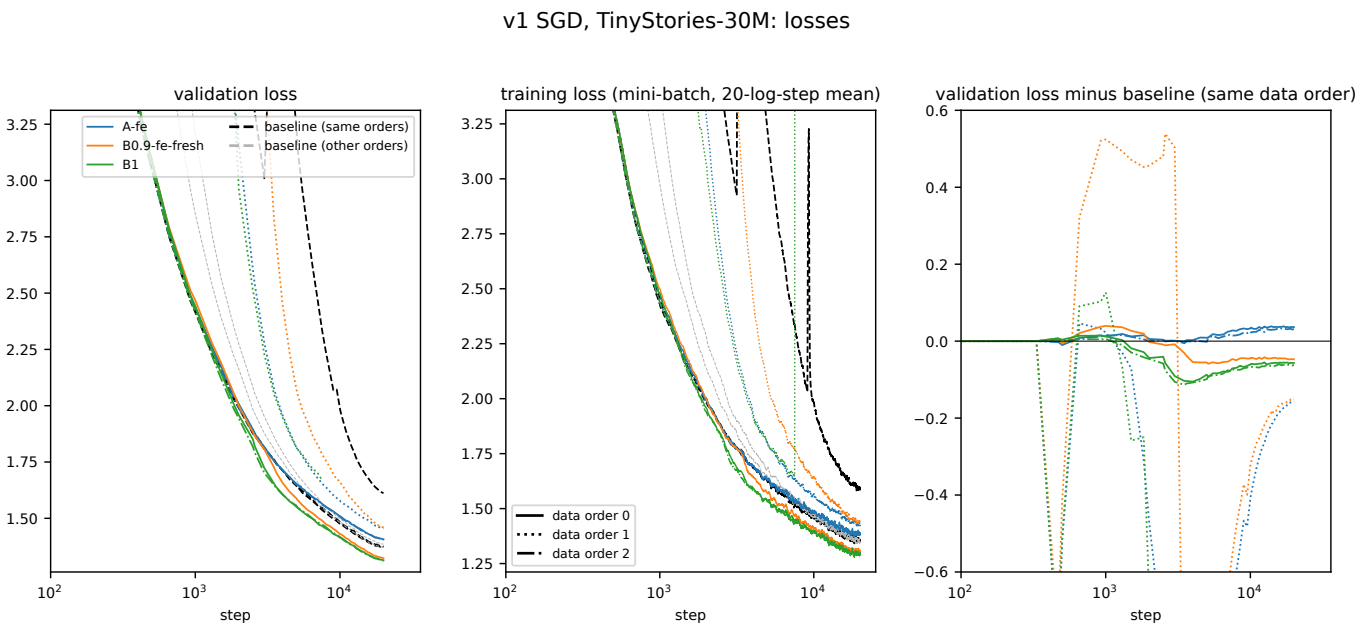


run	last step	final val	minus baseline	best val	minus baseline	reaches baseline best	speedup
adamw_A-fe_d0	10000	1.3454	+0.0063	1.3454	+0.0063	-	-
adamw_A_d0	10000	1.3567	+0.0175	1.3567	+0.0175	-	-
adamw_B0.5_d0	10000	1.4952	+0.1561	1.4618	+0.1227	-	-
adamw_B0.7_d0	10000	1.4660	+0.1269	1.4628	+0.1237	-	-
adamw_B0.9_d0	10000	1.4100	+0.0709	1.4100	+0.0709	-	-
adamw_B1_d0	10000	1.4003	+0.0612	1.4003	+0.0612	-	-
adamw_A_d1	10000	1.3615	+0.0153	1.3615	+0.0153	-	-
adamw_A_d2	10000	1.3611	+0.0165	1.3611	+0.0165	-	-

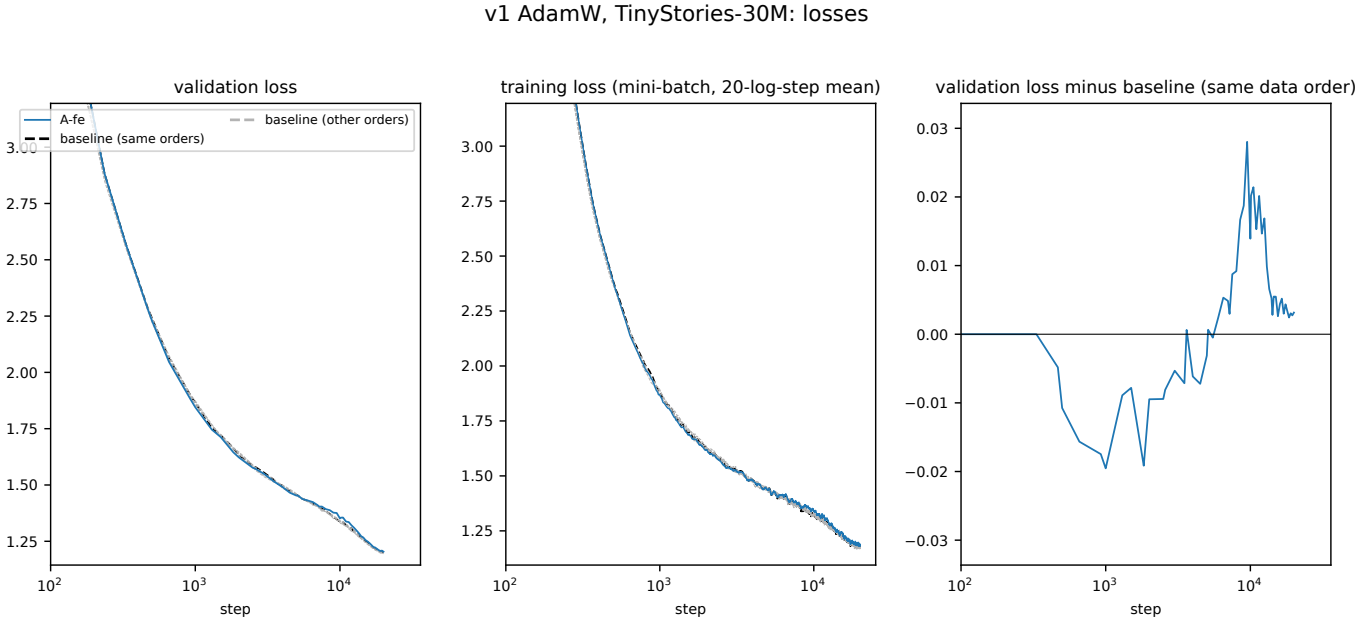


run	last step	final val	minus baseline	best val	minus baseline	reaches baseline best	speedup
muon_A_d0	10000	1.3012	-0.0037	1.3012	-0.0037	9250	1.08x
muon_B0.7_d0	10000	1.3267	+0.0218	1.3267	+0.0218	-	-
muon_B0.9_d0	10000	1.3433	+0.0384	1.3433	+0.0384	-	-
muon_B1_d0	10000	1.3290	+0.0241	1.3290	+0.0241	-	-

Figure 53: AdamW (top) and Muon (bottom) on TinyStories-10M, as in Fig. 51.

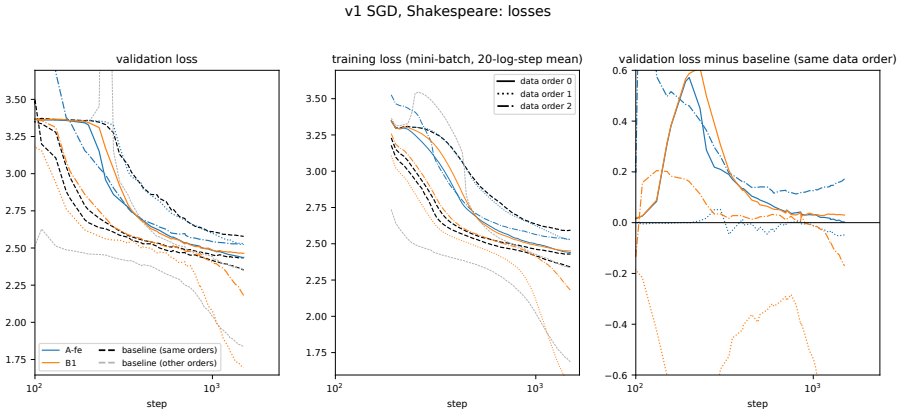


run	last step	final val	minus baseline	best val	minus baseline	reaches baseline best	speedup
sgd_A-fe_d0	20000	1.4066	+0.0367	1.4066	+0.0367	-	-
sgd_B0.9-fe-fresh_d0	20000	1.3229	-0.0471	1.3229	-0.0471	14000	1.43x
sgd_B1_d0	20000	1.3133	-0.0566	1.3133	-0.0566	13000	1.54x
sgd_A-fe_d1	20000	1.4559	-0.1559	1.4559	-0.1559	9000	2.22x
sgd_B0.9-fe-fresh_d1	20000	1.4601	-0.1517	1.4601	-0.1517	11500	1.74x
sgd_B1_d1	7500	1.6705	+0.0587	1.6705	+0.0587	-	-
sgd_A-fe_d2	20000	1.4051	+0.0317	1.4051	+0.0317	-	-
sgd_B1_d2	20000	1.3117	-0.0617	1.3117	-0.0617	13000	1.54x

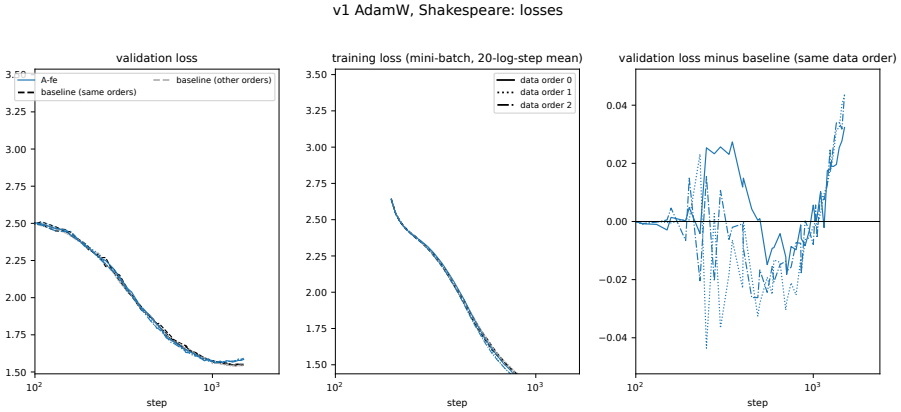


run	last step	final val	minus baseline	best val	minus baseline	reaches baseline best	speedup
adamw_A-fe_d0	20000	1.2040	+0.0032	1.2040	+0.0032	-	-

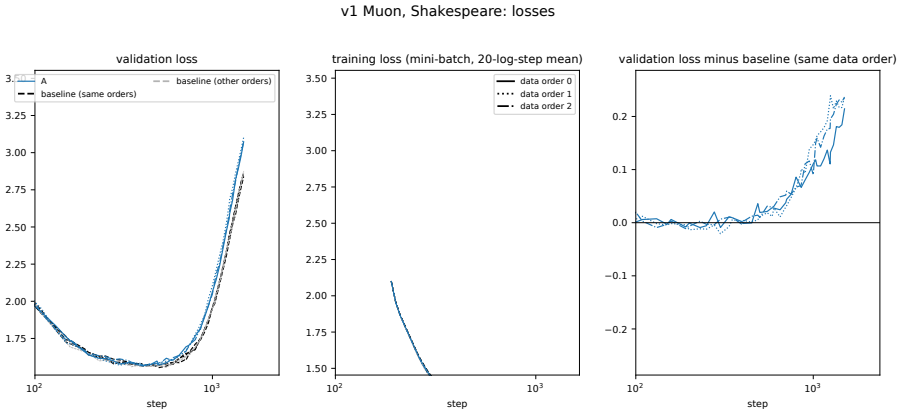
Figure 54: SGD (top) and AdamW (bottom) on TinyStories-30M, as in Fig. 51.



run	last step	final val	minus baseline	best val	minus baseline	reaches baseline best	speedup
sgd_A-fe_d0	1500	2.4363	+0.0024	2.4363	+0.0027	-	-
sgd_B1_d0	1500	2.4637	+0.0298	2.4637	+0.0301	-	-
sgd_A-fe_d1	1500	2.5300	-0.0481	2.5300	-0.0481	1100	1.36x
sgd_B1_d1	1500	1.6955	-0.8826	1.6955	-0.8826	250	6.00x
sgd_A-fe_d2	1500	2.5234	+0.1721	2.5234	+0.1721	-	-
sgd_B1_d2	1500	2.1810	-0.1703	2.1810	-0.1703	1200	1.25x

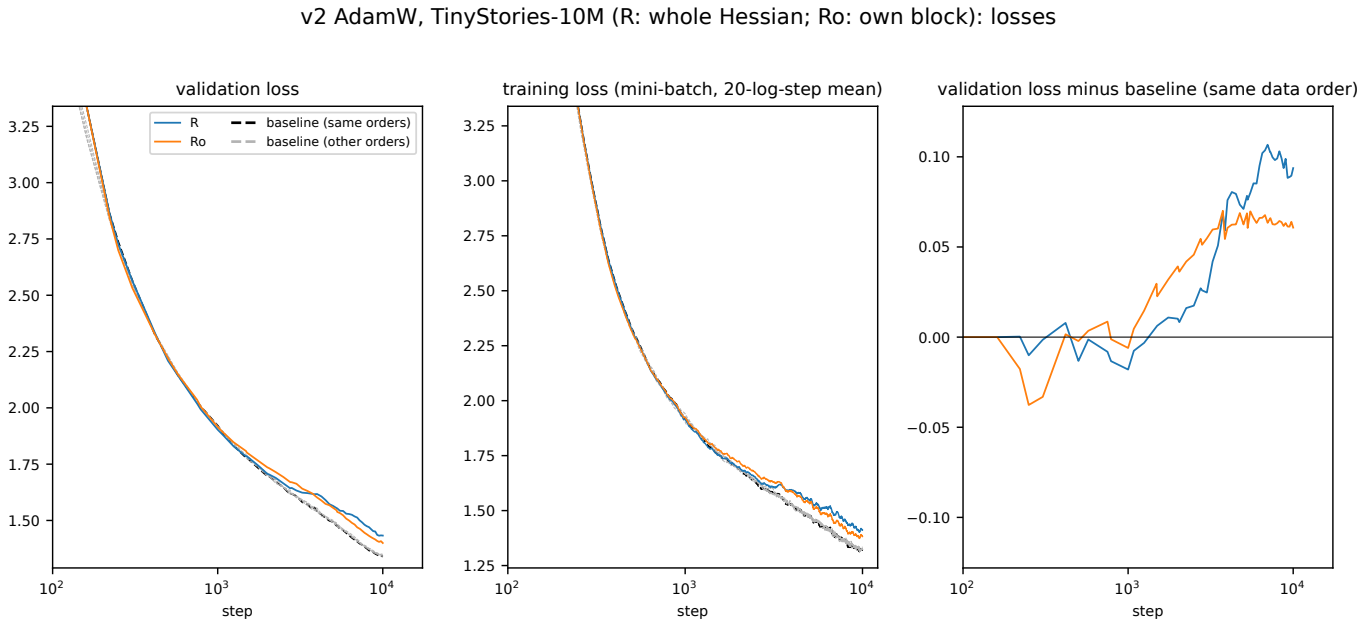


run	last step	final val	minus baseline	best val	minus baseline	reaches baseline best	speedup
adamw_A-fe_d0	1500	1.5848	+0.0323	1.5605	+0.0129	-	-
adamw_A-fe_d1	1500	1.5899	+0.0434	1.5627	+0.0171	-	-
adamw_A-fe_d2	1500	1.5927	+0.0437	1.5633	+0.0216	-	-

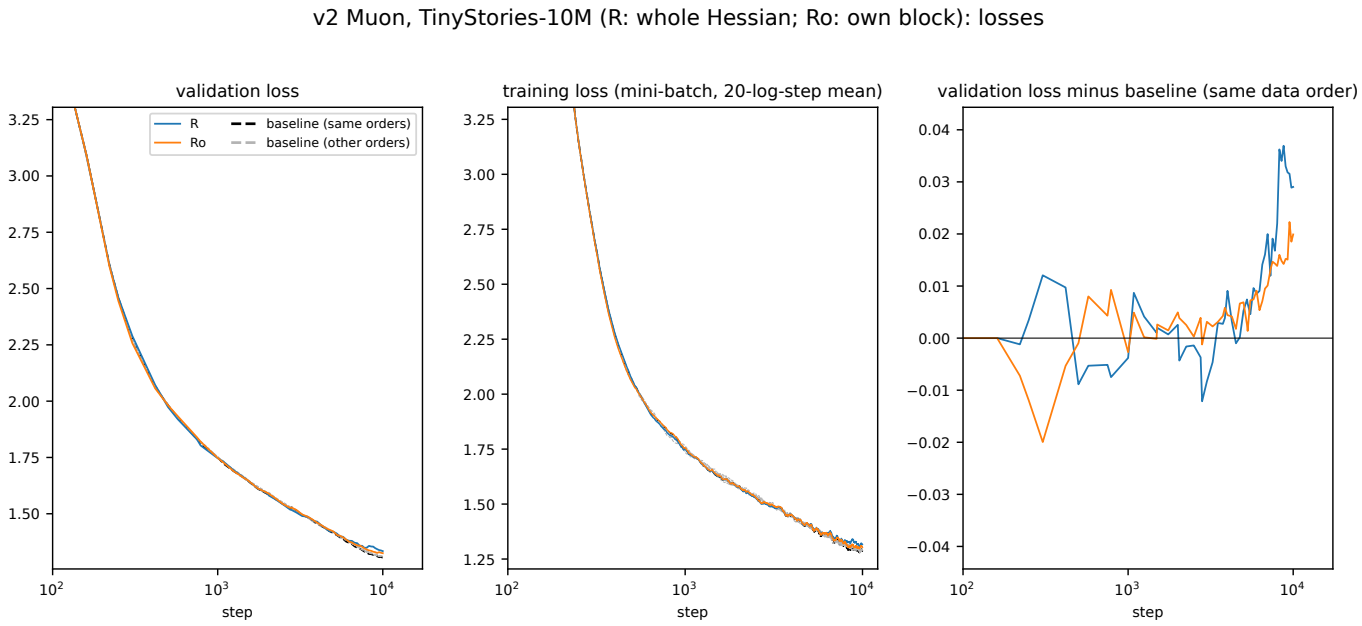


run	last step	final val	minus baseline	best val	minus baseline	reaches baseline best	speedup
muon_A_d0	1500	3.0729	+0.2143	1.5660	+0.0123	-	-
muon_A_d1	1500	3.1027	+0.2370	1.5684	+0.0055	-	-
muon_A_d2	1500	3.0756	+0.2366	1.5661	+0.0028	-	-

Figure 55: SGD (top), AdamW and Muon (bottom) on Shakespeare, as in Fig. 51; the runs overfit, so the tables also give the best validation loss.



run	last step	final val	minus baseline	best val	minus baseline	reaches baseline best	speedup
adamw_R_d0	10000	1.4329	+0.0937	1.4323	+0.0932	-	-
adamw_Ro_d0	10000	1.3999	+0.0607	1.3999	+0.0607	-	-



run	last step	final val	minus baseline	best val	minus baseline	reaches baseline best	speedup
muon_R_d0	10000	1.3339	+0.0290	1.3339	+0.0290	-	-
muon_Ro_d0	10000	1.3248	+0.0199	1.3248	+0.0199	-	-

Figure 56: v2 for AdamW (top) and Muon (bottom) on TinyStories-10M, with R_a and with the own block (Ro), as in Fig. 51.